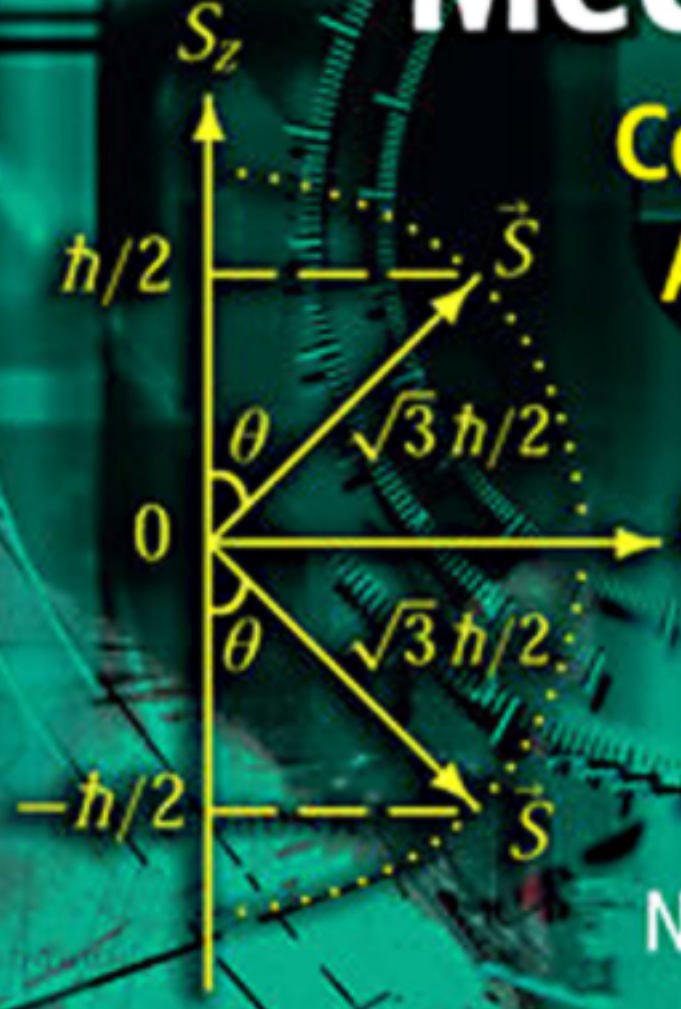
SECOND EDITION

Quantum Mechanics

Concepts and Applications



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Chapter 3

Postulates of Quantum Mechanics

3.1 Introduction

The formalism of quantum mechanics is based on a number of postulates. These postulates are in turn based on a wide range of experimental observations; the underlying physical ideas of these experimental observations have been briefly mentioned in Chapter 1. In this chapter we present a formal discussion of these postulates, and how they can be used to extract quantitative information about microphysical systems.

These postulates cannot be derived; they result from experiment. They represent the minimal set of assumptions needed to develop the theory of quantum mechanics. But how does one find out about the validity of these postulates? Their validity cannot be determined directly; only an indirect inferential statement is possible. For this, one has to turn to the theory built upon these postulates: if the theory works, the postulates will be valid; otherwise they will make no sense. Quantum theory not only works, but works extremely well, and this represents its experimental justification. It has a very penetrating qualitative as well as quantitative prediction power; this prediction power has been verified by a rich collection of experiments. So the accurate prediction power of quantum theory gives irrefutable evidence to the validity of the postulates upon which the theory is built.

3.2 The Basic Postulates of Quantum Mechanics

According to classical mechanics, the state of a particle is specified, at any time t, by two fundamental dynamical variables: the position $\vec{r}(t)$ and the momentum $\vec{p}(t)$. Any other physical quantity, relevant to the system, can be calculated in terms of these two dynamical variables. In addition, knowing these variables at a time t, we can predict, using for instance Hamilton's equations $dx/dt = \partial H/\partial p$ and $dp/dt = -\partial H/\partial x$, the values of these variables at any later time t'.

The quantum mechanical counterparts to these ideas are specified by postulates, which enable us to understand:

- how a quantum state is described mathematically at a given time t,
- how to calculate the various physical quantities from this quantum state, and

- knowing the system's state at a time t, how to find the state at any later time t'; that is, how to describe the time evolution of a system.

The answers to these questions are provided by the following set of five postulates.

Postulate 1: State of a system
The state of any physical system is specified, at each time t, by a state vector $|\psi(t)\rangle$ in a Hilbert space $\mathcal{H}$; $|\psi(t)\rangle$ contains (and serves as the basis to extract) all the needed information about the system. Any superposition of state vectors is also a state vector.

Postulate 2: Observables and operators
To every physically measurable quantity A, called an observable or dynamical variable, there corresponds a linear Hermitian operator $\hat{A}$ whose eigenvectors form a complete basis.

Postulate 3: Measurements and eigenvalues of operators
The measurement of an observable A may be represented formally by the action of $\hat{A}$ on a state vector $|\psi(t)\rangle$. The only possible result of such a measurement is one of the eigenvalues a_n (which are real) of the operator $\hat{A}$. If the result of a measurement of A on a state $|\psi(t)\rangle$ is a_n, the state of the system *immediately after* the measurement changes to $|\psi_n\rangle$:

$$\hat{A}|\psi(t)\rangle = a_n|\psi_n\rangle, \tag{3.1}$$

where $a_n = \langle\psi_n|\psi(t)\rangle$. **Note:** a_n is the component of $|\psi(t)\rangle$ when projected[1] onto the eigenvector $|\psi_n\rangle$.

Postulate 4: Probabilistic outcome of measurements

- **Discrete spectra**: When measuring an observable A of a system in a state $|\psi\rangle$, the probability of obtaining one of the nondegenerate eigenvalues a_n of the corresponding operator $\hat{A}$ is given by

$$P_n(a_n) = \frac{|\langle\psi_n|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \frac{|a_n|^2}{\langle\psi|\psi\rangle}, \tag{3.2}$$

  where $|\psi_n\rangle$ is the eigenstate of $\hat{A}$ with eigenvalue a_n. If the eigenvalue a_n is m-degenerate, P_n becomes

$$P_n(a_n) = \frac{\sum_{j=1}^{m} |\langle\psi_n^j|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \frac{\sum_{j=1}^{m} |a_n^{(j)}|^2}{\langle\psi|\psi\rangle}. \tag{3.3}$$

  The act of measurement changes the state of the system from $|\psi\rangle$ to $|\psi_n\rangle$. If the system is already in an eigenstate $|\psi_n\rangle$ of $\hat{A}$, a measurement of A yields with certainty the corresponding eigenvalue a_n: $\hat{A}|\psi_n\rangle = a_n|\psi_n\rangle$.

- **Continuous spectra:** The relation (3.2), which is valid for discrete spectra, can be extended to determine the probability density that a measurement of $\hat{A}$ yields a value between a and $a + da$ on a system which is initially in a state $|\psi\rangle$:

$$\frac{dP(a)}{da} = \frac{|\psi(a)|^2}{\langle\psi|\psi\rangle} = \frac{|\psi(a)|^2}{\int_{-\infty}^{+\infty} |\psi(a')|^2 da'}; \tag{3.4}$$

  for instance, the probability density for finding a particle between x and $x + dx$ is given by $dP(x)/dx = |\psi(x)|^2/\langle\psi|\psi\rangle$.

[1] To see this, we need only to expand $|\psi(t)\rangle$ in terms of the eigenvectors of $\hat{A}$ which form a complete basis: $|\psi(t)\rangle = \sum_n |\psi_n\rangle\langle\psi_n|\psi(t)\rangle = \sum_n a_n|\psi_n\rangle$.

Postulate 5: Time evolution of a system
The time evolution of the state vector $|\psi(t)\rangle$ of a system is governed by the time-dependent *Schrödinger equation*

$$i\hbar\frac{\partial|\psi(t)\rangle}{\partial t} = \hat{H}|\psi(t)\rangle, \tag{3.5}$$

where $\hat{H}$ is the Hamiltonian operator corresponding to the total energy of the system.

Remark
These postulates fall into two categories:

- The first four describe the system at a given time.
- The fifth shows how this description evolves in time.

In the rest of this chapter we are going to consider the physical implications of each one of the four postulates. Namely, we shall look at the state of a quantum system and its interpretation, the physical observables, measurements in quantum mechanics, and finally the time evolution of quantum systems.

3.3 The State of a System

To describe a system in quantum mechanics, we use a mathematical entity (a complex function) belonging to a Hilbert space, the state vector $|\psi(t)\rangle$, which contains all the information we need to know about the system and from which all needed physical quantities can be computed. As discussed in Chapter 2, the state vector $|\psi(t)\rangle$ may be represented in two ways:

- A wave function $\psi(\vec{r},t)$ in the position space: $\psi(\vec{r},t) = \langle\vec{r}|\psi(t)\rangle$.
- A momentum wave function $\Psi(\vec{p},t)$ in the momentum space: $\Psi(\vec{p},t) = \langle\vec{p}|\psi(t)\rangle$.

So, for instance, to describe the state of a one-dimensional particle in quantum mechanics we use a complex function $\psi(x,t)$ instead of two real real numbers $(x,\, p)$ in classical physics.

The wave functions to be used are only those that correspond to physical systems. What are the mathematical requirements that a wave function must satisfy to represent a physical system? Wave functions $\psi(x)$ that are physically acceptable must, along with their first derivatives $d\psi(x)/dx$, be *finite*, *continuous*, and *single-valued everywhere*. As will be discussed in Chapter 4, we will examine the underlying physics behind the continuity conditions of $\psi(x)$ and $d\psi(x)/dx$ (we will see that $\psi(x)$ and $d\psi(x)/dx$ must be be continuous because the probability density and the linear momentum are continuous functions of x).

3.3.1 Probability Density

What about the physical meaning of a wave function? Only the square of its norm, $|\psi(\vec{r},t)|^2$, has meaning. According to Born's probabilistic interpretation, the square of the norm of $\psi(\vec{r},t)$,

$$P(\vec{r},t) = |\psi(\vec{r},t)|^2, \tag{3.6}$$

represents a position probability density; that is, the quantity $|\psi(\vec{r},t)|^2 d^3r$ represents the probability of finding the particle at time t in a volume element d^3r located between $\vec{r}$ and $\vec{r}+d\vec{r}$. Therefore, the total probability of finding the system somewhere in space is equal to 1:

$$\int |\psi(\vec{r},t)|^2 d^3r = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dy \int_{-\infty}^{+\infty} |\psi(\vec{r},t)|^2 dz = 1. \tag{3.7}$$

A wave function $\psi(\vec{r},t)$ satisfying this relation is said to be *normalized*. We may mention that $\psi(\vec{r})$ has the physical dimensions of $1/\sqrt{L^3}$, where L is a length. Hence, the physical dimensions of $|\psi(\vec{r})|^2$ is $1/L^3$: $\left[|\psi(\vec{r})|^2\right] = 1/L^3$.

Note that the wave functions $\psi(\vec{r},t)$ and $e^{i\alpha}\psi(\vec{r},t)$, where α is a real number, represent the same state.

Example 3.1 (Physical and unphysical wave functions)
Which among the following functions represent physically acceptable wave functions: $f(x) = 3\sin \pi x$, $g(x) = 4 - |x|$, $h^2(x) = 5x$, and $e(x) = x^2$.

Solution
Among these functions only $f(x) = 3\sin \pi x$ represents a physically acceptable wave function, since $f(x)$ and its derivative are finite, continuous, single-valued everywhere, and integrable.

The other functions cannot be wave functions, since $g(x) = 4 - |x|$ is not continuous, not finite, and not square integrable; $h^2(x) = 5x$ is neither finite nor square integrable; and $e(x) = x^2$ is neither finite nor square integrable.

3.3.2 The Superposition Principle

The state of a system does not have to be represented by a *single* wave function; it can be represented by a *superposition* of two or more wave functions. An example from the macroscopic world is a vibrating string; its state can be represented by a single wave or by the superposition (linear combination) of many waves.

If $\psi_1(\vec{r},t)$ and $\psi_2(\vec{r},t)$ separately satisfy the Schrödinger equation, then the wave function $\psi(\vec{r},t) = \alpha_1\psi_1(\vec{r},t) + \alpha_2\psi_2(\vec{r},t)$ also satisfies the Schrödinger equation, where α_1 and α_2 are complex numbers. The Schrödinger equation is a linear equation. So in general, according to the superposition principle, the linear superposition of many wave functions (which describe the various permissible physical states of a system) gives a new wave function which represents a possible physical state of the system:

$$|\psi\rangle = \sum_i \alpha_i |\psi_i\rangle, \tag{3.8}$$

where the α_i are complex numbers. The quantity

$$P = \left| \sum_i \alpha_i |\psi_i\rangle \right|^2, \tag{3.9}$$

represents the probability for this superposition. If the states $|\psi_i\rangle$ are mutually *orthonormal*, the probability will be equal to the sum of the individual probabilities:

$$P = \left| \sum_i \alpha_i |\psi_i\rangle \right|^2 = \sum_i |\alpha_i|^2 = P_1 + P_2 + P_3 + \cdots, \tag{3.10}$$

where $P_i = |\alpha_i|^2$; P_i is the probability of finding the system in the state $|\psi_i\rangle$.

Example 3.2
Consider a system whose state is given in terms of an orthonormal set of three vectors: $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ as

$$|\psi\rangle = \frac{\sqrt{3}}{3}|\phi_1\rangle + \frac{2}{3}|\phi_2\rangle + \frac{\sqrt{2}}{3}|\phi_3\rangle.$$

(a) Verify that $|\psi\rangle$ is normalized. Then, calculate the probability of finding the system in any one of the states $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$. Verify that the total probability is equal to one.

(b) Consider now an ensemble of 810 identical systems, each one of them in the state $|\psi\rangle$. If measurements are done on all of them, how many systems will be found in each of the states $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$?

Solution
(a) Using the orthonormality condition $\langle\phi_j|\phi_k\rangle = \delta_{jk}$ where $j, k = 1, 2, 3$, we can verify that $|\psi\rangle$ is normalized:

$$\langle\psi|\psi\rangle = \frac{1}{3}\langle\phi_1|\phi_1\rangle + \frac{4}{9}\langle\phi_2|\phi_2\rangle + \frac{2}{9}\langle\phi_3|\phi_3\rangle = \frac{1}{3} + \frac{4}{9} + \frac{2}{9} = 1. \tag{3.11}$$

Since $|\psi\rangle$ is normalized, the probability of finding the system in $|\phi_1\rangle$ is given by

$$P_1 = |\langle\phi_1|\psi\rangle|^2 = \left| \frac{\sqrt{3}}{3}\langle\phi_1|\phi_1\rangle + \frac{2}{3}\langle\phi_1|\phi_2\rangle + \frac{\sqrt{2}}{3}\langle\phi_1|\phi_3\rangle \right|^2 = \frac{1}{3}, \tag{3.12}$$

since $\langle\phi_1|\phi_1\rangle = 1$ and $\langle\phi_1|\phi_2\rangle = \langle\phi_1|\phi_3\rangle = 0$.

Similarly, from the relations $\langle\phi_2|\phi_2\rangle = 1$ and $\langle\phi_2|\phi_1\rangle = \langle\phi_2|\phi_3\rangle = 0$, we obtain the probability of finding the system in $|\phi_2\rangle$:

$$P_2 = |\langle\phi_2|\psi\rangle|^2 = \left| \frac{2}{3}\langle\phi_2|\phi_2\rangle \right|^2 = \frac{4}{9}. \tag{3.13}$$

As for $\langle\phi_3|\phi_3\rangle = 1$ and $\langle\phi_3|\phi_1\rangle = \langle\phi_3|\phi_2\rangle = 0$, they lead to the probability of finding the system in $|\phi_3\rangle$:

$$P_3 = |\langle\phi_3|\psi\rangle|^2 = \left| \frac{\sqrt{2}}{3}\langle\phi_3|\phi_3\rangle \right|^2 = \frac{2}{9}. \tag{3.14}$$

As expected, the total probability is equal to one:

$$P = P_1 + P_2 + P_3 = \frac{1}{3} + \frac{4}{9} + \frac{2}{9} = 1. \tag{3.15}$$

(b) The number of systems that will be found in the state $|\phi_1\rangle$ is

$$N_1 = 810 \times P_1 = \frac{810}{3} = 270. \tag{3.16}$$

Likewise, the number of systems that will be found in states $|\phi_2\rangle$ and $|\phi_3\rangle$ are given, respectively, by

$$N_2 = 810 \times P_2 = \frac{810 \times 4}{9} = 360, \qquad N_3 = 810 \times P_3 = \frac{810 \times 2}{9} = 180. \tag{3.17}$$

3.4 Observables and Operators

An observable is a dynamical variable that can be measured; the dynamical variables encountered most in classical mechanics are the position, linear momentum, angular momentum, and energy. How do we mathematically represent these and other variables in quantum mechanics?

According to the second postulate, a *Hermitian operator* is associated with every *physical observable*. In the preceding chapter, we have seen that the position representation of the linear momentum operator is given in one-dimensional space by $\hat{P} = -i\hbar\partial/\partial x$ and in three-dimensional space by $\hat{\vec{P}} = -i\hbar\vec{\nabla}$.

In general, any function, $f(\vec{r}, \vec{p})$, which depends on the position and momentum variables, $\vec{r}$ and $\vec{p}$, can be "*quantized*" or made into a function of operators by replacing $\vec{r}$ and $\vec{p}$ with their corresponding operators:

$$f(\vec{r}, \vec{p}) \quad \longrightarrow \quad F(\hat{\vec{R}}, \hat{\vec{P}}) = f(\hat{\vec{R}}, -i\hbar\vec{\nabla}), \tag{3.18}$$

or $f(x, p) \rightarrow F(\hat{X}, \ -i\hbar\partial/\partial x)$. For instance, the operator corresponding to the Hamiltonian

$$H = \frac{1}{2m}\vec{p}^{\,2} + V(\vec{r}, t) \tag{3.19}$$

is given in the position representation by

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\hat{\vec{R}}, t), \tag{3.20}$$

where ∇^2 is the Laplacian operator; it is given in Cartesian coordinates by: $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$.

Since the momentum operator $\hat{\vec{P}}$ is Hermitian, and if the potential $V(\hat{\vec{R}}, t)$ is a real function, the Hamiltonian (3.19) is Hermitian. We saw in Chapter 2 that the eigenvalues of Hermitian operators are real. Hence, the spectrum of the Hamiltonian, which consists of the entire set of its eigenvalues, is real. This spectrum can be discrete, continuous, or a mixture of both. In the case of *bound* states, the Hamiltonian has a *discrete* spectrum of values and a *continuous* spectrum for *unbound* states. In general, an operator will have bound or unbound spectra in the same manner that the corresponding classical variable has bound or unbound orbits. As for $\hat{\vec{R}}$ and $\hat{\vec{P}}$, they have continuous spectra, since r and p may take a continuum of values.

Table 3.1 Some observables and their corresponding operators.

Observable	Corresponding operator
$\vec{r}$	$\hat{\vec{R}}$
$\vec{p}$	$\hat{\vec{P}} = -i\hbar\vec{\nabla}$
$T = \frac{p^2}{2m}$	$\hat{T} = -\frac{\hbar^2}{2m}\nabla^2$
$E = \frac{p^2}{2m} + V(\vec{r}, t)$	$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + \hat{V}(\hat{\vec{R}}, t)$
$\vec{L} = \vec{r} \times \vec{p}$	$\hat{\vec{L}} = -i\hbar\hat{\vec{R}} \times \vec{\nabla}$

According to Postulate 5, the total energy E for time-dependent systems is associated to the operator

$$\hat{H} = i\hbar\frac{\partial}{\partial t}. \tag{3.21}$$

This can be seen as follows. The wave function of a free particle of momentum $\vec{p}$ and total energy E is given by $\psi(\vec{r}, t) = Ae^{i(\vec{p}\cdot\vec{r} - Et)/\hbar}$, where A is a constant. The time derivative of $\psi(\vec{r}, t)$ yields

$$i\hbar\frac{\partial\psi(\vec{r}, t)}{\partial t} = E\psi(\vec{r}, t). \tag{3.22}$$

Let us look at the eigenfunctions and eigenvalues of the momentum operator $\hat{\vec{P}}$. The eigenvalue equation

$$-i\hbar\vec{\nabla}\psi(\vec{r}) = \vec{p}\psi(\vec{r}) \tag{3.23}$$

yields the eigenfunction $\psi(\vec{r})$ corresponding to the eigenvalue $\vec{p}$ such that $|\psi(\vec{r})|^2 d^3r$ is the probability of finding the particle with a momentum $\vec{p}$ in the volume element d^3r centered about $\vec{r}$. The solution to the eigenvalue equation (3.23) is

$$\psi(\vec{r}) = Ae^{i\vec{p}\cdot\vec{r}/\hbar}, \tag{3.24}$$

where A is a normalization constant. Since $\vec{p} = \hbar\vec{k}$ is the eigenvalue of the operator $\hat{\vec{P}}$, the eigenfunction (3.24) reduces to $\psi(\vec{r}) = Ae^{i\vec{k}\cdot\vec{r}}$; hence the eigenvalue equation (3.23) becomes

$$\hat{\vec{P}}\psi(\vec{r}) = \hbar\vec{k}\psi(\vec{r}). \tag{3.25}$$

To summarize, there is a one-to-one correspondence between observables and operators (Table 3.1).

Example 3.3 (Orbital angular momentum)
Find the operator representing the classical orbital angular momentum.

Solution
The classical expression for the orbital angular momentum of a particle whose position and linear momentum are $\vec{r}$ and $\vec{p}$ is given by $\vec{L} = \vec{r} \times \vec{p} = l_x\vec{i} + l_y\vec{j} + l_z\vec{k}$, where $l_x = yp_z - zp_y$, $l_y = zp_x - xp_z$, $l_z = xp_y - yp_x$.

To find the operator representing the classical angular momentum, we need simply to replace $\vec{r}$ and $\vec{p}$ with their corresponding operators $\hat{\vec{R}}$ and $\hat{\vec{P}} = -i\hbar\vec{\nabla}$: $\hat{\vec{L}} = -i\hbar\hat{\vec{R}} \times \vec{\nabla}$. This leads to

$$\hat{L}_x = \hat{Y}\hat{P}_z - \hat{Z}\hat{P}_y = -i\hbar\left(\hat{Y}\frac{\partial}{\partial z} - \hat{Z}\frac{\partial}{\partial y}\right), \tag{3.26}$$

$$\hat{L}_y = \hat{Z}\hat{P}_x - \hat{X}\hat{P}_z = -i\hbar\left(\hat{Z}\frac{\partial}{\partial x} - \hat{X}\frac{\partial}{\partial Z}\right), \tag{3.27}$$

$$\hat{L}_z = \hat{X}\hat{P}_y - \hat{Y}\hat{P}_x = -i\hbar\left(\hat{X}\frac{\partial}{\partial y} - \hat{Y}\frac{\partial}{\partial x}\right). \tag{3.28}$$

Recall that in classical mechanics the position and momentum components commute, $xp_x = p_x x$, and so do the components of the angular momentum, $l_x l_y = l_y l_x$. In quantum mechanics, however, this is not the case, since $\hat{X}\hat{P}_x = \hat{P}_x\hat{X} + i\hbar$ and, as will be shown in Chapter 5, $\hat{L}_x\hat{L}_y = \hat{L}_y\hat{L}_x + i\hbar\hat{L}_z$, and so on.

3.5 Measurement in Quantum Mechanics

Quantum theory is about the results of measurement; it says nothing about what might happen in the physical world outside the context of measurement. So the emphasis is on measurement.

3.5.1 How Measurements Disturb Systems

In classical physics it is possible to perform measurements on a system without disturbing it significantly. In quantum mechanics, however, the measurement process perturbs the system significantly. While carrying out measurements on classical systems, this perturbation does exist, but it is small enough that it can be neglected. In atomic and subatomic systems, however, the act of measurement induces nonnegligible or significant disturbances.

As an illustration, consider an experiment that measures the position of a hydrogenic electron. For this, we need to bombard the electron with electromagnetic radiation (photons). If we want to determine the position accurately, the wavelength of the radiation must be sufficiently short. Since the electronic orbit is of the order of 10^{-10} m, we must use a radiation whose wavelength is smaller than 10^{-10} m. That is, we need to bombard the electron with photons of energies higher than

$$h\nu = h\frac{c}{\lambda} = h\frac{3 \times 10^8}{10^{-10}} \sim 10^4 \text{ eV}. \tag{3.29}$$

When such photons strike the electron, not only will they perturb it, they will knock it completely off its orbit; recall that the ionization energy of the hydrogen atom is about 13.5 eV. Thus, the mere act of measuring the position of the electron disturbs it appreciably.

Let us now discuss the general concept of measurement in quantum mechanics. *The act of measurement generally changes the state of the system*. In theory we can represent the measuring device by an operator so that, after carrying out the measurement, the system will be in one of the eigenstates of the operator. Consider a system which is in a state $|\psi\rangle$. Before measuring an observable A, the state $|\psi\rangle$ can be represented by a linear superposition of eigenstates $|\psi_n\rangle$

of the corresponding operator $\hat{A}$:

$$|\psi\rangle = \sum_n |\psi_n\rangle\langle\psi_n|\psi\rangle = \sum_n a_n|\psi_n\rangle. \tag{3.30}$$

According to Postulate 4, the act of measuring A changes the state of the system from $|\psi\rangle$ to one of the eigenstates $|\psi_n\rangle$ of the operator $\hat{A}$, and the result obtained is the eigenvalue a_n. The only *exception* to this rule is when the *system is already in one of the eigenstates of the observable being measured.* For instance, if the system is in the eigenstate $|\psi_n\rangle$, a measurement of the observable A yields with certainty (i.e., with probability = 1) the value a_n without changing the state $|\psi_n\rangle$.

Before a measurement, we do not know in advance with certainty in which eigenstate, among the various states $|\psi_n\rangle$, a system will be after the measurement; only a probabilistic outcome is possible. Postulate 4 states that the probability of finding the system in one particular nondegenerate eigenstate $|\psi_n\rangle$ is given by

$$P_n = \frac{|\langle\psi_n|\psi\rangle|^2}{\langle\psi|\psi\rangle}. \tag{3.31}$$

Note that the wave function does not predict the results of individual measurements; it instead determines the probability distribution, $P = |\psi|^2$, over measurements on many identical systems in the same state.

Finally, we may state that quantum mechanics is the mechanics applicable to objects for which measurements necessarily interfere with the state of the system. Quantum mechanically, we cannot ignore the effects of the measuring equipment on the system, for they are important. In general, certain measurements cannot be performed without major disturbances to other properties of the quantum system. In conclusion, *it is the effects of the interference by the equipment on the system which is the essence of quantum mechanics*.

3.5.2 Expectation Values

The expectation value $\langle\hat{A}\rangle$ of $\hat{A}$ with respect to a state $|\psi\rangle$ is defined by

$$\langle\hat{A}\rangle = \frac{\langle\psi|\hat{A}|\psi\rangle}{\langle\psi|\psi\rangle}. \tag{3.32}$$

For instance, the energy of a system is given by the expectation value of the Hamiltonian: $E = \langle\hat{H}\rangle = \langle\psi|\hat{H}|\psi\rangle/\langle\psi|\psi\rangle$.

In essence, the expectation value $\langle\hat{A}\rangle$ represents the average result of measuring $\hat{A}$ on the state $|\ \psi\rangle$. To see this, using the complete set of eigenvectors $|\psi_n\rangle$ of $\hat{A}$ as a basis (i.e., $\hat{A}$ is diagonal in ψ_n), we can rewrite $\langle\hat{A}\rangle$ as follows:

$$\langle\hat{A}\rangle = \frac{1}{\langle\psi|\psi\rangle}\sum_{nm}\langle\psi|\psi_m\rangle\langle\psi_m|\hat{A}|\psi_n\rangle\langle\psi_n|\psi\rangle = \sum_n a_n\frac{|\langle\psi_n|\psi\rangle|^2}{\langle\psi|\psi\rangle}, \tag{3.33}$$

where we have used $\langle\psi_m|\hat{A}|\psi_n\rangle = a_n\delta_{nm}$. Since the quantity $|\langle\psi_n|\psi\rangle|^2/\langle\psi|\psi\rangle$ gives the probability P_n of finding the value a_n after measuring the observable A, we can indeed interpret $\langle\hat{A}\rangle$ as an *average* of a series of measurements of A:

$$\boxed{\langle\hat{A}\rangle = \sum_n a_n\frac{|\langle\psi_n|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \sum_n a_n P_n.} \tag{3.34}$$

That is, the expectation value of an observable is obtained by adding all permissible eigenvalues a_n, with each a_n multiplied by the corresponding probability P_n.

The relation (3.34), which is valid for *discrete* spectra, can be extended to a *continuous* distribution of probabilities $P(a)$ as follows:

$$\langle \hat{A} \rangle = \frac{\int_{-\infty}^{+\infty} a\,|\psi(a)|^2 da}{\int_{-\infty}^{+\infty} |\psi(a)|^2 da} = \int_{-\infty}^{+\infty} a\,dP(a). \tag{3.35}$$

The expectation value of an observable can be obtained physically as follows: prepare a very large number of *identical* systems each in the *same* state $|\psi\rangle$. The observable A is then measured on all these identical systems; the results of these measurements are $a_1, a_2, \ldots, a_n, \ldots$; the corresponding probabilities of occurrence are $P_1, P_2, \ldots, P_n, \ldots$. The average value of all these repeated measurements is called the expectation value of $\hat{A}$ with respect to the state $|\psi\rangle$.

Note that the process of obtaining different results when measuring the same observable on many identically prepared systems is contrary to classical physics, where these measurements must give the same outcome. In quantum mechanics, however, we can predict only the probability of obtaining a certain value for an observable.

Example 3.4

Consider a system whose state is given in terms of a complete and orthonormal set of five vectors $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle, |\phi_4\rangle, |\phi_5\rangle$ as follows:

$$|\psi\rangle = \frac{1}{\sqrt{19}}|\phi_1\rangle + \frac{2}{\sqrt{19}}|\phi_2\rangle + \sqrt{\frac{2}{19}}|\phi_3\rangle + \sqrt{\frac{3}{19}}|\phi_4\rangle + \sqrt{\frac{5}{19}}|\phi_5\rangle,$$

where $|\phi_n\rangle$ are eigenstates to the system's Hamiltonian, $\hat{H}|\phi_n\rangle = n\varepsilon_0|\phi_n\rangle$ with $n = 1, 2, 3, 4, 5$, and where ε_0 has the dimensions of energy.

(a) If the energy is measured on a large number of identical systems that are all initially in the same state $|\psi\rangle$, what values would one obtain and with what probabilities?

(b) Find the average energy of one such system.

Solution

First, note that $|\psi\rangle$ is not normalized:

$$\langle\psi|\psi\rangle = \sum_{n=1}^{5} a_n^2 \langle\phi_n|\phi_n\rangle = \sum_{n=1}^{5} a_n^2 = \frac{1}{19} + \frac{4}{19} + \frac{2}{19} + \frac{3}{19} + \frac{5}{19} = \frac{15}{19}, \tag{3.36}$$

since $\langle\phi_j|\phi_k\rangle = \delta_{jk}$ with $j, k = 1, 2, 3, 4, 5$.

(a) Since $E_n = \langle\phi_n|\hat{H}|\phi_n\rangle = n\varepsilon_0$ ($n = 1, 2, 3, 4, 5$), the various measurements of the energy of the system yield the values $E_1 = \varepsilon_0$, $E_2 = 2\varepsilon_0$, $E_3 = 3\varepsilon_0$, $E_4 = 4\varepsilon_0$, $E_5 = 5\varepsilon_0$ with the following probabilities:

$$P_1(E_1) = \frac{|\langle\phi_1|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \left|\frac{1}{\sqrt{19}}\langle\phi_1|\phi_1\rangle\right|^2 \times \frac{19}{15} = \frac{1}{15}, \tag{3.37}$$

$$P_2(E_2) = \frac{|\langle\phi_2|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \left|\frac{2}{\sqrt{19}}\langle\phi_2|\phi_2\rangle\right|^2 \times \frac{19}{15} = \frac{4}{15}, \tag{3.38}$$

$$P_3(E_3) = \frac{|\langle\phi_3|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \left|\sqrt{\frac{2}{19}}\langle\phi_3|\phi_3\rangle\right|^2 \times \frac{19}{15} = \frac{2}{15}, \tag{3.39}$$

$$P_4(E_4) = \frac{|\langle\phi_4|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \left|\sqrt{\frac{3}{19}}\langle\phi_4|\phi_4\rangle\right|^2 \times \frac{19}{15} = \frac{3}{15}, \tag{3.40}$$

and

$$P_5(E_5) = \frac{|\langle\phi_5|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \left|\sqrt{\frac{5}{19}}\langle\phi_5|\phi_5\rangle\right|^2 \times \frac{19}{15} = \frac{5}{15}, \tag{3.41}$$

(b) The average energy of a system is given by

$$E = \sum_{j=1}^{5} P_j E_j = \frac{1}{15}\varepsilon_0 + \frac{8}{15}\varepsilon_0 + \frac{6}{15}\varepsilon_0 + \frac{12}{15}\varepsilon_0 + \frac{25}{15}\varepsilon_0 = \frac{52}{15}\varepsilon_0. \tag{3.42}$$

This energy can also be obtained from the expectation value of the Hamiltonian:

$$\begin{aligned} E &= \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle} = \frac{19}{15}\sum_{n=1}^{5} a_n^2\langle\phi_n|\hat{H}|\phi_n\rangle = \frac{19}{15}\left(\frac{1}{19} + \frac{8}{19} + \frac{6}{19} + \frac{12}{19} + \frac{25}{19}\right)\varepsilon_0 \\ &= \frac{52}{15}\varepsilon_0, \end{aligned} \tag{3.43}$$

where the values of the coefficients a_n^2 are listed in (3.36).

3.5.3 Complete Sets of Commuting Operators (CSCO)

Two observables A and B are said to be *compatible* when their corresponding operators commute, $[\hat{A}, \hat{B}] = 0$; observables corresponding to noncommuting operators are said to be *noncompatible*.

In what follows we are going to consider the task of measuring two observables A and B on a given system. Since the act of measurement generally *perturbs* the system, the result of measuring A and B therefore depends on the *order* in which they are carried out. Measuring A first and then B leads[2] in general to results that are different from those obtained by measuring B first and then A. How does this take place?

If $\hat{A}$ and $\hat{B}$ do not commute and if the system is in an eigenstate $|\psi_n^{(a)}\rangle$ of $\hat{A}$, a measurement of A yields with certainty a value a_n, since $\hat{A}|\psi_n^{(a)}\rangle = a_n|\psi_n^{(a)}\rangle$. Then, when we measure B, the state of the system will be left in one of the eigenstates of B. If we measure A again, we will find a value which will be different from a_n. What is this new value? We cannot answer this question with certainty: only a probabilistic outcome is possible. For this, we need to expand the eigenstates of B in terms of those of A, and thus provide a probabilistic answer as to the value of measuring A. *So if $\hat{A}$ and $\hat{B}$ do not commute, they cannot be measured simultaneously; the order in which they are measured matters.*

[2] The act of measuring A first and then B is represented by the action of product $\hat{B}\hat{A}$ of their corresponding operators on the state vector.

What happens when A and B commute? We can show that the results of their measurements will not depend on the order in which they are carried out. Before showing this, let us mention a useful theorem.

Theorem 3.1 *If two observables are compatible, their corresponding operators possess a set of common (or simultaneous) eigenstates (this theorem holds for both degenerate and nondegenerate eigenstates).*

Proof
We provide here a proof for the nondegenerate case only. If $|\psi_n\rangle$ is a nondegenerate eigenstate of $\hat{A}$, $\hat{A}|\psi_n\rangle = a_n|\psi_n\rangle$, we have

$$\langle\psi_m|[\hat{A}, \hat{B}]|\psi_n\rangle = (a_m - a_n)\langle\psi_m|\hat{B}|\psi_n\rangle = 0, \tag{3.44}$$

since $\hat{A}$ and $\hat{B}$ commute. So $\langle\psi_m|\hat{B}|\psi_n\rangle$ must vanish unless $a_n = a_m$. That is,

$$\langle\psi_m|\hat{B}|\psi_n\rangle = \langle\psi_n|\hat{B}|\psi_n\rangle \propto \delta_{nm}. \tag{3.45}$$

Hence the $|\psi_n\rangle$ are *joint* or *simultaneous* eigenstates of $\hat{A}$ and $\hat{B}$ (this completes the proof).

Denoting the simultaneous eigenstate of $\hat{A}$ and $\hat{B}$ by $|\psi_{n_1}^{(a)}, \psi_{n_2}^{(b)}\rangle$, we have

$$\hat{A}|\psi_{n_1}^{(a)}, \psi_{n_2}^{(b)}\rangle = a_{n_1}|\psi_{n_1}^{(a)}, \psi_{n_2}^{(b)}\rangle, \tag{3.46}$$

$$\hat{B}|\psi_{n_1}^{(a)}, \psi_{n_2}^{(b)}\rangle = b_{n_2}|\psi_{n_1}^{(a)}, \psi_{n_2}^{(b)}\rangle. \tag{3.47}$$

Theorem 3.1 can be generalized to the case of many mutually compatible observables A, B, C, ... These compatible observables possess a *complete* set of *joint eigenstates*

$$|\psi_n\rangle = |\psi_{n_1}^{(a)}, \psi_{n_2}^{(b)}, \psi_{n_3}^{(c)}, \ldots\rangle. \tag{3.48}$$

The completeness and orthonormality conditions of this set are

$$\sum_{n_1}\sum_{n_2}\sum_{n_3}\cdots|\psi_{n_1}^{(a)}, \psi_{n_2}^{(b)}, \psi_{n_3}^{(c)}, \ldots\rangle\langle\psi_{n_1}^{(a)}, \psi_{n_2}^{(b)}, \psi_{n_3}^{(c)}, \ldots| = 1; \tag{3.49}$$

$$\langle\psi_{n'}|\psi_n\rangle = \delta_{n'n} = \delta_{n_1'n_1}\delta_{n_2'n_2}\delta_{n_3'n_3}\cdots. \tag{3.50}$$

Let us now show why, when two observables A and B are compatible, the order in which we carry out their measurements is irrelevant. Measuring A first, we would find a value a_n and would leave the system in an eigenstate of A. According to Theorem 3.1, this eigenstate is also an eigenstate of B. Thus a measurement of B yields with certainty b_n without affecting the state of the system. In this way, if we measure A again, we obtain with certainty the same initial value a_n. Similarly, another measurement of B will yield b_n and will leave the system in the same joint eigenstate of A and B. Thus, if two observables A and B are compatible, and if the system is initially in an eigenstate of one of their operators, their measurements not only yield precise values (eigenvalues) but they will not depend on the order in which the measurements were performed. In this case, A and B are said to be *simultaneously measurable*. *So compatible observables can be measured simultaneously with arbitrary accuracy; noncompatible observables cannot*.

What happens if an operator, say $\hat{A}$, has *degenerate* eigenvalues? The specification of one eigenvalue does not uniquely determine the state of the system. Among the degenerate

eigenstates of $\hat{A}$, only a subset of them are also eigenstates of $\hat{B}$. Thus, the set of states that are joint eigenstates of both $\hat{A}$ and $\hat{B}$ is not complete. To resolve the degeneracy, we can introduce a third operator $\hat{C}$ which commutes with both $\hat{A}$ and $\hat{B}$; then we can construct a set of joint eigenstates of $\hat{A}$, $\hat{B}$, and $\hat{C}$ that is complete. If the degeneracy persists, we may introduce a fourth operator $\hat{D}$ that commutes with the previous three and then look for their joint eigenstates which form a complete set. Continuing in this way, we will ultimately exhaust all the operators (that is, there are no more independent operators) which commute with each other. When that happens, we have then obtained a *complete set of commuting operators* (CSCO). Only then will the state of the system be specified unambiguously, for the joint eigenstates of the CSCO are determined uniquely and will form a complete set (recall that a complete set of eigenvectors of an operator is called a basis). We should, at this level, state the following definition.

Definition: A set of Hermitian operators, $\hat{A}$, $\hat{B}$, $\hat{C}$, ..., is called a CSCO if the operators mutually commute and if the set of their common eigenstates is complete and not degenerate (i.e., unique).

The complete commuting set may sometimes consist of only one operator. *Any operator with nondegenerate eigenvalues constitutes, all by itself, a CSCO*. For instance, the position operator $\hat{X}$ of a one-dimensional, spinless particle provides a complete set. Its momentum operator $\hat{P}$ is also a complete set; together, however, $\hat{X}$ and $\hat{P}$ cannot form a CSCO, for they do not commute. In three-dimensional problems, the three-coordinate position operators $\hat{X}$, $\hat{Y}$, and $\hat{Z}$ form a CSCO; similarly, the components of the momentum operator $\hat{P}_x$, $\hat{P}_y$, and $\hat{P}_z$ also form a CSCO. In the case of spherically symmetric three-dimensional potentials, the set $\hat{H}$, $\hat{\vec{L}}^2$, $\hat{L}_z$ forms a CSCO. Note that in this case of spherical symmetry, we need three operators to form a CSCO because $\hat{H}$, $\hat{\vec{L}}^2$, and $\hat{L}_z$ are all degenerate; hence the complete and unique determination of the wave function cannot be achieved with one operator or with two.

In summary, when a given operator, say $\hat{A}$, is degenerate, the wave function cannot be determined uniquely unless we introduce one or more additional operators so as to form a complete commuting set.

3.5.4 Measurement and the Uncertainty Relations

We have seen in Chapter 2 that the uncertainty condition pertaining to the measurement of any two observables A and B is given by

$$\Delta A \Delta B \geq \frac{1}{2}|\langle[\hat{A},\ \hat{B}]\rangle|, \tag{3.51}$$

where $\Delta A = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2}$.

Let us illustrate this on the joint measurement of the position and momentum observables. Since these observables are not compatible, their simultaneous measurement with infinite accuracy is not possible; that is, since $[\hat{X}, \hat{P}] = i\hbar$ there exists no state which is a simultaneous eigenstate of $\hat{X}$ and $\hat{P}$. For the case of the position and momentum operators, the relation (3.51) yields

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \tag{3.52}$$

This condition shows that the position and momentum of a microscopic system cannot be measured with infinite accuracy both at once. If the position is measured with an uncertainty Δx,

the uncertainty associated with its momentum measurement cannot be smaller than $\hbar/2\Delta x$. This is due to *the interference between the two measurements*. If we measure the position first, we perturb the system by changing its state to an eigenstate of the position operator; then the measurement of the momentum throws the system into an eigenstate of the momentum operator.

Another interesting application of the uncertainty relation (3.51) is to the orbital angular momentum of a particle. Since its components satisfy the commutator $[\hat{L}_x,\ \hat{L}_y] = i\hbar\hat{L}_z$, we obtain

$$\Delta L_x \Delta L_y \geq \frac{1}{2}\hbar|\langle\hat{L}_z\rangle|. \tag{3.53}$$

We can obtain the other two inequalities by means of a cyclic permutation of x, y, and z. If $\langle\hat{L}_z\rangle = 0$, $\hat{L}_x$ and $\hat{L}_y$ will have sharp values simultaneously. This occurs when the particle is in an s state. In fact, when a particle is in an s state, we have $\langle\hat{L}_x\rangle = \langle\hat{L}_y\rangle = \langle\hat{L}_z\rangle = 0$; hence all the components of orbital angular momentum will have sharp values simultaneously.

3.6 Time Evolution of the System's State

3.6.1 Time Evolution Operator

We want to examine here how quantum states evolve in time. That is, given the initial state $|\psi(t_0)\rangle$, how does one find the state $|\psi(t)\rangle$ at any later time t? The two states can be related by means of a linear operator $\hat{U}(t, t_0)$ such that

$$|\psi(t)\rangle = \hat{U}(t, t_0)|\psi(t_0)\rangle \qquad (t > t_0); \tag{3.54}$$

$\hat{U}(t, t_0)$ is known as the *time evolution operator* or *propagator*. From (3.54), we infer that

$$\hat{U}(t_0, t_0) = \hat{I}, \tag{3.55}$$

where $\hat{I}$ is the unit (identity) operator.

The issue now is to find $\hat{U}(t, t_0)$. For this, we need simply to substitute (3.54) into the time-dependent Schrödinger equation (3.5):

$$i\hbar\frac{\partial}{\partial t}\left(\hat{U}(t, t_0)|\psi(t_0)\rangle\right) = \hat{H}\left(\hat{U}(t, t_0)|\psi(t_0)\rangle\right) \tag{3.56}$$

or

$$\frac{\partial\hat{U}(t, t_0)}{\partial t} = -\frac{i}{\hbar}\hat{H}\hat{U}(t, t_0). \tag{3.57}$$

The integration of this differential equation depends on whether or not the Hamiltonian depends on time. If it does not depend on time, and taking into account the initial condition (3.55), we can easily ascertain that the integration of (3.57) leads to

$$\boxed{\hat{U}(t, t_0) = e^{-i(t-t_0)\hat{H}/\hbar} \qquad \text{and} \qquad |\psi(t)\rangle = e^{-i(t-t_0)\hat{H}/\hbar}|\psi(t_0)\rangle.} \tag{3.58}$$

We will show in Section 3.7 that the operator $\hat{U}(t, t_0) = e^{-i(t-t_0)\hat{H}/\hbar}$ represents a finite time translation.

If, on the other hand, $\hat{H}$ depends on time the integration of (3.57) becomes less trivial. We will deal with this issue in Chapter 10 when we look at time-dependent potentials or at the

time-dependent perturbation theory. In this chapter, and in all chapters up to Chapter 10, we will consider only Hamiltonians that do not depend on time.

Note that $\hat{U}(t, t_0)$ is a unitary operator, since

$$\hat{U}(t, t_0)\hat{U}^{\dagger}(t, t_0) = \hat{U}(t, t_0)\hat{U}^{-1}(t, t_0) = e^{-i(t-t_0)\hat{H}/\hbar}e^{i(t-t_0)\hat{H}/\hbar} = \hat{I} \tag{3.59}$$

or $\hat{U}^{\dagger} = \hat{U}^{-1}$.

3.6.2 Stationary States: Time-Independent Potentials

In the position representation, the time-dependent Schrödinger equation (3.5) for a particle of mass m moving in a time-dependent potential $\hat{V}(\vec{r}, t)$ can be written as follows:

$$\boxed{i\hbar\frac{\partial\Psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\Psi(\vec{r}, t) + \hat{V}(\vec{r}, t)\Psi(\vec{r}, t).} \tag{3.60}$$

Now, let us consider the particular case of *time-independent* potentials: $\hat{V}(\vec{r}, t) = \hat{V}(\vec{r})$. In this case the Hamiltonian operator will also be time independent, and hence the Schrödinger equation will have solutions that are *separable*, i.e., solutions that consist of a product of two functions, one depending only on $\vec{r}$ and the other only on time:

$$\Psi(\vec{r}, t) = \psi(\vec{r})f(t). \tag{3.61}$$

Substituting (3.61) into (3.60) and dividing both sides by $\psi(\vec{r})f(t)$, we obtain

$$i\hbar\frac{1}{f(t)}\frac{df(t)}{dt} = \frac{1}{\psi(\vec{r})}\left[-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r}) + \hat{V}(\vec{r})\psi(\vec{r})\right]. \tag{3.62}$$

Since the left-hand side depends only on time and the right-hand side depends only on $\vec{r}$, both sides must be equal to a constant; this constant, which we denote by E, has the dimensions of energy. We can therefore break (3.62) into two separate differential equations, one depending on time only,

$$i\hbar\frac{df(t)}{dt} = Ef(t), \tag{3.63}$$

and the other on the space variable $\vec{r}$,

$$\boxed{\left[-\frac{\hbar^2}{2m}\nabla^2 + \hat{V}(\vec{r})\right]\psi(\vec{r}) = E\psi(\vec{r}).} \tag{3.64}$$

This equation is known as the *time-independent* Schrödinger equation for a particle of mass m moving in a time-independent potential $\hat{V}(\vec{r})$.

The solutions to (3.63) can be written as $f(t) = e^{-iEt/\hbar}$; hence the state (3.61) becomes

$$\Psi(\vec{r}, t) = \psi(\vec{r})e^{-iEt/\hbar}. \tag{3.65}$$

This particular solution of the Schrödinger equation (3.60) for a *time-independent potential* is called a *stationary state*. Why is this state called *stationary*? The reason is obvious: the probability density is stationary, i.e., it does not depend on time:

$$|\Psi(\vec{r}, t)|^2 = |\psi(\vec{r})e^{-iEt/\hbar}|^2 = |\psi(\vec{r})|^2. \tag{3.66}$$

Note that such a state has a precise value for the energy, $E = \hbar\omega$.

In summary, stationary states, which are given by the solutions of (3.64), exist only for time-independent potentials. The set of energy levels that are solutions to this equation are called the *energy spectrum* of the system. The states corresponding to discrete and continuous spectra are called *bound* and *unbound* states, respectively. We will consider these questions in detail in Chapter 4.

The most general solution to the time-dependent Schrödinger equation (3.60) can be written as an expansion in terms of the stationary states $\psi_n(\vec{r})\exp(-iE_nt/\hbar)$:

$$\Psi(\vec{r},t) = \sum_n c_n \psi_n(\vec{r}) \exp\left(-\frac{iE_nt}{\hbar}\right), \tag{3.67}$$

where $c_n = \langle\psi_n|\Psi(t=0)\rangle = \int \psi_n^*(\vec{r})\psi(\vec{r})\,d^3r$. The general solution (3.67) is not a stationary state, because a linear superposition of stationary states is not necessarily a stationary state.

Remark
The time-dependent and time-independent Schrödinger equations are given in one dimension by (see (3.60) and (3.64))

$$\boxed{i\hbar\frac{\partial\Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + \hat{V}(x,t)\Psi(x,t),} \tag{3.68}$$

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \hat{V}(x)\psi(x) = E\psi(x).} \tag{3.69}$$

3.6.3 Schrödinger Equation and Wave Packets

Can we derive the Schrödinger equation (3.5) formally from first principles? No, we cannot; we can only postulate it. What we can do, however, is to provide an educated guess on the formal steps leading to it. *Wave packets* offer the formal tool to achieve that. We are going to show how to start from a wave packet and end up with the Schrödinger equation.

As seen in Chapter 1, the wave packet representing a particle of energy E and momentum p moving in a potential V is given by

$$\begin{aligned}\Psi(x,t) &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{+\infty}\tilde{\phi}(p)\exp\left[\frac{i}{\hbar}(px-Et)\right]dp \\ &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{+\infty}\tilde{\phi}(p)\exp\left[\frac{i}{\hbar}\left(px-\left(\frac{p^2}{2m}+V\right)t\right)\right]dp;\end{aligned} \tag{3.70}$$

recall that wave packets unify the corpuscular (E and p) and the wave (k and ω) features of particles: $k = p/\hbar$, $\hbar\omega = E = p^2/(2m) + V$. A partial time derivative of (3.70) yields

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{+\infty}\tilde{\phi}(p)\left(\frac{p^2}{2m}+V\right)\exp\left[\frac{i}{\hbar}\left(px-\left(\frac{p^2}{2m}+V\right)t\right)\right]dp. \tag{3.71}$$

Since $p^2/(2m) = -(\hbar^2/2m)\partial^2/\partial x^2$ and assuming that V is constant, we can take the term $-(\hbar^2/2m)\partial^2/\partial x^2 + V$ outside the integral sign, for it does not depend on p:

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t) = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V\right)\frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{+\infty}\tilde{\phi}(p)\exp\left[\frac{i}{\hbar}\left(px - \left(\frac{p^2}{2m} + V\right)t\right)\right]dp. \tag{3.72}$$

This can be written as

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t) = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V\right]\Psi(x,t). \tag{3.73}$$

Now, since this equation is valid for spatially varying potentials $V = V(x)$, we see that we have ended up with the Schrödinger equation (3.68).

3.6.4 The Conservation of Probability

Since the Hamiltonian operator is Hermitian, we can show that the norm $\langle\Psi(t)|\Psi(t)\rangle$, which is given by

$$\langle\Psi(t)|\Psi(t)\rangle = \int |\Psi(\vec{r},t)|^2\, d^3r, \tag{3.74}$$

is time independent. This means, if $|\Psi(t)\rangle$ is normalized, it stays normalized for all subsequent times. This is a direct consequence of the hermiticity of $\hat{H}$.

To prove that $\langle\Psi(t)|\Psi(t)\rangle$ is constant, we need simply to show that its time derivative is zero. First, the time derivative of $\langle\Psi(t)|\Psi(t)\rangle$ is

$$\frac{d}{dt}\langle\Psi(t)|\Psi(t)\rangle = \left(\frac{d}{dt}\langle\Psi(t)|\right)|\Psi(t)\rangle + \langle\Psi(t)|\left(\frac{d|\Psi(t)\rangle}{dt}\right), \tag{3.75}$$

where $d|\Psi(t)\rangle/dt$ and $d\langle\Psi(t)|/dt$ can be obtained from (3.5):

$$\frac{d}{dt}|\Psi(t)\rangle = -\frac{i}{\hbar}\hat{H}|\Psi(t)\rangle, \tag{3.76}$$

$$\frac{d}{dt}\langle\Psi(t)| = \frac{i}{\hbar}\langle\Psi(t)|\hat{H}^\dagger = \frac{i}{\hbar}\langle\Psi(t)|\hat{H}. \tag{3.77}$$

Inserting these two equations into (3.75), we end up with

$$\frac{d}{dt}\langle\Psi(t)|\Psi(t)\rangle = \left(\frac{i}{\hbar} - \frac{i}{\hbar}\right)\langle\Psi(t)|\hat{H}|\Psi(t)\rangle = 0. \tag{3.78}$$

Thus, the probability density $\langle\Psi|\Psi\rangle$ does not evolve in time.

In what follows we are going to calculate the probability density in the position representation. For this, we need to invoke the time-dependent Schrödinger equation

$$i\hbar\frac{\partial\Psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\Psi(\vec{r},t) + \hat{V}(\vec{r},t)\Psi(\vec{r},t) \tag{3.79}$$

and its complex conjugate

$$-i\hbar\frac{\partial\Psi^*(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\Psi^*(\vec{r},t) + \hat{V}(\vec{r},t)\Psi^*(\vec{r},t). \tag{3.80}$$

Multiplying both sides of (3.79) by $\Psi^*(\vec{r},t)$ and both sides of (3.80) by $\Psi(\vec{r},t)$, and subtracting the two resulting equations, we obtain

$$i\hbar\frac{\partial}{\partial t}\left[\Psi^*(\vec{r},t)\Psi(\vec{r},t)\right] = -\frac{\hbar^2}{2m}\left[\Psi^*(\vec{r},t)\nabla^2\Psi(\vec{r},t) - \Psi\nabla^2\Psi^*\right]. \tag{3.81}$$

We can rewrite this equation as

$$\boxed{\frac{\partial\rho(\vec{r},t)}{\partial t} + \vec{\nabla}\cdot\vec{J} = 0,} \tag{3.82}$$

where $\rho(\vec{r},t)$ and $\vec{J}$ are given by

$$\boxed{\rho(\vec{r},t) = \Psi^*(\vec{r},t)\Psi(\vec{r},t), \qquad \vec{J}(\vec{r},t) = \frac{i\hbar}{2m}\left(\Psi\vec{\nabla}\Psi^* - \Psi^*\vec{\nabla}\Psi\right);} \tag{3.83}$$

$\rho(\vec{r},t)$ is called the *probability density*, while $\vec{J}(\vec{r},t)$ is the *probability current density*, or simply the *current density*, or even the particle density flux. By analogy with charge conservation in electrodynamics, equation (3.82) is interpreted as the *conservation of probability*.

Let us find the relationship between the density operators $\hat{\rho}(t)$ and $\hat{\rho}(t_0)$. Since $|\Psi(t)\rangle = \hat{U}(t,t_0)|\Psi(t_0)\rangle$ and $\langle\Psi(t)| = \langle\Psi(t_0)|\hat{U}^\dagger(t,t_0)$, we have

$$\hat{\rho}(t) = |\Psi(t)\rangle\langle\Psi(t)| = \hat{U}(t,t_0)|\Psi(0)\rangle\langle\Psi(0)|\hat{U}^\dagger(t,t_0). \tag{3.84}$$

This is known as the *density operator* for the state $|\Psi(t)\rangle$. Hence knowing $\hat{\rho}(t_0)$ we can calculate $\hat{\rho}(t)$ as follows:

$$\hat{\rho}(t) = \hat{U}(t,t_0)\hat{\rho}(t_0)\hat{U}^\dagger(t,t_0). \tag{3.85}$$

3.6.5 Time Evolution of Expectation Values

We want to look here at the time dependence of the expectation value of a linear operator; if the state $|\Psi(t)\rangle$ is normalized, the expectation value is given by

$$\langle\hat{A}\rangle = \langle\Psi(t)|\hat{A}|\Psi(t)\rangle. \tag{3.86}$$

Using (3.76) and (3.77), we can write $d\langle\hat{A}\rangle/dt$ as follows:

$$\frac{d}{dt}\langle\hat{A}\rangle = \frac{1}{i\hbar}\langle\Psi(t)|\hat{A}\hat{H} - \hat{H}\hat{A}|\Psi(t)\rangle + \langle\Psi(t)|\frac{\partial A}{\partial t}|\Psi(t)\rangle \tag{3.87}$$

or

$$\boxed{\frac{d}{dt}\langle\hat{A}\rangle = \frac{1}{i\hbar}\langle[\hat{A},\hat{H}]\rangle + \langle\frac{\partial\hat{A}}{\partial t}\rangle.} \tag{3.88}$$

Two important results stem from this relation. First, if the observable A does not depend explicitly on time, the term $\partial\hat{A}/\partial t$ will vanish, so the rate of change of the expectation value of $\hat{A}$ is given by $\langle[\hat{A},\hat{H}]\rangle/i\hbar$. Second, besides not depending explicitly on time, if the observable A commutes with the Hamiltonian, the quantity $d\langle\hat{A}\rangle/dt$ will then be zero; hence the expectation

value $\langle \hat{A} \rangle$ will be constant in time. So if $\hat{A}$ commutes with the Hamiltonian and is not dependent on time, the observable A is said to be a *constant of the motion*; that is, *the expectation value of an operator that does not depend on time and that commutes with the Hamiltonian is constant in time*:

$$\boxed{\text{If} \quad [\hat{H}, \hat{A}] = 0 \quad \text{and} \quad \frac{\partial \hat{A}}{\partial t} = 0 \implies \frac{d\langle \hat{A} \rangle}{dt} = 0 \implies \langle \hat{A} \rangle = \text{constant.}} \tag{3.89}$$

For instance, we can verify that the energy, the linear momentum, and the angular momentum of an isolated system are conserved: $d\langle \hat{H} \rangle/dt = 0$, $d\langle \hat{\vec{P}} \rangle/dt = 0$, and $d\langle \hat{\vec{L}} \rangle/dt = 0$. This implies that the expectation values of $\hat{H}$, $\hat{\vec{P}}$, and $\hat{\vec{L}}$ are constant. Recall from classical physics that the conservation of energy, linear momentum, and angular momentum are consequences of the following symmetries, respectively: homogeneity of time, homogeneity of space, and isotropy of space. We will show in the following section that these symmetries are associated, respectively, with invariances in time translation, space translation, and space rotation.

As an example, let us consider the time evolution of the expectation value of the density operator $\hat{\rho}(t) = |\Psi(t)\rangle\langle\Psi(t)|$; see (3.84). From (3.5), which leads to $\partial|\Psi(t)\rangle/\partial t = (1/i\hbar)\hat{H}|\Psi(t)\rangle$ and $\partial\langle\Psi(t)|/\partial t = -(1/i\hbar)\langle\Psi(t)|\hat{H}$, we have

$$\frac{\partial \hat{\rho}(t)}{\partial t} = \frac{1}{i\hbar}\hat{H}|\Psi(t)\rangle\langle\Psi(t)| - \frac{1}{i\hbar}|\Psi(t)\rangle\langle\Psi(t)|\hat{H} = -\frac{1}{i\hbar}[\hat{\rho}(t), \hat{H}]. \tag{3.90}$$

A substitution of this relation into (3.88) leads to

$$\frac{d}{dt}\langle\hat{\rho}(t)\rangle = \frac{1}{i\hbar}\langle[\hat{\rho}(t), \hat{H}]\rangle + \langle\frac{\partial\hat{\rho}(t)}{\partial t}\rangle = \frac{1}{i\hbar}\langle[\hat{\rho}(t), \hat{H}]\rangle - \frac{1}{i\hbar}\langle[\hat{\rho}(t), \hat{H}]\rangle = 0. \tag{3.91}$$

So the density operator is a constant of the motion. In fact, we can easily show that

$$\begin{aligned} \langle[\hat{\rho}(t), \hat{H}]\rangle &= \langle\Psi(t)|[|\Psi(t)\rangle\langle\Psi(t)|, \hat{H}]|\Psi(t)\rangle \\ &= \langle\Psi(t)|\Psi(t)\rangle\langle\Psi(t)|\hat{H}|\Psi(t)\rangle - \langle\Psi(t)|\hat{H}|\Psi(t)\rangle\langle\Psi(t)|\Psi(t)\rangle \\ &= 0, \end{aligned} \tag{3.92}$$

which, when combined with (3.90), yields $\langle\partial\hat{\rho}(t)/\partial t\rangle = 0$.

Finally, we should note that the constants of motion are nothing but observables that can be measured simultaneously with the energy to arbitrary accuracy. If a system has a complete set of commuting operators (CSCO), the number of these operators is given by the total number of constants of the motion.

3.7 Symmetries and Conservation Laws

We are interested here in symmetries that leave the Hamiltonian of an *isolated* system invariant. We will show that for each such symmetry there corresponds an observable which is a constant of the motion. The invariance principles relevant to our study are the time translation invariance and the space translation invariance. We may recall from classical physics that whenever a system is invariant under space translations, its total momentum is conserved; and whenever it is invariant under rotations, its total angular momentum is also conserved.

To prepare the stage for symmetries and conservation laws in quantum mechanics, we are going to examine the properties of infinitesimal and finite unitary transformations that are most essential to these invariance principles.

3.7.1 Infinitesimal Unitary Transformations

In Chapter 2 we saw that the transformations of a state vector $|\psi\rangle$ and an operator $\hat{A}$ under an infinitesimal unitary transformation $U_\varepsilon(\hat{G}) = \hat{I} + i\varepsilon\hat{G}$ are given by

$$|\psi'\rangle = (\hat{I} + i\varepsilon\hat{G})|\psi\rangle = |\psi\rangle + \delta|\psi\rangle, \tag{3.93}$$

$$\hat{A}' = (\hat{I} + i\varepsilon\hat{G})\hat{A}(\hat{I} - i\varepsilon\hat{G}) \simeq \hat{A} + i\varepsilon[\hat{G}, \hat{A}], \tag{3.94}$$

where ε and $\hat{G}$ are called the parameter and the generator of the transformation, respectively.

Let us consider two important applications of infinitesimal unitary transformations: time and space translations.

3.7.1.1 Time Translations: $\hat{G} = \hat{H}/\hbar$

The application of $\hat{U}_{\delta t}(\hat{H}) = \hat{I} + (i/\hbar)\delta t\, \hat{H}$ on a state $|\psi(t)\rangle$ gives

$$\left(\hat{I} + \frac{i}{\hbar}\,\delta t\, \hat{H}\right)|\psi(t)\rangle = |\psi(t)\rangle + \left(\frac{i}{\hbar}\delta t\right)\hat{H}|\psi(t)\rangle. \tag{3.95}$$

Since $\hat{H}|\psi(t)\rangle = i\hbar\partial|\psi(t)\rangle/\partial t$ we have

$$\boxed{\left(\hat{I} + \frac{i}{\hbar}\,\delta t\, \hat{H}\right)|\psi(t)\rangle = |\psi(t)\rangle - \delta t\frac{\partial|\psi(t)\rangle}{\partial t} \simeq |\psi(t-\delta t)\rangle,} \tag{3.96}$$

because $|\psi(t)\rangle - \delta t\, \partial|\psi(t)\rangle/\partial t$ is nothing but the first-order Taylor expansion of $|\psi(t-\delta t)\rangle$. We conclude from (3.96) that the application of $\hat{U}_{\delta t}(\hat{H})$ to $|\psi(t)\rangle$ generates a state $|\psi(t-\delta t)\rangle$ which consists simply of a *time translation* of $|\psi(t)\rangle$ by an amount equal to δt. *The Hamiltonian in* $(\hat{I} + (i/\hbar)\delta t\, \hat{H})$ *is thus the generator of infinitesimal time translations*. Note that this translation preserves the shape of the state $|\psi(t)\rangle$, for its overall shape is merely translated in time by δt.

3.7.1.2 Spatial Translations: $\hat{G} = \hat{P}_x/\hbar$

The application of $\hat{U}_\varepsilon(\hat{P}_x) = \hat{I} + (i/\hbar)\varepsilon\hat{P}_x$ to $\psi(x)$ gives

$$\left(\hat{I} + \frac{i}{\hbar}\varepsilon\hat{P}_x\right)\psi(x) = \psi(x) + \left(\frac{i}{\hbar}\varepsilon\right)\hat{P}_x\psi(x). \tag{3.97}$$

Since $\hat{P}_x = -i\hbar\partial/\partial x$ and since the first-order Taylor expansion of $\psi(x+\varepsilon)$ is given by $\psi(x+\varepsilon) = \psi(x) + \varepsilon\partial\psi(x)/\partial x$, we have

$$\boxed{\left(\hat{I} + \frac{i}{\hbar}\varepsilon\hat{P}_x\right)\psi(x) = \psi(x) + \varepsilon\frac{\partial\psi(x)}{\partial x} \simeq \psi(x+\varepsilon).} \tag{3.98}$$

So, when $\hat{U}_\varepsilon(\hat{P}_x)$ acts on a wave function, it translates it spatially by an amount equal to ε.

Using $[\hat{X},\ \hat{P}_x] = i\hbar$ we infer from (3.94) that the position operator $\hat{X}$ transforms as follows:

$$\hat{X}' = \left(\hat{I} + \frac{i}{\hbar}\varepsilon\hat{P}_x\right)\hat{X}\left(\hat{I} - \frac{i}{\hbar}\varepsilon\hat{P}_x\right) \simeq \hat{X} + \frac{i}{\hbar}\varepsilon[\hat{P}_x, \hat{X}] = \hat{X} + \varepsilon. \tag{3.99}$$

The relations (3.98) and (3.99) show that the *linear momentum operator in* $(\hat{I} + (i/\hbar)\varepsilon\hat{P}_x)$ *is a generator of infinitesimal spatial translations*.

3.7.2 Finite Unitary Transformations

In Chapter 2 we saw that a *finite* unitary transformation can be constructed by performing a succession of infinitesimal transformations. For instance, by applying a single infinitesimal time translation N times in steps of τ/N, we can generate a finite time translation

$$\hat{U}_\tau(\hat{H}) = \lim_{N\to+\infty} \prod_{k=1}^{N} \left(\hat{I} + \frac{i}{\hbar}\frac{\tau}{N}\hat{H}\right) = \lim_{N\to+\infty} \left(\hat{I} + \frac{i}{\hbar}\tau\hat{H}\right)^N = \exp\left(\frac{i}{\hbar}\tau\hat{H}\right), \quad (3.100)$$

where the Hamiltonian is the generator of finite time translations. We should note that the time evolution operator $\hat{U}(t, t_0) = e^{-i(t-t_0)\hat{H}/\hbar}$, displayed in (3.58), represents a finite unitary transformation where $\hat{H}$ is the generator of the time translation.

By analogy with (3.96) we can show that the application of $\hat{U}_\tau(\hat{H})$ to $|\psi(t)\rangle$ yields

$$\boxed{\hat{U}_\tau(\hat{H})|\psi(t)\rangle = \exp\left(\frac{i}{\hbar}\tau\hat{H}\right)|\psi(t)\rangle = |\psi(t-\tau)\rangle,} \quad (3.101)$$

where $|\psi(t-\tau)\rangle$ is merely a time translation of $|\psi(t)\rangle$.

Similarly, we can infer from (3.98) that the application of $\hat{U}_{\vec{a}}(\hat{\vec{P}}) = \exp(i\vec{a}\cdot\hat{\vec{P}}/\hbar)$ to a wave function causes it to be translated in space by a vector $\vec{a}$:

$$\boxed{\hat{U}_{\vec{a}}(\hat{\vec{P}})\psi(\vec{r}) = \exp\left(\frac{i}{\hbar}\vec{a}\cdot\hat{\vec{P}}\right)\psi(\vec{r}) = \psi(\vec{r}+\vec{a}).} \quad (3.102)$$

To calculate the transformed position vector operator $\hat{\vec{R}}'$, let us invoke a relation we derived in Chapter 2:

$$\hat{A}' = e^{i\alpha\hat{G}}\hat{A}e^{-i\alpha\hat{G}} = \hat{A} + i\alpha[\hat{G}, \hat{A}] + \frac{(i\alpha)^2}{2!}[\hat{G}, [\hat{G}, \hat{A}]] + \frac{(i\alpha)^3}{3!}[\hat{G}, [\hat{G}, [\hat{G}, \hat{A}]]] + \cdots. \quad (3.103)$$

An application of this relation to the spatial translation operator $\hat{U}_{\vec{a}}(\hat{\vec{P}})$ yields

$$\hat{\vec{R}}' = \exp\left(\frac{i}{\hbar}\vec{a}\cdot\hat{\vec{P}}\right)\hat{\vec{R}}\exp\left(-\frac{i}{\hbar}\vec{a}\cdot\hat{\vec{P}}\right) = \hat{\vec{R}} + \frac{i}{\hbar}[\vec{a}\cdot\hat{\vec{P}}, \hat{\vec{R}}] = \hat{\vec{R}} + \vec{a}. \quad (3.104)$$

In deriving this, we have used the fact that $[\vec{a}\cdot\hat{\vec{P}}, \hat{\vec{R}}] = -i\hbar\vec{a}$ and that the other commutators are zero, notably $[\vec{a}\cdot\hat{\vec{P}}, [\vec{a}\cdot\hat{\vec{P}}, \hat{\vec{R}}]] = 0$. From (3.102) and (3.104), we see that *the linear momentum in* $\exp(i\vec{a}\cdot\hat{\vec{P}}/\hbar)$ *is a generator of finite spatial translations*.

3.7.3 Symmetries and Conservation Laws

We want to show here that every invariance principle of $\hat{H}$ is connected with a conservation law.

The Hamiltonian of a system transforms under a unitary transformation $e^{i\alpha\hat{G}}$ as follows; see (3.103):

$$\hat{H}' = e^{i\alpha\hat{G}}\hat{H}e^{-i\alpha\hat{G}} = \hat{H} + i\alpha[\hat{G}, \hat{H}] + \frac{(i\alpha)^2}{2!}[\hat{G}, [\hat{G}, \hat{H}]] + \frac{(i\alpha)^3}{3!}[\hat{G}, [\hat{G}, [\hat{G}, \hat{H}]]] + \cdots. \quad (3.105)$$

If $\hat{H}$ commutes with $\hat{G}$, it also commutes with the unitary transformation $\hat{U}_\alpha(\hat{G}) = e^{i\alpha\hat{G}}$. In this case we may infer two important conclusions. On the one hand, there is an *invariance principle*: the Hamiltonian is invariant under the transformation $\hat{U}_\alpha(\hat{G})$, since

$$\hat{H}' = e^{i\alpha\hat{G}}\hat{H}e^{-i\alpha\hat{G}} = e^{i\alpha\hat{G}}e^{-i\alpha\hat{G}}\hat{H} = \hat{H}. \tag{3.106}$$

On the other hand, if in addition to $[\hat{G},\ \hat{H}] = 0$, the operator $\hat{G}$ does not depend on time explicitly, there is a *conservation law*: equation (3.88) shows that $\hat{G}$ is a *constant of the motion*, since

$$\frac{d}{dt}\langle\hat{G}\rangle = \frac{1}{i\hbar}\langle[\hat{G}, \hat{H}]\rangle + \langle\frac{\partial\hat{G}}{\partial t}\rangle = 0. \tag{3.107}$$

We say that $\hat{G}$ is conserved.

So whenever the Hamiltonian is invariant under a unitary transformation, the generator of the transformation is conserved. We may say, in general, that *for every invariance symmetry of the Hamiltonian, there corresponds a conservation law*.

3.7.3.1 Conservation of Energy and Linear Momentum

Let us consider two interesting applications pertaining to the invariance of the Hamiltonian of an *isolated* system with respect to time translations and to space translations. First, let us consider time translations. As shown in (3.58), time translations are generated in the case of time-independent Hamiltonians by the evolution operator $\hat{U}(t, t_0) = e^{-i(t-t_0)\hat{H}/\hbar}$. Since $\hat{H}$ commutes with the generator of the time translation (which is given by $\hat{H}$ itself), it is invariant under time translations. *As $\hat{H}$ is invariant under time translations, the energy of an isolated system is conserved*. We should note that if the system is invariant under time translations, this means there is a symmetry of time homogeneity. Time homogeneity implies that the time-displaced state $\psi(t-\tau)$, like $\psi(t)$, satisfies the Schrödinger equation.

The second application pertains to the spatial translations, or to transformations under $\hat{U}_{\vec{a}}(\hat{\vec{P}}) = \exp(i\vec{a}\cdot\hat{\vec{P}}/\hbar)$, of an isolated system. The linear momentum is invariant under $\hat{U}_{\vec{a}}(\hat{\vec{P}})$ and the position operator transforms according to (3.104):

$$\hat{\vec{P}}\,' = \hat{\vec{P}}, \qquad \hat{\vec{R}}\,' = \hat{\vec{R}} + \vec{a}. \tag{3.108}$$

For instance, since the Hamiltonian of a free particle does not depend on the coordinates, it commutes with the linear momentum $[\hat{H}, \hat{\vec{P}}] = 0$. The Hamiltonian is then invariant under spatial translations, since

$$\hat{H}' = \exp\left(\frac{i}{\hbar}\vec{a}\cdot\hat{\vec{P}}\right)\hat{H}\exp\left(-\frac{i}{\hbar}\vec{a}\cdot\hat{\vec{P}}\right) = \exp\left(\frac{i}{\hbar}\vec{a}\cdot\hat{\vec{P}}\right)\exp\left(-\frac{i}{\hbar}\vec{a}\cdot\hat{\vec{P}}\right)\hat{H} = \hat{H}. \tag{3.109}$$

Since $[\hat{H}, \hat{\vec{P}}] = 0$ and since the linear momentum operator does not depend explicitly on time, we infer from (3.88) that $\hat{\vec{P}}$ is a constant of the motion, since

$$\frac{d}{dt}\langle\hat{\vec{P}}\rangle = \frac{1}{i\hbar}\langle[\hat{\vec{P}}, \hat{H}]\rangle + \langle\frac{\partial\hat{\vec{P}}}{\partial t}\rangle = 0. \tag{3.110}$$

So if $[\hat{H},\ \hat{\vec{P}}] = 0$ the *Hamiltonian will be invariant under spatial translations and the linear momentum will be conserved*. A more general case where the linear momentum is a constant

of the motion is provided by an isolated system, for its total linear momentum is conserved. Note that the invariance of the system under spatial translations means there is a symmetry of spatial homogeneity. The requirement for the homogeneity of space implies that the spatially displaced wave function $\psi(\vec{r}+\vec{a})$, much like $\psi(\vec{r})$, satisfies the Schrödinger equation.

In summary, the symmetry of time homogeneity gives rise to the conservation of energy, whereas the symmetry of space homogeneity gives rise to the conservation of linear momentum.

In Chapter 7 we will see that the symmetry of space isotropy, or the invariance of the Hamiltonian with respect to space rotations, leads to conservation of the angular momentum.

Parity operator

The unitary transformations we have considered so far, time translations and space translations, are *continuous*. We may consider now a *discrete* unitary transformation, the *parity*. As seen in Chapter 2, the parity transformation consists of an inversion or reflection through the origin of the coordinate system:

$$\hat{\mathcal{P}}\psi(\vec{r}) = \psi(-\vec{r}). \tag{3.111}$$

If the parity operator commutes with the system's Hamiltonian,

$$[\hat{H},\ \hat{\mathcal{P}}] = 0, \tag{3.112}$$

the *parity will be conserved*, and hence a constant of the motion. In this case the Hamiltonian and the parity operator have simultaneous eigenstates. For instance, we will see in Chapter 4 that the wave functions of a particle moving in a symmetric potential, $\hat{V}(\vec{r}) = \hat{V}(-\vec{r})$, have definite parities: they can be only even or odd. Similarly, we can ascertain that the parity of an isolated system is a constant of the motion.

3.8 Connecting Quantum to Classical Mechanics

3.8.1 Poisson Brackets and Commutators

To establish a connection between quantum mechanics and classical mechanics, we may look at the time evolution of observables.

Before describing the time evolution of a dynamical variable within the context of classical mechanics, let us review the main ideas of the mathematical tool relevant to this description, the *Poisson bracket*. The Poisson bracket between two dynamical variables A and B is defined in terms of the generalized coordinates q_i and the momenta p_i of the system:

$$\{A, B\} = \sum_j \left(\frac{\partial A}{\partial q_j}\frac{\partial B}{\partial p_j} - \frac{\partial A}{\partial p_j}\frac{\partial B}{\partial q_j} \right). \tag{3.113}$$

Since the variables q_i are independent of p_i, we have $\partial q_j/\partial p_k = 0$, $\partial p_j/\partial q_k = 0$; thus we can show that

$$\{q_j, q_k\} = \{p_j, p_k\} = 0, \qquad \{q_j, p_k\} = \delta_{jk}. \tag{3.114}$$

Using (3.113) we can easily infer the following properties of the Poisson brackets:

- Antisymmetry

$$\{A,\ B\} = -\{B,\ A\} \tag{3.115}$$

- Linearity

$$\{A,\ \alpha B + \beta C + \gamma D + \cdots\} = \alpha\{A,\ B\} + \beta\{A,\ C\} + \gamma\{A,\ D\} + \cdots \tag{3.116}$$

- Complex conjugate

$$\{A,\ B\}^* = \{A^*,\ B^*\} \tag{3.117}$$

- Distributivity

$$\{A,\ BC\} = \{A,\ B\}C + B\{A,\ C\}, \qquad \{AB,\ C\} = A\{B,\ C\} + \{A,\ C\}B \tag{3.118}$$

- Jacobi identity

$$\{A, \{B,\ C\}\} + \{B,\ \{C,\ A\}\} + \{C,\ \{A,\ B\}\} = 0 \tag{3.119}$$

- Using $df^n(x)/dx = nf^{n-1}(x)df(x)/dx$, we can show that

$$\{A,\ B^n\} = nB^{n-1}\{A,\ B\}, \qquad \{A^n,\ B\} = nA^{n-1}\{A,\ B\} \tag{3.120}$$

These properties are similar to the properties of the quantum mechanical commutators seen in Chapter 2.

The total time derivative of a dynamical variable A is given by

$$\frac{dA}{dt} = \sum_j \left(\frac{\partial A}{\partial q_j}\frac{\partial q_j}{\partial t} + \frac{\partial A}{\partial p_j}\frac{\partial p_j}{\partial t}\right) + \frac{\partial A}{\partial t} = \sum_j \left(\frac{\partial A}{\partial q_j}\frac{\partial H}{\partial p_j} - \frac{\partial A}{\partial p_j}\frac{\partial H}{\partial p_j}\right) + \frac{\partial A}{\partial t}; \tag{3.121}$$

in deriving this relation we have used the Hamilton equations of classical mechanics:

$$\frac{dq_j}{dt} = \frac{\partial H}{\partial p_j}, \qquad \frac{dp_j}{dt} = -\frac{\partial H}{\partial q_j}, \tag{3.122}$$

where H is the Hamiltonian of the system. The total time evolution of a dynamical variable A is thus given by the following equation of motion:

$$\frac{dA}{dt} = \{A, H\} + \frac{\partial A}{\partial t}. \tag{3.123}$$

Note that if A does not depend explicitly on time, its time evolution is given simply by $dA/dt = \{A, H\}$. If $dA/dt = 0$ or $\{A, H\} = 0$, A is said to be a *constant of the motion*.

Comparing the classical relation (3.123) with its quantum mechanical counterpart (3.88),

$$\frac{d}{dt}\langle\hat{A}\rangle = \frac{1}{i\hbar}\langle[\hat{A}, \hat{H}]\rangle + \langle\frac{\partial\hat{A}}{\partial t}\rangle, \tag{3.124}$$

we see that they are identical only if we identify the Poisson bracket $\{A, H\}$ with the commutator $[\hat{A}, \hat{H}]/(i\hbar)$. We may thus infer the following general rule. The Poisson bracket of any pair of classical variables can be obtained from the commutator between the corresponding pair of quantum operators by dividing it by $i\hbar$:

$$\boxed{\frac{1}{i\hbar}[\hat{A}, \hat{B}] \longrightarrow \{A, B\}_{classical}.} \tag{3.125}$$

Note that the expressions of classical mechanics can be derived from their quantum counterparts, but the opposite is not possible. That is, dividing quantum mechanical expressions by $i\hbar$ leads to their classical analog, but multiplying classical mechanical expressions by $i\hbar$ doesn't necessarily lead to their quantum counterparts.

Example 3.5

(a) Evaluate the Poisson bracket $\{x, p\}$ between the position, x, and momentum, p, variables.

(b) Compare the commutator $\left[\hat{X}, \hat{P}\right]$ with Poisson bracket $\{x, p\}$ calculated in Part (a).

Solution

(a) Applying the general relation

$$\{A, B\} = \sum_j \left(\frac{\partial A}{\partial x_j}\frac{\partial B}{\partial p_j} - \frac{\partial A}{\partial p_j}\frac{\partial B}{\partial x_j} \right) \tag{3.126}$$

to x and p, we can readily evaluate the given Poisson bracket:

$$\begin{aligned} \{x, p\} &= \frac{\partial(x)}{\partial x}\frac{\partial(p)}{\partial p} - \frac{\partial(x)}{\partial p}\frac{\partial(p)}{\partial x} \\ &= \frac{\partial(x)}{\partial x}\frac{\partial(p)}{\partial p} \\ &= 1. \end{aligned} \tag{3.127}$$

(b) Using the fact that $[\hat{X}, \hat{P}] = i\hbar$, we see that

$$\frac{1}{i\hbar}[\hat{X}, \hat{P}] = 1, \tag{3.128}$$

which is equal to the Poisson bracket (3.127); that is,

$$\frac{1}{i\hbar}[\hat{X}, \hat{P}] = \{x, p\}_{classical} = 1. \tag{3.129}$$

This result is in agreement with Eq. (3.125).

3.8.2 The Ehrenfest Theorem

If quantum mechanics is to be more general than classical mechanics, it must contain classical mechanics as a limiting case. To illustrate this idea, let us look at the time evolution of the expectation values of the position and momentum operators, $\hat{\vec{R}}$ and $\hat{\vec{P}}$, of a particle moving in a potential $\hat{V}(\vec{r})$, and then compare these relations with their classical counterparts.

Since the position and the momentum observables do not depend explicitly on time, within the context of wave mechanics, the terms $\langle \partial \hat{\vec{R}}/\partial t \rangle$ and $\langle \partial \hat{\vec{P}}/\partial t \rangle$ are zero. Hence, inserting

$\hat{H} = \hat{\vec{P}}^2/(2m) + \hat{V}(\hat{\vec{R}}, t)$ into (3.88) and using the fact that $\hat{\vec{R}}$ commutes with $\hat{V}(\hat{\vec{R}}, t)$, we can write

$$\frac{d}{dt}\langle\hat{\vec{R}}\rangle = \frac{1}{i\hbar}\langle[\hat{\vec{R}}, \hat{H}]\rangle = \frac{1}{i\hbar}\langle[\hat{\vec{R}}, \frac{\hat{\vec{P}}^2}{2m} + \hat{V}(\hat{\vec{R}}, t)]\rangle = \frac{1}{2im\hbar}\langle[\hat{\vec{R}}, \hat{\vec{P}}^2]\rangle. \tag{3.130}$$

Since

$$[\hat{\vec{R}}, \hat{\vec{P}}^2] = 2i\hbar\hat{\vec{P}}, \tag{3.131}$$

we have

$$\boxed{\frac{d}{dt}\langle\hat{\vec{R}}\rangle = \frac{1}{m}\langle\hat{\vec{P}}\rangle.} \tag{3.132}$$

As for $d\langle\hat{\vec{P}}\rangle/dt$, we can infer its expression from a treatment analogous to $d\langle\hat{\vec{R}}\rangle/dt$. Using

$$[\hat{\vec{P}}, \hat{V}(\hat{\vec{R}}, t)] = -i\hbar\vec{\nabla}\hat{V}(\vec{R}, t), \tag{3.133}$$

we can write

$$\boxed{\frac{d}{dt}\langle\hat{\vec{P}}\rangle = \frac{1}{i\hbar}\langle[\hat{\vec{P}}, \hat{V}(\hat{\vec{R}}, t)]\rangle = -\langle\vec{\nabla}\hat{V}(\hat{\vec{R}}, t)\rangle.} \tag{3.134}$$

The two relations (3.132) and (3.134), expressing the time evolution of the expectation values of the position and momentum operators, are known as the *Ehrenfest theorem*, or Ehrenfest equations. Their respective forms are reminiscent of the Hamilton–Jacobi equations of classical mechanics,

$$\frac{d\vec{r}}{dt} = \frac{\vec{p}}{m}, \qquad \frac{d\vec{p}}{dt} = -\vec{\nabla}V(\vec{r}), \tag{3.135}$$

which reduce to Newton's equation of motion for a *classical* particle of mass m, position $\vec{r}$, and momentum $\vec{p}$:

$$\frac{d\vec{p}}{dt} = m\frac{d^2\vec{r}}{dt^2} = -\vec{\nabla}V(\vec{r}). \tag{3.136}$$

Notice $\hbar$ has completely disappeared in the Ehrenfest equations (3.132) and (3.134). These two equations certainly establish a connection between quantum mechanics and classical mechanics. We can, within this context, view the center of the wave packet as moving like a classical particle when subject to a potential $V(\vec{r})$.

3.8.3 Quantum Mechanics and Classical Mechanics

In Chapter 1 we focused mainly on those experimental observations which confirm the failure of classical physics at the microscopic level. We should bear in mind, however, that classical physics works perfectly well within the realm of the macroscopic world. Thus, if the theory of quantum mechanics is to be considered more general than classical physics, it must yield accurate results not only on the microscopic scale but at the classical limit as well.

How does one decide on when to use classical or quantum mechanics to describe the motion of a given system? That is, how do we know when a classical description is good enough or when a quantum description becomes a must? The answer is provided by comparing the size of those quantities of the system that have the dimensions of an action with the Planck constant, h. Since, as shown in (3.125), the quantum relations are characterized by h, we can state that

if the value of the action of a system is too large compared to h, this system can be accurately described by means of classical physics. Otherwise, the use of a quantal description becomes unavoidable. One should recall that, for microscopic systems, the size of action variables is of the order of h; for instance, the angular momentum of the hydrogen atom is $L = n\hbar$, where n is finite.

Another equivalent way of defining the classical limit is by means of "*length*." Since $\lambda = h/p$ the classical domain can be specified by the limit $\lambda \to 0$. This means that, when the de Broglie wavelength of a system is too small compared to its size, the system can be described accurately by means of classical physics.

In summary, the classical limit can be described as the limit $h \to 0$ or, equivalently, as the limit $\lambda \to 0$. In these limits the results of quantum mechanics should be similar to those of classical physics:

$$\lim_{h\to 0} \text{ Quantum Mechanics} \longrightarrow \text{Classical Mechanics}, \tag{3.137}$$

$$\lim_{\lambda\to 0} \text{ Quantum Mechanics} \longrightarrow \text{Classical Mechanics}. \tag{3.138}$$

Classical mechanics can thus be regarded as the short wavelength limit of quantum mechanics. In this way, quantum mechanics contains classical mechanics as a limiting case. So, in the limit of $h \to 0$ or $\lambda \to 0$, quantum dynamical quantities should have, as proposed by Bohr, a one-to-one correspondence with their classical counterparts. This is the essence of the *correspondence principle*.

But how does one reconcile, in the classical limit, the probabilistic nature of quantum mechanics with the determinism of classical physics? The answer is quite straightforward: quantum *fluctuations* must become negligible or even vanish when $h \to 0$, for Heisenberg's uncertainty principle would acquire the status of *certainty*; when $h \to 0$, the fluctuations in the position and momentum will vanish, $\Delta x \to 0$ and $\Delta p \to 0$. Thus, the position and momentum can be measured simultaneously with arbitrary accuracy. This implies that the probabilistic assessments of dynamical quantities by quantum mechanics must give way to exact calculations (these ideas will be discussed further when we study the WKB method in Chapter 9).

So, for those cases where the action variables of a system are too large compared to h (or, equivalently, when the lengths of this system are too large compared to its de Broglie wavelength), quantum mechanics gives the same results as classical mechanics.

In the rest of this text, we will deal with the various applications of the Schrödinger equation. We start, in Chapter 4, with the simple case of one-dimensional systems and later on consider more realistic systems.

3.9 Solved Problems

Problem 3.1

A particle of mass m, which moves freely inside an infinite potential well of length a, has the following initial wave function at $t = 0$:

$$\psi(x, 0) = \frac{A}{\sqrt{a}} \sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{3}{5a}} \sin\left(\frac{3\pi x}{a}\right) + \frac{1}{\sqrt{5a}} \sin\left(\frac{5\pi x}{a}\right),$$

where A is a real constant.

(a) Find A so that $\psi(x,0)$ is normalized.

(b) If measurements of the energy are carried out, what are the values that will be found and what are the corresponding probabilities? Calculate the average energy.

(c) Find the wave function $\psi(x,t)$ at any later time t.

(d) Determine the probability of finding the system at a time t in the state $\varphi(x,t) = \sqrt{2/a}\sin(5\pi x/a)\exp(-iE_5 t/\hbar)$; then determine the probability of finding it in the state $\chi(x,t) = \sqrt{2/a}\sin(2\pi x/a)\exp(-iE_2 t/\hbar)$.

Solution

Since the functions

$$\phi_n(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi x}{a}\right) \tag{3.139}$$

are orthonormal,

$$\langle\phi_n|\phi_m\rangle = \int_0^a \phi_n^*(x)\phi_m(x)\,dx = \frac{2}{a}\int_0^a \sin\left(\frac{n\pi x}{a}\right)\sin\left(\frac{m\pi x}{a}\right)dx = \delta_{nm}, \tag{3.140}$$

it is more convenient to write $\psi(x,0)$ in terms of $\phi_n(x)$:

$$\begin{aligned}\psi(x,0) &= \frac{A}{\sqrt{a}}\sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{3}{5a}}\sin\left(\frac{3\pi x}{a}\right) + \frac{1}{\sqrt{5a}}\sin\left(\frac{5\pi x}{a}\right)\\ &= \frac{A}{\sqrt{2}}\phi_1(x) + \sqrt{\frac{3}{10}}\phi_3(x) + \frac{1}{\sqrt{10}}\phi_5(x).\end{aligned} \tag{3.141}$$

(a) Since $\langle\phi_n|\phi_m\rangle = \delta_{nm}$ the normalization of $\psi(x,0)$ yields

$$1 = \langle\psi|\psi\rangle = \frac{A^2}{2} + \frac{3}{10} + \frac{1}{10}, \tag{3.142}$$

or $A = \sqrt{6/5}$; hence

$$\psi(x,0) = \sqrt{\frac{3}{5}}\phi_1(x) + \sqrt{\frac{3}{10}}\phi_3(x) + \frac{1}{\sqrt{10}}\phi_5(x). \tag{3.143}$$

(b) Since the second derivative of (3.139) is given by $d^2\phi_n(x)/dx^2 = -(n^2\pi^2/a^2)\phi_n(x)$, and since the Hamiltonian of a free particle is $\hat{H} = -(\hbar^2/2m)d^2/dx^2$, the expectation value of $\hat{H}$ with respect to $\phi_n(x)$ is

$$E_n = \langle\phi_n|\hat{H}|\phi_n\rangle = -\frac{\hbar^2}{2m}\int_0^a \phi_n^*(x)\frac{d^2\phi_n(x)}{dx^2}dx = \frac{n^2\pi^2\hbar^2}{2ma^2}. \tag{3.144}$$

If a measurement is carried out on the system, we would obtain $E_n = n^2\pi^2\hbar^2/(2ma^2)$ with a corresponding probability of $P_n(E_n) = |\langle\phi_n|\psi\rangle|^2$. Since the initial wave function (3.143) contains only three eigenstates of $\hat{H}$, $\phi_1(x)$, $\phi_3(x)$, and $\phi_5(x)$, the results of the energy measurements along with the corresponding probabilities are

$$E_1 = \langle\phi_1|\hat{H}|\phi_1\rangle = \frac{\pi^2\hbar^2}{2ma^2}, \qquad P_1(E_1) = |\langle\phi_1|\psi\rangle|^2 = \frac{3}{5}, \tag{3.145}$$

$$E_3 = \langle\phi_3|\hat{H}|\phi_3\rangle = \frac{9\pi^2\hbar^2}{2ma^2}, \qquad P_3(E_3) = |\langle\phi_3|\psi\rangle|^2 = \frac{3}{10}, \tag{3.146}$$

$$E_5 = \langle\phi_5|\hat{H}|\phi_5\rangle = \frac{25\pi^2\hbar^2}{2ma^2}, \qquad P_5(E_5) = |\langle\phi_5|\psi\rangle|^2 = \frac{1}{10}. \tag{3.147}$$

The average energy is

$$E = \sum_n P_n E_n = \frac{3}{5}E_1 + \frac{3}{10}E_3 + \frac{1}{10}E_5 = \frac{29\pi^2\hbar^2}{10ma^2}. \tag{3.148}$$

(c) As the initial state $\psi(x,0)$ is given by (3.143), the wave function $\psi(x,t)$ at any later time t is

$$\psi(x,t) = \sqrt{\frac{3}{5}}\phi_1(x)e^{-iE_1t/\hbar} + \sqrt{\frac{3}{10}}\phi_3(x)e^{-iE_3t/\hbar} + \frac{1}{\sqrt{10}}\phi_5(x)e^{-iE_5t/\hbar}, \tag{3.149}$$

where the expressions of E_n are listed in (3.144) and $\phi_n(x)$ in (3.139).

(d) First, let us express $\varphi(x,t)$ in terms of $\phi_n(x)$:

$$\varphi(x,t) = \sqrt{\frac{2}{a}}\sin\left(\frac{5\pi x}{a}\right)e^{-iE_5t/\hbar} = \phi_5(x)e^{-iE_5t/\hbar}. \tag{3.150}$$

The probability of finding the system at a time t in the state $\varphi(x,t)$ is

$$P = |\langle\varphi|\psi\rangle|^2 = \left|\int_0^a \varphi^*(x,t)\psi(x,t)\,dx\right|^2 = \frac{1}{10}\left|\int_0^a \phi_5^*(x)\phi_5(x)\,dx\right|^2 = \frac{1}{10}, \tag{3.151}$$

since $\langle\varphi|\phi_1\rangle = \langle\varphi|\phi_3\rangle = 0$ and $\langle\varphi|\phi_5\rangle = \exp(iE_5t/\hbar)$.

Similarly, since $\chi(x,t) = \sqrt{2/a}\sin(2\pi x/a)\exp(-iE_2t/\hbar) = \phi_2(x)\exp(-iE_2t/\hbar)$, we can easily show that the probability for finding the system in the state $\chi(x,t)$ is zero:

$$P = |\langle\chi|\psi\rangle|^2 = \left|\int_0^a \chi^*(x,t)\psi(x,t)\,dx\right|^2 = 0, \tag{3.152}$$

since $\langle\chi|\phi_1\rangle = \langle\chi|\phi_3\rangle = \langle\chi|\phi_5\rangle = 0$.

Problem 3.2

A particle of mass m, which moves freely inside an infinite potential well of length a, is initially in the state $\psi(x,0) = \sqrt{3/5a}\sin(3\pi x/a) + (1/\sqrt{5a})\sin(5\pi x/a)$.

(a) Find $\psi(x,t)$ at any later time t.

(b) Calculate the probability density $\rho(x,t)$ and the current density, $\vec{J}(x,t)$.

(c) Verify that the probability is conserved, i.e., $\partial\rho/\partial t + \vec{\nabla}\cdot\vec{J}(x,t) = 0$.

Solution

(a) Since $\psi(x,0)$ can be expressed in terms of $\phi_n(x) = \sqrt{2/a}\sin(n\pi x/a)$ as

$$\psi(x,0) = \sqrt{\frac{3}{5a}}\sin\left(\frac{3\pi x}{a}\right) + \frac{1}{\sqrt{5a}}\sin\left(\frac{5\pi x}{a}\right) = \sqrt{\frac{3}{10}}\phi_3(x) + \frac{1}{\sqrt{10}}\phi_5(x), \tag{3.153}$$

we can write

$$\begin{aligned}\psi(x,t) &= \sqrt{\frac{3}{5a}}\sin\left(\frac{3\pi x}{a}\right)e^{-iE_3t/\hbar} + \frac{1}{\sqrt{5a}}\sin\left(\frac{5\pi x}{a}\right)e^{-iE_5t/\hbar} \\ &= \sqrt{\frac{3}{10}}\phi_3(x)e^{-iE_3t/\hbar} + \frac{1}{\sqrt{10}}\phi_5(x)e^{-iE_5t/\hbar},\end{aligned} \tag{3.154}$$

where the expressions for E_n are listed in (3.144): $E_n = n^2\pi^2\hbar^2/(2ma^2)$.

(b) Since $\rho(x,t) = \psi^*(x,t)\psi(x,t)$, where $\psi(x,t)$ is given by (3.154), we can write

$$\rho(x,t) = \frac{3}{10}\phi_3^2(x) + \frac{\sqrt{3}}{10}\phi_3(x)\phi_5(x)\left[e^{i(E_3-E_5)t/\hbar} + e^{-i(E_3-E_5)t/\hbar}\right] + \frac{1}{10}\phi_5^2(x). \quad (3.155)$$

From (3.144) we have $E_3 - E_5 = 9E_1 - 25E_1 = -16E_1 = -8\pi^2\hbar^2/(ma^2)$. Thus, $\rho(x,t)$ becomes

$$\begin{aligned}\rho(x,t) &= \frac{3}{10}\phi_3^2(x) + \frac{\sqrt{3}}{5}\phi_3(x)\phi_5(x)\cos\left(\frac{16E_1t}{\hbar}\right) + \frac{1}{10}\phi_5^2(x) \\ &= \frac{3}{5a}\sin^2\left(\frac{3\pi x}{a}\right) + \frac{2\sqrt{3}}{5a}\sin\left(\frac{3\pi x}{a}\right)\sin\left(\frac{5\pi x}{a}\right)\cos\left(\frac{16E_1t}{\hbar}\right) \\ &\quad + \frac{1}{5a}\sin^2\left(\frac{5\pi x}{a}\right). \end{aligned} \quad (3.156)$$

Since the system is one-dimensional, the action of the gradient operator on $\psi(x,t)$ and $\psi^*(x,t)$ is given by $\vec{\nabla}\psi(x,t) = (d\psi(x,t)/dx)\vec{i}$ and $\vec{\nabla}\psi^*(x,t) = (d\psi^*(x,t)/dx)\vec{i}$. We can thus write the current density $\vec{J}(x,t) = (i\hbar/2m)\left(\psi(x,t)\vec{\nabla}\psi^*(x,t) - \psi^*(x,t)\vec{\nabla}\psi(x,t)\right)$ as

$$\vec{J}(x,t) = \frac{i\hbar}{2m}\left(\psi(x,t)\frac{d\psi^*(x,t)}{dx} - \psi^*(x,t)\frac{d\psi(x,t)}{dx}\right)\vec{i}. \quad (3.157)$$

Using (3.154) we have

$$\frac{d\psi(x,t)}{dx} = \frac{3\pi}{a}\sqrt{\frac{3}{5a}}\cos\left(\frac{3\pi x}{a}\right)e^{-iE_3t/\hbar} + \frac{5\pi}{a}\frac{1}{\sqrt{5a}}\cos\left(\frac{5\pi x}{a}\right)e^{-iE_5t/\hbar}, \quad (3.158)$$

$$\frac{d\psi^*(x,t)}{dx} = \frac{3\pi}{a}\sqrt{\frac{3}{5a}}\cos\left(\frac{3\pi x}{a}\right)e^{iE_3t/\hbar} + \frac{5\pi}{a}\frac{1}{\sqrt{5a}}\cos\left(\frac{5\pi x}{a}\right)e^{iE_5t/\hbar}. \quad (3.159)$$

A straightforward calculation yields

$$\begin{aligned}\psi\frac{d\psi^*}{dx} - \psi^*\frac{d\psi}{dx} = &-2i\pi\frac{\sqrt{3}}{5a^2}\left[5\sin\left(\frac{3\pi x}{a}\right)\cos\left(\frac{5\pi x}{a}\right) - 3\sin\left(\frac{5\pi x}{a}\right)\cos\left(\frac{3\pi x}{a}\right)\right] \\ &\times \sin\left(\frac{E_3 - E_5}{\hbar}t\right). \end{aligned} \quad (3.160)$$

Inserting this into (3.157) and using $E_3 - E_5 = -16E_1$, we have

$$\vec{J}(x,t) = -\frac{\pi\hbar}{m}\frac{\sqrt{3}}{5a^2}\left[5\sin\left(\frac{3\pi x}{a}\right)\cos\left(\frac{5\pi x}{a}\right) - 3\sin\left(\frac{5\pi x}{a}\right)\cos\left(\frac{3\pi x}{a}\right)\right]\sin\left(\frac{16E_1t}{\hbar}\right)\vec{i}. \quad (3.161)$$

(c) Performing the time derivative of (3.156) and using the expression $32\sqrt{3}E_1/(5a\hbar) = 16\pi^2\hbar\sqrt{3}/(5ma^3)$, since $E_1 = \pi^2\hbar^2/(2ma^2)$, we obtain

$$\begin{aligned}\frac{\partial\rho}{\partial t} &= -\frac{32\sqrt{3}E_1}{5a\hbar}\sin\left(\frac{3\pi x}{a}\right)\sin\left(\frac{5\pi x}{a}\right)\sin\left(\frac{16E_1t}{\hbar}\right) \\ &= -\frac{16\pi^2\hbar\sqrt{3}}{5ma^3}\sin\left(\frac{3\pi x}{a}\right)\sin\left(\frac{5\pi x}{a}\right)\sin\left(\frac{16E_1t}{\hbar}\right), \end{aligned} \quad (3.162)$$

Now, taking the divergence of (3.161), we end up with

$$\vec{\nabla}\cdot\vec{J}(x,t)=\frac{dJ(x,t)}{dx}=\frac{16\pi^2\hbar\sqrt{3}}{5ma^3}\sin\left(\frac{3\pi x}{a}\right)\sin\left(\frac{5\pi x}{a}\right)\sin\left(\frac{16E_1t}{\hbar}\right). \tag{3.163}$$

The addition of (3.162) and (3.163) confirms the conservation of probability:

$$\frac{\partial\rho}{\partial t}+\vec{\nabla}\cdot\vec{J}(x,t)=0. \tag{3.164}$$

Problem 3.3
Consider a one-dimensional particle which is confined within the region $0\le x\le a$ and whose wave function is $\Psi(x,t)=\sin(\pi x/a)\exp(-i\omega t)$.

(a) Find the potential $V(x)$.

(b) Calculate the probability of finding the particle in the interval $a/4\le x\le 3a/4$.

Solution

(a) Since the first time derivative and the second x derivative of $\Psi(x,t)$ are given by $\partial\Psi(x,t)/\partial t=-i\omega\Psi(x,t)$ and $\partial^2\Psi(x,t)/\partial x^2=-(\pi^2/a^2)\Psi(x,t)$, the Schrödinger equation (3.68) yields

$$i\hbar(-i\omega)\Psi(x,t)=\frac{\hbar^2}{2m}\frac{\pi^2}{a^2}\Psi(x,t)+\hat{V}(x,t)\Psi(x,t). \tag{3.165}$$

Hence $V(x,t)$ is time independent and given by $V(x)=\hbar\omega-\hbar^2\pi^2/(2ma^2)$.

(b) The probability of finding the particle in the interval $a/4\le x\le 3a/4$ can be obtained from (3.4):

$$P=\frac{\int_{a/4}^{3a/4}|\psi(x)|^2dx}{\int_0^a|\psi(x)|^2dx}=\frac{\int_{a/4}^{3a/4}\sin^2(\pi x/a)\,dx}{\int_0^a\sin^2(\pi x/a)\,dx}=\frac{2+\pi}{2\pi}=0.82 \tag{3.166}$$

Problem 3.4
A system is initially in the state $|\psi_0\rangle=[\sqrt{2}|\phi_1\rangle+\sqrt{3}|\phi_2\rangle+|\phi_3\rangle+|\phi_4\rangle]/\sqrt{7}$, where $|\phi_n\rangle$ are eigenstates of the system's Hamiltonian such that $\hat{H}|\phi_n\rangle=n^2\mathcal{E}_0|\phi_n\rangle$.

(a) If energy is measured, what values will be obtained and with what probabilities?

(b) Consider an operator $\hat{A}$ whose action on $|\phi_n\rangle$ is defined by $\hat{A}|\phi_n\rangle=(n+1)a_0|\phi_n\rangle$. If A is measured, what values will be obtained and with what probabilities?

(c) Suppose that a measurement of the energy yields $4\mathcal{E}_0$. If we measure A immediately afterwards, what value will be obtained?

Solution

(a) A measurement of the energy yields $E_n=\langle\phi_n|\hat{H}|\phi_n\rangle=n^2\mathcal{E}_0$, that is

$$E_1=\mathcal{E}_0,\quad E_2=4\mathcal{E}_0,\quad E_3=9\mathcal{E}_0,\quad E_4=16\mathcal{E}_0. \tag{3.167}$$

Since $|\ \psi_0\rangle$ is normalized, $\langle\psi_0\mid\psi_0\rangle=(2+3+1+1)/7=1$, and using (3.2), we can write the probabilities corresponding to (3.167) as $P(E_n)=|\langle\phi_n|\psi_0\rangle|^2/\langle\psi_0\mid\psi_0\rangle=|\langle\phi_n|\psi_0\rangle|^2$; hence,

using the fact that $\langle\phi_n|\phi_m\rangle = \delta_{nm}$, we have

$$P(E_1) = \left|\sqrt{\frac{2}{7}}\langle\phi_1|\phi_1\rangle\right|^2 = \frac{2}{7}, \qquad P(E_2) = \left|\sqrt{\frac{3}{7}}\langle\phi_2|\phi_2\rangle\right|^2 = \frac{3}{7}, \tag{3.168}$$

$$P(E_3) = \left|\frac{1}{\sqrt{7}}\langle\phi_3|\phi_3\rangle\right|^2 = \frac{1}{7}, \qquad P(E_4) = \left|\frac{1}{\sqrt{7}}\langle\phi_4|\phi_4\rangle\right|^2 = \frac{1}{7}. \tag{3.169}$$

(b) Similarly, a measurement of the observable $\hat{A}$ yields $a_n = \langle\phi_n|\hat{A}|\phi_n\rangle = (n+1)a_0$; that is,

$$a_1 = 2a_0, \quad a_2 = 3a_0, \quad a_3 = 4a_0, \quad a_4 = 5a_0. \tag{3.170}$$

Again, using (3.2) and since $|\ \psi_0\rangle$ is normalized, we can ascertain that the probabilities corresponding to the values (3.170) are given by $P(a_n) = |\langle\phi_n|\psi_0\rangle|^2 / \langle\psi_0\ |\ \psi_0\rangle = |\langle\phi_n|\psi_0\rangle|^2$, or

$$P(a_1) = \left|\sqrt{\frac{2}{7}}\langle\phi_1|\phi_1\rangle\right|^2 = \frac{2}{7}, \qquad P(a_2) = \left|\sqrt{\frac{3}{7}}\langle\phi_2|\phi_2\rangle\right|^2 = \frac{3}{7}, \tag{3.171}$$

$$P(a_3) = \left|\frac{1}{\sqrt{7}}\langle\phi_3|\phi_3\rangle\right|^2 = \frac{1}{7}, \qquad P(a_4) = \left|\frac{1}{\sqrt{7}}\langle\phi_4|\phi_4\rangle\right|^2 = \frac{1}{7}. \tag{3.172}$$

(c) An energy measurement that yields $4\mathcal{E}_0$ implies that the system is left in the state $|\phi_2\rangle$. A measurement of the observable A immediately afterwards leads to

$$\langle\phi_2|\hat{A}|\phi_2\rangle = 3a_0\langle\phi_2|\phi_2\rangle = 3a_0. \tag{3.173}$$

Problem 3.5

(a) Assuming that the system of Problem 3.4 is initially in the state $|\phi_3\rangle$, what values for the energy and the observable A will be obtained if we measure: (i)H first then A, (ii) A first then H?

(b) Compare the results obtained in (i) and (ii) and infer whether $\hat{H}$ and $\hat{A}$ are compatible. Calculate $[\hat{H},\ \hat{A}]|\phi_3\rangle$.

Solution

(a) (i) The measurement of H first then A is represented by $\hat{A}\hat{H}|\phi_3\rangle$. Using the relations $\hat{H}|\phi_n\rangle = n^2\mathcal{E}_0|\phi_n\rangle$ and $\hat{A}|\phi_n\rangle = na_0|\phi_{n+1}\rangle$, we have

$$\hat{A}\hat{H}|\phi_3\rangle = 9\mathcal{E}_0\hat{A}|\phi_3\rangle = 27\mathcal{E}_0a_0|\phi_4\rangle. \tag{3.174}$$

(ii) Measuring A first and then H, we will obtain

$$\hat{H}\hat{A}|\phi_3\rangle = 3a_0\hat{H}|\phi_4\rangle = 48\mathcal{E}_0a_0|\phi_4\rangle. \tag{3.175}$$

(b) Equations (3.174) and (3.175) show that the actions of $\hat{A}\hat{H}$ and $\hat{H}\hat{A}$ yield different results. This means that $\hat{H}$ and $\hat{A}$ do not commute; hence they are not compatible. We can thus write

$$[\hat{H},\ \hat{A}]|\phi_3\rangle = (48-27)\mathcal{E}_0a_0|\phi_4\rangle = 17\mathcal{E}_0a_0|\phi_4\rangle. \tag{3.176}$$

Problem 3.6
Consider a physical system whose Hamiltonian H and initial state $|\psi_0\rangle$ are given by

$$H = \mathcal{E}\begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad |\psi_0\rangle = \frac{1}{\sqrt{5}}\begin{pmatrix} 1-i \\ 1-i \\ 1 \end{pmatrix},$$

where $\mathcal{E}$ has the dimensions of energy.

(a) What values will we obtain when measuring the energy and with what probabilities?

(b) Calculate $\langle \hat{H} \rangle$, the expectation value of the Hamiltonian.

Solution

(a) The results of the energy measurement are given by the eigenvalues of H. A diagonalization of H yields a nondegenerate eigenenergy $E_1 = \mathcal{E}$ and a doubly degenerate value $E_2 = E_3 = -\mathcal{E}$ whose respective eigenvectors are given by

$$|\phi_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}, \qquad |\phi_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} -i \\ 1 \\ 0 \end{pmatrix}, \qquad |\phi_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}; \tag{3.177}$$

these eigenvectors are orthogonal since H is Hermitian. Note that the initial state $|\psi_0\rangle$ can be written in terms of $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$ as follows:

$$|\psi_0\rangle = \frac{1}{\sqrt{5}}\begin{pmatrix} 1-i \\ 1-i \\ 1 \end{pmatrix} = \sqrt{\frac{2}{5}}|\phi_1\rangle + \sqrt{\frac{2}{5}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle. \tag{3.178}$$

Since $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$ are orthonormal, the probability of measuring $E_1 = \mathcal{E}$ is given by

$$P_1(E_1) = |\langle\phi_1|\psi_0\rangle|^2 = \left|\sqrt{\frac{2}{5}}\langle\phi_1|\phi_1\rangle\right|^2 = \frac{2}{5}. \tag{3.179}$$

Now, since the other eigenvalue is doubly degenerate, $E_2 = E_3 = -\mathcal{E}$, the probability of measuring $-\mathcal{E}$ can be obtained from (3.3):

$$P_2(E_2) = |\langle\phi_2|\psi_0\rangle|^2 + |\langle\phi_3|\psi_0\rangle|^2 = \frac{2}{5} + \frac{1}{5} = \frac{3}{5}. \tag{3.180}$$

(b) From (3.179) and (3.180), we have

$$\langle \hat{H} \rangle = P_1E_1 + P_2E_2 = \frac{2}{5}\mathcal{E} - \frac{3}{5}\mathcal{E} = -\frac{1}{5}\mathcal{E}. \tag{3.181}$$

We can obtain the same result by calculating the expectation value of $\hat{H}$ with respect to $|\psi_0\rangle$. Since $\langle\psi_0|\psi_0\rangle = 1$, we have $\langle \hat{H} \rangle = \langle\psi_0|\hat{H}|\psi_0\rangle/\langle\psi_0|\psi_0\rangle = \langle\psi_0|\hat{H}|\psi_0\rangle$:

$$\langle \hat{H} \rangle = \langle\psi_0|\hat{H}|\psi_0\rangle = \frac{\mathcal{E}}{5}\begin{pmatrix} 1+i & 1+i & 1 \end{pmatrix}\begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} 1-i \\ 1-i \\ 1 \end{pmatrix} = -\frac{1}{5}\mathcal{E}. \tag{3.182}$$

Problem 3.7
Consider a system whose Hamiltonian H and an operator A are given by the matrices

$$H = \mathcal{E}_0 \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad A = a \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

where $\mathcal{E}_0$ has the dimensions of energy.

(a) If we measure the energy, what values will we obtain?

(b) Suppose that when we measure the energy, we obtain a value of $-\mathcal{E}_0$. Immediately afterwards, we measure A. What values will we obtain for A and what are the probabilities corresponding to each value?

(c) Calculate the uncertainty ΔA.

Solution

(a) The possible energies are given by the eigenvalues of H. A diagonalization of H yields three nondegenerate eigenenergies $E_1 = 0$, $E_2 = -\mathcal{E}_0$, and $E_3 = 2\mathcal{E}_0$. The respective eigenvectors are

$$|\phi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad |\phi_2\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \qquad |\phi_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}; \tag{3.183}$$

these eigenvectors are orthonormal.

(b) If a measurement of the energy yields $-\mathcal{E}_0$, this means that the system is left in the state $|\phi_2\rangle$. When we measure the next observable, A, the system is in the state $|\phi_2\rangle$. The result we obtain for A is given by any of the eigenvalues of A. A diagonalization of A yields three nondegenerate values: $a_1 = -\sqrt{17}a$, $a_2 = 0$, and $a_3 = \sqrt{17}a$; their respective eigenvectors are given by

$$|a_1\rangle = \frac{1}{\sqrt{34}} \begin{pmatrix} 4 \\ -\sqrt{17} \\ 1 \end{pmatrix}, \qquad |a_2\rangle = \frac{1}{\sqrt{17}} \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix}, \qquad |a_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 4 \\ \sqrt{17} \\ 1 \end{pmatrix}. \tag{3.184}$$

Thus, when measuring A on a system which is in the state $|\phi_2\rangle$, the probability of finding $-\sqrt{17}a$ is given by

$$P_1(a_1) = |\langle a_1|\phi_2\rangle|^2 = \left| \frac{1}{\sqrt{34}} \begin{pmatrix} 4 & -\sqrt{17} & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right|^2 = \frac{1}{34}. \tag{3.185}$$

Similarly, the probabilities of measuring 0 and $\sqrt{17}a$ are

$$P_2(a_2) = |\langle a_2|\phi_2\rangle|^2 = \left| \frac{1}{\sqrt{17}} \begin{pmatrix} 1 & 0 & -4 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right|^2 = \frac{16}{17}, \tag{3.186}$$

$$P_3(a_3) = |\langle a_3|\phi_2\rangle|^2 = \left| \frac{1}{\sqrt{34}} \begin{pmatrix} 4 & \sqrt{17} & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right|^2 = \frac{1}{34}. \tag{3.187}$$

(c) Since the system, when measuring A is in the state $|\phi_2\rangle$, the uncertainty ΔA is given by $\Delta A = \sqrt{\langle\phi_2|A^2|\phi_2\rangle - \langle\phi_2|A|\phi_2\rangle^2}$, where

$$\langle\phi_2|A|\phi_2\rangle = a \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0, \tag{3.188}$$

$$\langle\phi_2|A^2|\phi_2\rangle = a^2 \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = a^2. \tag{3.189}$$

Thus we have $\Delta A = a$.

Problem 3.8

Consider a system whose state and two observables are given by

$$|\psi(t)\rangle = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \qquad A = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

(a) What is the probability that a measurement of A at time t yields -1?

(b) Let us carry out a set of two measurements where B is measured first and then, immediately afterwards, A is measured. Find the probability of obtaining a value of 0 for B and a value of 1 for A.

(c) Now we measure A first then, immediately afterwards, B. Find the probability of obtaining a value of 1 for A and a value of 0 for B.

(d) Compare the results of (b) and (c). Explain.

(e) Which among the sets of operators $\{\hat{A}\}$, $\{\hat{B}\}$, and $\{\hat{A}, \hat{B}\}$ form a complete set of commuting operators (CSCO)?

Solution

(a) A measurement of A yields any of the eigenvalues of A which are given by $a_1 = -1$, $a_2 = 0$, $a_3 = 1$; the respective (normalized) eigenstates are

$$|a_1\rangle = \frac{1}{2} \begin{pmatrix} -1 \\ \sqrt{2} \\ -1 \end{pmatrix}, \qquad |a_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \qquad |a_3\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}. \tag{3.190}$$

The probability of obtaining $a_1 = -1$ is

$$P(-1) = \frac{|\langle a_1|\psi(t)\rangle|^2}{\langle\psi(t)|\psi(t)\rangle} = \frac{1}{6} \left| \frac{1}{2} \begin{pmatrix} -1 & \sqrt{2} & -1 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} \right|^2 = \frac{1}{3}, \tag{3.191}$$

where we have used the fact that $\langle\psi(t)|\psi(t)\rangle = \begin{pmatrix} -1 & 2 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} = 6$.

(b) A measurement of B yields a value which is equal to any of the eigenvalues of B: $b_1 = -1$, $b_2 = 0$, and $b_3 = 1$; their corresponding eigenvectors are

$$|b_1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \qquad |b_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad |b_3\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \tag{3.192}$$

Since the system was in the state $|\psi(t)\rangle$, the probability of obtaining the value $b_2 = 0$ for B is

$$P(b_2) = \frac{|\langle b_2|\psi(t)\rangle|^2}{\langle\psi(t)|\psi(t)\rangle} = \frac{1}{6}\left|\begin{pmatrix} 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}\right|^2 = \frac{2}{3}. \tag{3.193}$$

We deal now with the measurement of the other observable, A. The observables A and B do not have common eigenstates, since they do not commute. After measuring B (the result is $b_2 = 0$), the system is left, according to Postulate 3, in a state $|\phi\rangle$ which can be found by projecting $|\psi(t)\rangle$ onto $|b_2\rangle$:

$$|\phi\rangle = |b_2\rangle\langle b_2|\psi(t)\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}. \tag{3.194}$$

The probability of finding 1 when we measure A is given by

$$P(a_3) = \frac{|\langle a_3|\phi\rangle|^2}{\langle\phi|\phi\rangle} = \frac{1}{4}\left|\frac{1}{2}\begin{pmatrix} 1 & \sqrt{2} & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}\right|^2 = \frac{1}{2}, \tag{3.195}$$

since $\langle\phi|\phi\rangle = 4$. In summary, when measuring B then A, the probability of finding a value of 0 for B and 1 for A is given by the product of the probabilities (3.193) and (3.195):

$$P(b_2, a_3) = P(b_2)P(a_3) = \frac{2}{3}\frac{1}{2} = \frac{1}{3}. \tag{3.196}$$

(c) Next we measure A first then B. Since the system is in the state $|\psi(t)\rangle$, the probability of measuring $a_3 = 1$ for A is given by

$$P'(a_3) = \frac{|\langle a_3|\psi(t)\rangle|^2}{\langle\psi(t)|\psi(t)\rangle} = \frac{1}{6}\left|\frac{1}{2}\begin{pmatrix} 1 & \sqrt{2} & 1 \end{pmatrix}\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}\right|^2 = \frac{1}{3}, \tag{3.197}$$

where we have used the expression (3.190) for $|a_3\rangle$.

We then proceed to the measurement of B. The state of the system just after measuring A (with a value $a_3 = 1$) is given by a projection of $|\psi(t)\rangle$ onto $|a_3\rangle$:

$$|\chi\rangle = |a_3\rangle\langle a_3|\psi(t)\rangle = \frac{1}{4}\begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}\begin{pmatrix} 1 & \sqrt{2} & 1 \end{pmatrix}\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} = \frac{\sqrt{2}}{2}\begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}. \tag{3.198}$$

So the probability of finding a value of $b_2 = 0$ when measuring B is given by

$$P'(b_2) = \frac{|\langle b_2|\chi\rangle|^2}{\langle\chi|\chi\rangle} = \frac{1}{2}\left|\frac{\sqrt{2}}{2}\begin{pmatrix} 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}\right|^2 = \frac{1}{2}, \tag{3.199}$$

since $\langle\chi|\chi\rangle = 2$.

So when measuring A then B, the probability of finding a value of 1 for A and 0 for B is given by the product of the probabilities (3.199) and (3.197):

$$P(a_3, b_2) = P'(a_3)P'(b_2) = \frac{1}{3}\frac{1}{2} = \frac{1}{6}. \tag{3.200}$$

(d) The probabilities $P(b_2, a_3)$ and $P(a_3, b_2)$, as shown in (3.196) and (3.200), are different. This is expected, since A and B do not commute. The result of the successive measurements of A and B therefore depends on the order in which they are carried out. The probability of obtaining 0 for B then 1 for A is equal to $\frac{1}{3}$. On the other hand, the probability of obtaining 1 for A then 0 for B is equal to $\frac{1}{6}$. However, if the observables A and B commute, the result of the measurements will not depend on the order in which they are carried out (this idea is illustrated in the following solved problem).

(e) As stated in the text, any operator with non-degenerate eigenvalues constitutes, all by itself, a CSCO. Hence each of $\{\hat{A}\}$ and $\{\hat{B}\}$ forms a CSCO, since their eigenvalues are not degenerate. However, the set $\{\hat{A}, \hat{B}\}$ does not form a CSCO since the opertators $\{\hat{A}\}$ and $\{\hat{B}\}$ do not commute.

Problem 3.9

Consider a system whose state and two observables A and B are given by

$$|\psi(t)\rangle = \frac{1}{6}\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}, \qquad A = \frac{1}{\sqrt{2}}\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & i \\ 0 & -i & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$

(a) We perform a measurement where A is measured first and then, immediately afterwards, B is measured. Find the probability of obtaining a value of 0 for A and a value of 1 for B.

(b) Now we measure B first then, immediately afterwards, A. Find the probability of obtaining a value of 1 for B and a value of 0 for A.

(c) Compare the results of (b) and (c). Explain.

(d) Which among the sets of operators $\{\hat{A}\}$, $\{\hat{B}\}$, and $\{\hat{A}, \hat{B}\}$ form a complete set of commuting operators (CSCO)?

Solution

(a) A measurement of A yields any of the eigenvalues of A which are given by $a_1 = 0$ (not degenerate) and $a_2 = a_3 = 2$ (doubly degenerate); the respective (normalized) eigenstates are

$$|a_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -i \\ 1 \end{pmatrix}, \qquad |a_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ i \\ 1 \end{pmatrix}, \qquad |a_3\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \tag{3.201}$$

The probability that a measurement of A yields $a_1 = 0$ is given by

$$P(a_1) = \frac{|\langle a_1|\psi(t)\rangle|^2}{\langle\psi(t)|\psi(t)\rangle} = \frac{36}{17}\left|\frac{1}{\sqrt{2}}\frac{1}{6}\begin{pmatrix} 0 & i & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}\right|^2 = \frac{8}{17}, \tag{3.202}$$

where we have used the fact that $\langle\psi(t)|\psi(t)\rangle = \frac{1}{36}\begin{pmatrix} 1 & 0 & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} = \frac{17}{36}$.

Since the system was initially in the state $|\psi(t)\rangle$, after a measurement of A yields $a_1 = 0$, the system is left, as mentioned in Postulate 3, in the following state:

$$|\phi\rangle = |a_1\rangle\langle a_1|\psi(t)\rangle = \frac{1}{2}\frac{1}{6}\begin{pmatrix} 0 \\ -i \\ 1 \end{pmatrix}\begin{pmatrix} 0 & i & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 0 \\ -i \\ 1 \end{pmatrix}. \tag{3.203}$$

As for the measurement of B, we obtain any of the eigenvalues $b_1 = -1$, $b_2 = b_3 = 1$; their corresponding eigenvectors are

$$|b_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ i \\ 1 \end{pmatrix}, \qquad |b_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -i \\ 1 \end{pmatrix}, \qquad |b_3\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \tag{3.204}$$

Since the system is now in the state $|\phi\rangle$, the probability of obtaining the (doubly degenerate) value $b_2 = b_3 = 1$ for B is

$$\begin{aligned} P(b_2) &= \frac{|\langle b_2|\phi\rangle|^2}{\langle\phi|\phi\rangle} + \frac{|\langle b_3|\phi\rangle|^2}{\langle\phi|\phi\rangle} \\ &= \frac{1}{2}\left|\frac{1}{\sqrt{2}}\begin{pmatrix} 0 & i & 1 \end{pmatrix}\begin{pmatrix} 0 \\ -i \\ 1 \end{pmatrix}\right|^2 + \frac{1}{2}\left|\begin{pmatrix} 1 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ -i \\ 1 \end{pmatrix}\right|^2 \\ &= 1. \end{aligned} \tag{3.205}$$

The reason $P(b_2) = 1$ is because the new state $|\phi\rangle$ is an eigenstate of B; in fact $|\phi\rangle = \sqrt{2}/3|b_2\rangle$.

In sum, when measuring A then B, the probability of finding a value of 0 for A and 1 for B is given by the product of the probabilities (3.202) and (3.205):

$$P(a_1, b_2) = P(a_1)P(b_2) = \frac{8}{17}. \tag{3.206}$$

(b) Next we measure B first then A. Since the system is in the state $|\psi(t)\rangle$ and since the value $b_2 = b_3 = 1$ is doubly degenerate, the probability of measuring 1 for B is given by

$$\begin{aligned} P'(b_2) &= \frac{|\langle b_2|\psi(t)\rangle|^2}{\langle\psi(t)|\psi(t)\rangle} + \frac{|\langle b_3|\psi(t)\rangle|^2}{\langle\psi(t)|\psi(t)\rangle} \\ &= \frac{36}{17}\frac{1}{36}\left[\left|\frac{1}{\sqrt{2}}\begin{pmatrix} 0 & i & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}\right|^2 + \left|\begin{pmatrix} 1 & 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}\right|^2\right] \\ &= \frac{9}{17}. \end{aligned} \tag{3.207}$$

We now proceed to the measurement of A. The state of the system immediately after measuring B (with a value $b_2 = b_3 = 1$) is given by a projection of $|\psi(t)\rangle$ onto $|b_2\rangle$, and $|b_3\rangle$

$$\begin{aligned} |\chi\rangle &= |b_2\rangle\langle b_2|\psi(t)\rangle + |b_3\rangle\langle b_3|\psi(t)\rangle \\ &= \frac{1}{12}\begin{pmatrix} 0 \\ -i \\ 1 \end{pmatrix}\begin{pmatrix} 0 & i & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} + \frac{1}{6}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} \\ &= \frac{1}{6}\begin{pmatrix} 1 \\ -2i \\ 2i \end{pmatrix}. \end{aligned} \tag{3.208}$$

So the probability of finding a value of $a_1 = 0$ when measuring A is given by

$$P'(a_1) = \frac{|\langle a_1|\chi\rangle|^2}{\langle\chi|\chi\rangle} = \frac{36}{9}\left|\frac{1}{6\sqrt{2}}\begin{pmatrix} 0 & i & 1 \end{pmatrix}\begin{pmatrix} 1 \\ -2i \\ 2i \end{pmatrix}\right|^2 = \frac{8}{9}, \tag{3.209}$$

since $\langle\chi|\chi\rangle = \frac{9}{36}$.

Therefore, when measuring B then A, the probability of finding a value of 1 for B and 0 for A is given by the product of the probabilities (3.207) and (3.209):

$$P(b_2, a_3) = P'(b_2)P'(a_1) = \frac{9}{17}\frac{8}{9} = \frac{8}{17}. \tag{3.210}$$

(c) The probabilities $P(a_1, b_2)$ and $P(b_2, a_1)$, as shown in (3.206) and (3.210), are equal. This is expected since A and B do commute. The result of the successive measurements of A and B does not depend on the order in which they are carried out.

(d) Neither $\{\hat{A}\}$ nor $\{\hat{B}\}$ forms a CSCO since their eigenvalues are degenerate. The set $\{\hat{A}, \hat{B}\}$, however, does form a CSCO since the opertators $\{\hat{A}\}$ and $\{\hat{B}\}$ commute. The set of eigenstates that are common to $\{\hat{A}, \hat{B}\}$ are given by

$$|a_2,\ b_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ i \\ 1 \end{pmatrix}, \qquad |a_1,\ b_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -i \\ 1 \end{pmatrix}, \qquad |a_3,\ b_3\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \tag{3.211}$$

Problem 3.10

Consider a physical system which has a number of observables that are represented by the following matrices:

$$A = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix},\ B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 3 & 0 \end{pmatrix},\ C = \begin{pmatrix} 0 & 3 & 0 \\ 3 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix},\ D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$

(a) Find the results of the measurements of these observables.

(b) Which among these observables are compatible? Give a basis of eigenvectors common to these observables.

(c) Which among the sets of operators $\{\hat{A}\}$, $\{\hat{B}\}$, $\{\hat{C}\}$, $\{\hat{D}\}$ and their various combinations, such as $\{\hat{A}, \hat{B}\}$, $\{\hat{A}, \hat{C}\}$, $\{\hat{B}, \hat{C}\}$, $\{\hat{A}, \hat{D}\}$, $\{\hat{A}, \hat{B}, \hat{C}\}$, form a complete set of commuting operators (CSCO)?

Solution

(a) The measurements of A, B, C and D yield $a_1 = -1$, $a_2 = 3$, $a_3 = 5$, $b_1 = -3$, $b_2 = 1$, $b_3 = 3$, $c_1 = -1/\sqrt{2}$, $c_2 = 0$, $c_3 = 1/\sqrt{2}$, $d_1 = -1$, $d_2 = d_3 = 1$; the respective eigenvectors of A, B, C and D are

$$|a_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \qquad |a_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad |a_3\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \tag{3.212}$$

$$|b_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \qquad |b_2\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad |b_3\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \tag{3.213}$$

$$|c_1\rangle = \frac{1}{\sqrt{26}}\begin{pmatrix} 3 \\ -\sqrt{13} \\ 2 \end{pmatrix}, \quad |c_2\rangle = \frac{1}{\sqrt{13}}\begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix}, \quad |c_3\rangle = \frac{1}{\sqrt{26}}\begin{pmatrix} 3 \\ \sqrt{13} \\ 2 \end{pmatrix}, \tag{3.214}$$

$$|d_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ i \\ 1 \end{pmatrix}, \qquad |d_2\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad |d_3\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ i \end{pmatrix}. \tag{3.215}$$

(b) We can verify that, among the observables A, B, C, and D, only A and B are compatible, since the matrices A and B commute; the rest do not commute with one another (neither A nor B commutes with C or D; C and D do not commute).

From (3.212) and (3.213) we see that the three states $|a_1, b_1\rangle$, $|a_2, b_3\rangle$, $|a_3, b_2\rangle$,

$$|a_1, b_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \qquad |a_2, b_3\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad |a_3, b_2\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \tag{3.216}$$

form a common, complete basis for A and B, since $\hat{A}|a_n, b_m\rangle = a_n|a_n, b_m\rangle$ and $\hat{B}|a_n, b_m\rangle = b_m|a_n, b_m\rangle$.

(c) First, since the eigenvalues of the operators $\{\hat{A}\}$, $\{\hat{B}\}$, and $\{\hat{C}\}$ are all nondegenerate, each one of $\{\hat{A}\}$, $\{\hat{B}\}$, and $\{\hat{C}\}$ forms separately a CSCO. Additionally, since two eigenvalues of $\{\hat{D}\}$ are degenerate ($d_2 = d_3 = 1$), the operator $\{\hat{D}\}$ does not form a CSCO.

Now, among the various combinations $\{\hat{A}, \hat{B}\}$, $\{\hat{A}, \hat{C}\}$, $\{\hat{B}, \hat{C}\}$, $\{\hat{A}, \hat{D}\}$, and $\{\hat{A}, \hat{B}, \hat{C}\}$, only $\{\hat{A}, \hat{B}\}$ forms a CSCO, because $\{\hat{A}\}$ and $\{\hat{B}\}$ are the only operators that commute; the set of their joint eigenvectors are given by $|a_1, b_1\rangle$, $|a_2, b_3\rangle$, $|a_3, b_2\rangle$.

Problem 3.11

Consider a system whose initial state $|\psi(0)\rangle$ and Hamiltonian are given by

$$|\psi(0)\rangle = \frac{1}{5}\begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}, \qquad H = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 5 \\ 0 & 5 & 0 \end{pmatrix}.$$

(a) If a measurement of the energy is carried out, what values would we obtain and with what probabilities?

(b) Find the state of the system at a later time t; you may need to expand $|\psi(0)\rangle$ in terms of the eigenvectors of H.

(c) Find the total energy of the system at time $t = 0$ and any later time t; are these values different?

(d) Does $\{\hat{H}\}$ form a complete set of commuting operators?

Solution

(a) A measurement of the energy yields the values $E_1 = -5$, $E_2 = 3$, $E_3 = 5$; the respective (orthonormal) eigenvectors of these values are

$$|\phi_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \qquad |\phi_2\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad |\phi_3\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}. \tag{3.217}$$

The probabilities of finding the values $E_1 = -5$, $E_2 = 3$, $E_3 = 5$ are given by

$$P(E_1) = |\langle\phi_1|\psi(0)\rangle|^2 = \left|\frac{1}{5\sqrt{2}}\begin{pmatrix} 0 & -1 & 1 \end{pmatrix}\begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}\right|^2 = \frac{8}{25}, \tag{3.218}$$

$$P(E_2) = |\langle\phi_2|\psi(0)\rangle|^2 = \left|\frac{1}{5}\begin{pmatrix} 1 & 0 & 0 \end{pmatrix}\begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}\right|^2 = \frac{9}{25}, \tag{3.219}$$

$$P(E_3) = |\langle\phi_3|\psi(0)\rangle|^2 = \left|\frac{1}{5\sqrt{2}}\begin{pmatrix} 0 & 1 & 1 \end{pmatrix}\begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}\right|^2 = \frac{8}{25}. \tag{3.220}$$

(b) To find $|\psi(t)\rangle$ we need to expand $|\psi(0)\rangle$ in terms of the eigenvectors (3.217):

$$|\psi(0)\rangle = \frac{1}{5}\begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix} = \frac{2\sqrt{2}}{5}|\phi_1\rangle + \frac{3}{5}|\phi_2\rangle + \frac{2\sqrt{2}}{5}|\phi_3\rangle; \tag{3.221}$$

hence

$$|\psi(t)\rangle = \frac{2\sqrt{2}}{5}e^{-iE_1t}|\phi_1\rangle + \frac{3}{5}e^{-iE_2t}|\phi_2\rangle + \frac{2\sqrt{2}}{5}e^{-iE_3t}|\phi_3\rangle = \frac{1}{5}\begin{pmatrix} 3e^{-3it} \\ -4i\sin 5t \\ 4\cos 5t \end{pmatrix}. \tag{3.222}$$

(c) We can calculate the energy at time $t = 0$ in three quite different ways. The first method uses the bra-ket notation. Since $\langle\psi(0)|\psi(0)\rangle = 1$, $\langle\phi_n|\phi_m\rangle = \delta_{nm}$ and since $\hat{H}|\phi_n\rangle = E_n|\phi_n\rangle$, we have

$$\begin{aligned} E(0) = \langle\psi(0)|\hat{H}|\psi(0)\rangle &= \frac{8}{25}\langle\phi_1|\hat{H}|\phi_1\rangle + \frac{9}{25}\langle\phi_2|\hat{H}|\phi_2\rangle + \frac{8}{25}\langle\phi_3|\hat{H}|\phi_3\rangle \\ &= \frac{8}{25}(-5) + \frac{9}{25}(3) + \frac{8}{25}(5) = \frac{27}{25}. \end{aligned} \tag{3.223}$$

The second method uses matrix algebra:

$$E(0) = \langle\psi(0)|\hat{H}|\psi(0)\rangle = \frac{1}{25}\begin{pmatrix} 3 & 0 & 4 \end{pmatrix}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 5 \\ 0 & 5 & 0 \end{pmatrix}\begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix} = \frac{27}{25}. \tag{3.224}$$

The third method uses the probabilities:

$$E(0) = \sum_{n=1}^{2} P(E_n)E_n = \frac{8}{25}(-5) + \frac{9}{25}(3) + \frac{8}{25}(5) = \frac{27}{25}. \tag{3.225}$$

The energy at a time t is

$$\begin{aligned} E(t) = \langle\psi(t)|\hat{H}|\psi(t)\rangle &= \frac{8}{25}e^{iE_1t}e^{-iE_1t}\langle\phi_1|\hat{H}|\phi_1\rangle + \frac{9}{25}e^{iE_2t}e^{-iE_2t}\langle\phi_2|\hat{H}|\phi_2\rangle \\ &+ \frac{8}{25}e^{iE_3t}e^{-iE_3t}\langle\phi_3|\hat{H}|\phi_3\rangle = \frac{8}{25}(-5) + \frac{9}{25}(3) + \frac{8}{25}(5) = \frac{27}{25} = E(0). \end{aligned} \tag{3.226}$$

As expected, $E(t) = E(0)$ since $d\langle\hat{H}\rangle/dt = 0$.

(d) Since none of the eigenvalues of $\hat{H}$ is degenerate, the eigenvectors $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ form a compete (orthonormal) basis. Thus $\{\hat{H}\}$ forms a complete set of commuting operators.

Problem 3.12

(a) Calculate the Poisson bracket between the x and y components of the classical orbital angular momentum.

(b) Calculate the commutator between the x and y components of the orbital angular momentum operator.

(c) Compare the results obtained in (a) and (b).

Solution

(a) Using the definition (3.113) we can write the Poisson bracket $\{l_x, l_y\}$ as

$$\{l_x, l_y\} = \sum_{j=1}^{3} \left(\frac{\partial l_x}{\partial q_j} \frac{\partial l_y}{\partial p_j} - \frac{\partial l_x}{\partial p_j} \frac{\partial l_y}{\partial q_j} \right), \tag{3.227}$$

where $q_1 = x$, $q_2 = y$, $q_3 = z$, $p_1 = p_x$, $p_2 = p_y$, and $p_3 = p_z$. Since $l_x = yp_z - zp_y$, $l_y = zp_x - xp_z$, $l_z = xp_y - yp_x$, the only partial derivatives that survive are $\partial l_x/\partial z = -p_y$, $\partial l_y/\partial p_z = -x$, $\partial l_x/\partial p_z = y$, and $\partial l_y/\partial z = p_x$. Thus, we have

$$\{l_x, l_y\} = \frac{\partial l_x}{\partial z} \frac{\partial l_y}{\partial p_z} - \frac{\partial l_x}{\partial p_z} \frac{\partial l_y}{\partial z} = xp_y - yp_x = l_z. \tag{3.228}$$

(b) The components of $\hat{\vec{L}}$ are listed in (3.26) to (3.28): $\hat{L}_x = \hat{Y}\hat{P}_z - \hat{Z}\hat{P}_y$, $\hat{L}_y = \hat{Z}\hat{P}_x - \hat{X}\hat{P}_z$, and $\hat{L}_Z = \hat{X}\hat{P}_y - \hat{Y}\hat{P}_x$. Since $\hat{X}$, $\hat{Y}$, and $\hat{Z}$ mutually commute and so do $\hat{P}_x$, $\hat{P}_y$, and $\hat{P}_z$, we have

$$\begin{aligned}
[\hat{L}_x, \hat{L}_y] &= [\hat{Y}\hat{P}_z - \hat{Z}\hat{P}_y,\ \hat{Z}\hat{P}_x - \hat{X}\hat{P}_z] \\
&= [\hat{Y}\hat{P}_z, \hat{Z}\hat{P}_x] - [\hat{Y}\hat{P}_z, \hat{X}\hat{P}_z] - [\hat{Z}\hat{P}_y, \hat{Z}\hat{P}_x] + [\hat{Z}\hat{P}_y, \hat{X}\hat{P}_z] \\
&= \hat{Y}[\hat{P}_z, \hat{Z}]\hat{P}_x + \hat{X}[\hat{Z}, \hat{P}_z]\hat{P}_y = i\hbar(\hat{X}\hat{P}_y - \hat{Y}\hat{P}_x) \\
&= i\hbar\hat{L}_z.
\end{aligned} \tag{3.229}$$

(c) A comparison of (3.228) and (3.229) shows that

$$\{l_x, l_y\} = l_z \longrightarrow [\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z. \tag{3.230}$$

Problem 3.13

Consider a charged oscillator, of positive charge q and mass m, which is subject to an oscillating electric field $E_0 \cos \omega t$; the particle's Hamiltonian is $\hat{H} = P^2/(2m) + k\hat{X}^2/2 + qE_0\hat{X}\cos\omega t$.

(a) Calculate $d\langle\hat{X}\rangle/dt$, $d\langle\hat{P}\rangle/dt$, $d\langle\hat{H}\rangle/dt$.

(b) Solve the equation for $d\langle\hat{X}\rangle/dt$ and obtain $\langle\hat{X}\rangle(t)$ such that $\langle\hat{X}\rangle(0) = x_0$.

Solution

(a) Since the position operator $\hat{X}$ does not depend explicitly on time (i.e., $\partial\hat{X}/\partial t = 0$), equation (3.88) yields

$$\frac{d}{dt}\langle\hat{X}\rangle = \frac{1}{i\hbar}\langle[\hat{X}, \hat{H}]\rangle = \frac{1}{i\hbar}\langle\left[\hat{X}, \frac{P^2}{2m}\right]\rangle = \frac{\langle\hat{P}\rangle}{m}. \tag{3.231}$$

Now, since $[\hat{P}, \hat{X}] = -i\hbar$, $[\hat{P}, \hat{X}^2] = -2i\hbar\hat{X}$ and $\partial\hat{P}/\partial t = 0$, we have

$$\frac{d}{dt}\langle\hat{P}\rangle = \frac{1}{i\hbar}\langle[\hat{P}, \hat{H}]\rangle = \frac{1}{i\hbar}\langle\left[\hat{P}, \frac{1}{2}k\hat{X}^2 + qE_0\hat{X}\cos\omega t\right]\rangle = -k\langle\hat{X}\rangle - qE_0\cos\omega t, \tag{3.232}$$

$$\frac{d}{dt}\langle\hat{H}\rangle = \frac{1}{i\hbar}\langle[\hat{H}, \hat{H}]\rangle + \langle\frac{\partial\hat{H}}{\partial t}\rangle = \langle\frac{\partial\hat{H}}{\partial t}\rangle = -qE_0\omega\langle\hat{X}\rangle\sin\omega t. \tag{3.233}$$

(b) To find $\langle\hat{X}\rangle$ we need to take a time derivative of (3.231) and then make use of (3.232):

$$\frac{d^2}{dt^2}\langle\hat{X}\rangle = \frac{1}{m}\frac{d}{dt}\langle\hat{P}\rangle = -\frac{k}{m}\langle\hat{X}\rangle - \frac{qE_0}{m}\cos\omega t. \tag{3.234}$$

The solution of this equation is

$$\langle\hat{X}\rangle(t) = \langle\hat{X}\rangle(0)\cos\left(\sqrt{\frac{k}{m}}t\right) - \frac{qE_0}{m\omega}\sin\omega t + A, \tag{3.235}$$

where A is a constant which can be determined from the initial conditions; since $\langle\hat{X}\rangle(0) = x_0$ we have $A = 0$, and hence

$$\langle\hat{X}\rangle(t) = x_0\cos\left(\sqrt{\frac{k}{m}}t\right) - \frac{qE_0}{m\omega}\sin\omega t. \tag{3.236}$$

Problem 3.14

Consider a one-dimensional free particle of mass m whose position and momentum at time $t = 0$ are given by x_0 and p_0, respectively.

(a) Calculate $\langle\hat{P}\rangle(t)$ and show that $\langle\hat{X}\rangle(t) = p_0t^2/m + x_0$.

(b) Show that $d\langle\hat{X}^2\rangle/dt = 2\langle\hat{P}\hat{X}\rangle/m + i\hbar/m$ and $d\langle\hat{P}^2\rangle/dt = 0$.

(c) Show that the position and momentum fluctuations are related by $d^2(\Delta x)^2/dt^2 = 2(\Delta p)^2/m^2$ and that the solution to this equation is given by $(\Delta x)^2 = (\Delta p)_0^2t^2/m^2 + (\Delta x)_0^2$ where $(\Delta x)_0$ and $(\Delta p)_0$ are the initial fluctuations.

Solution

(a) From the Ehrenfest equations $d\langle\hat{P}\rangle/dt = \langle[\hat{P}, \hat{V}(x,t)]\rangle/i\hbar$ as shown in (3.134), and since for a free particle $\hat{V}(x,t) = 0$, we see that $d\langle\hat{P}\rangle/dt = 0$. As expected this leads to $\langle\hat{P}\rangle(t) = p_0$, since the linear momentum of a free particle is conserved. Inserting $\langle\hat{P}\rangle = p_0$ into Ehrenfest's other equation $d\langle\hat{X}\rangle/dt = \langle\hat{P}\rangle/m$ (see (3.132)), we obtain

$$\frac{d\langle\hat{X}\rangle}{dt} = \frac{1}{m}p_0. \tag{3.237}$$

The solution of this equation with the initial condition $\langle\hat{X}\rangle(0) = x_0$ is

$$\langle\hat{X}\rangle(t) = \frac{p_0}{m}t + x_0. \tag{3.238}$$

(b) First, the proof of $d\langle\hat{P}^2\rangle/dt = 0$ is straightforward. Since $[\hat{P}^2, \hat{H}] = [\hat{P}^2, \hat{P}^2/2m] = 0$ and $\partial\hat{P}^2/\partial t = 0$ (the momentum operator does not depend on time), (3.124) yields

$$\frac{d}{dt}\langle\hat{P}^2\rangle = \frac{1}{i\hbar}\langle[\hat{P}^2, \hat{H}]\rangle + \langle\frac{\partial\hat{P}^2}{\partial t}\rangle = 0. \tag{3.239}$$

For $d\langle\hat{X}^2\rangle/dt$ we have

$$\frac{d}{dt}\langle\hat{X}^2\rangle = \frac{1}{i\hbar}\langle[\hat{X}^2, \hat{H}]\rangle = \frac{1}{2im\hbar}\langle[\hat{X}^2, \hat{P}^2]\rangle, \tag{3.240}$$

since $\partial\hat{X}^2/\partial t = 0$. Using $[\hat{X}, \hat{P}] = i\hbar$, we obtain

$$\begin{aligned}[\hat{X}^2, \hat{P}^2] &= \hat{P}[\hat{X}^2, \hat{P}] + [\hat{X}^2, \hat{P}]\hat{P} \\ &= \hat{P}\hat{X}[\hat{X}, \hat{P}] + \hat{P}[\hat{X}, \hat{P}]\hat{X} + \hat{X}[\hat{X}, \hat{P}]\hat{P} + [\hat{X}, \hat{P}]\hat{X}\hat{P} \\ &= 2i\hbar(\hat{P}\hat{X} + \hat{X}\hat{P}) = 2i\hbar(2\hat{P}\hat{X} + i\hbar);\end{aligned} \tag{3.241}$$

hence

$$\frac{d}{dt}\langle\hat{X}^2\rangle = \frac{2}{m}\langle\hat{P}\hat{X}\rangle + \frac{i\hbar}{m}. \tag{3.242}$$

(c) As the position fluctuation is given by $(\Delta x)^2 = \langle\hat{X}^2\rangle - \langle\hat{X}\rangle^2$, we have

$$\frac{d(\Delta x)^2}{dt} = \frac{d\langle\hat{X}^2\rangle}{dt} - 2\langle\hat{X}\rangle\frac{d\langle\hat{X}\rangle}{dt} = \frac{2}{m}\langle\hat{P}\hat{X}\rangle + \frac{i\hbar}{m} - \frac{2}{m}\langle\hat{X}\rangle\langle\hat{P}\rangle. \tag{3.243}$$

In deriving this expression we have used (3.242) and $d\langle\hat{X}\rangle/dt = \langle\hat{P}\rangle/m$. Now, since $d(\langle\hat{X}\rangle\langle\hat{P}\rangle)/dt = \langle\hat{P}\rangle d\langle\hat{X}\rangle/dt = \langle\hat{P}\rangle^2/m$ and

$$\frac{d\langle\hat{P}\hat{X}\rangle}{dt} = \frac{1}{i\hbar}\langle[\hat{P}\hat{X},\ \hat{H}]\rangle = \frac{1}{2im\hbar}\langle[\hat{P}\hat{X},\ \hat{P}^2]\rangle = \frac{1}{m}\langle\hat{P}^2\rangle, \tag{3.244}$$

we can write the second time derivative of (3.243) as follows:

$$\frac{d^2(\Delta x)^2}{dt^2} = \frac{2}{m}\left(\frac{d\langle\hat{P}\hat{X}\rangle}{dt} - \frac{d\langle\hat{X}\rangle\langle\hat{P}\rangle}{dt}\right) = \frac{2}{m^2}\left(\langle\hat{P}^2\rangle - \langle\hat{P}\rangle^2\right) = \frac{2}{m^2}(\Delta p)_0^2, \tag{3.245}$$

where $(\Delta p)_0^2 = \langle\hat{P}^2\rangle - \langle\hat{P}\rangle^2 = \langle\hat{P}^2\rangle_0 - \langle\hat{P}\rangle_0^2$; the momentum of the free particle is a constant of the motion. We can verify that the solution of the differential equation (3.245) is given by

$$(\Delta x)^2 = \frac{1}{m^2}(\Delta p)_0^2 t^2 + (\Delta x)_0^2. \tag{3.246}$$

This fluctuation is similar to the spreading of a Gaussian wave packet we derived in Chapter 1.

3.10 Exercises

Exercise 3.1
A particle in an infinite potential box with walls at $x = 0$ and $x = a$ (i.e., the potential is infinite for $x < 0$ and $x > a$ and zero in between) has the following wave function at some initial time:

$$\psi(x) = \frac{1}{\sqrt{5a}} \sin\left(\frac{\pi x}{a}\right) + \frac{2}{\sqrt{5a}} \sin\left(\frac{3\pi x}{a}\right).$$

(a) Find the possible results of the measurement of the system's energy and the corresponding probabilities.

(b) Find the form of the wave function after such a measurement.

(c) If the energy is measured again immediately afterwards, what are the relative probabilities of the possible outcomes?

Exercise 3.2
Let $\psi_n(x)$ denote the orthonormal stationary states of a system corresponding to the energy E_n. Suppose that the normalized wave function of the system at time $t = 0$ is $\psi(x, 0)$ and suppose that a measurement of the energy yields the value E_1 with probability 1/2, E_2 with probability 3/8, and E_3 with probability 1/8.

(a) Write the most general expansion for $\psi(x, 0)$ consistent with this information.

(b) What is the expansion for the wave function of the system at time t, $\psi(x, t)$?

(c) Show that the expectation value of the Hamiltonian does not change with time.

Exercise 3.3
Consider a neutron which is confined to an infinite potential well of width $a = 8\,\text{fm}$. At time $t = 0$ the neutron is assumed to be in the state

$$\Psi(x, 0) = \sqrt{\frac{4}{7a}} \sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{2}{7a}} \sin\left(\frac{2\pi x}{a}\right) + \sqrt{\frac{8}{7a}} \sin\left(\frac{3\pi x}{a}\right).$$

(a) If an energy measurement is carried out on the system, what are the values that will be found for the energy and with what probabilities? Express your answer in MeV (the mass of the neutron is $mc^2 \simeq 939\,\text{MeV}$, $\hbar c \simeq 197\,\text{MeV fm}$).

(b) If this measurement is repeated on many identical systems, what is the average value of the energy that will be found? Again, express your answer in MeV.

(c) Using the uncertainty principle, estimate the order of magnitude of the neutron's speed in this well as a function of the speed of light c.

Exercise 3.4
Consider the dimensionless harmonic oscillator Hamiltonian

$$\hat{H} = \frac{1}{2}\hat{P}^2 + \frac{1}{2}\hat{X}^2, \qquad \text{with} \quad \hat{P} = -i\frac{d}{dx}.$$

(a) Show that the two wave functions $\psi_0(x) = e^{-x^2/2}$ and $\psi_1(x) = xe^{-x^2/2}$ are eigenfunctions of $\hat{H}$ with eigenvalues $1/2$ and $3/2$, respectively.

(b) Find the value of the coefficient α such that $\psi_2(x) = \left(1 + \alpha x^2\right) e^{-x^2/2}$ is orthogonal to $\psi_0(x)$. Then show that $\psi_2(x)$ is an eigenfunction of $\hat{H}$ with eigenvalue $5/2$.

Exercise 3.5
Consider that the wave function of a dimensionless harmonic oscillator, whose Hamiltonian is $\hat{H} = \frac{1}{2}\hat{P}^2 + \frac{1}{2}\hat{X}^2$, is given at time $t = 0$ by

$$\psi(x,0) = \frac{1}{\sqrt{8\pi}}\phi_0(x) + \frac{1}{\sqrt{18\pi}}\phi_2(x) = \frac{1}{\sqrt{8\pi}}e^{-x^2/2} + \frac{1}{\sqrt{18\pi}}\left(1-2x^2\right)e^{-x^2/2}.$$

(a) Find the expression of the oscillator's wave function at any later time t.

(b) Calculate the probability P_0 to find the system in an eigenstate of energy $1/2$ and the probability P_2 of finding the system in an eigenstate of energy $5/2$.

(c) Calculate the probability density, $\rho(x,t)$, and the current density, $\vec{J}(x,t)$.

(d) Verify that the probability is conserved; that is, show that $\partial\rho/\partial t + \vec{\nabla}\cdot\vec{J}(x,t) = 0$.

Exercise 3.6
A particle of mass m, in an infinite potential well of length a, has the following initial wave function at $t = 0$:

$$\psi(x,0) = \sqrt{\frac{3}{5a}}\sin\left(\frac{3\pi x}{a}\right) + \frac{1}{\sqrt{5a}}\sin\left(\frac{5\pi x}{a}\right), \tag{3.247}$$

and an energy spectrum $E_n = -\hbar^2\pi^2 n^2/(2ma^2)$.

Find $\psi(x,t)$ at any later time t, then calculate $\frac{\partial\rho}{\partial t}$ and the probability current density vector $\vec{J}(x,t)$ and verify that $\frac{\partial\rho}{\partial t} + \vec{\nabla}\cdot\vec{J}(x,t) = 0$. Recall that $\rho = \psi^*(x,t)\psi(x,t)$ and $\vec{J}(x,t) = \frac{i\hbar}{2m}\left(\psi(x,t)\vec{\nabla}\psi^*(x,t) - \psi^*(x,t)\vec{\nabla}\psi(x,t)\right)$.

Exercise 3.7
Consider a system whose initial state at $t = 0$ is given in terms of a complete and orthonormal set of three vectors: $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ as follows: $|\psi(0)\rangle = 1/\sqrt{3}|\phi_1\rangle + A|\phi_2\rangle + 1/\sqrt{6}|\phi_3\rangle$, where A is a real constant.

(a) Find A so that $|\psi(0)\rangle$ is normalized.

(b) If the energies corresponding to $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ are given by E_1, E_2, and E_3, respectively, write down the state of the system $|\psi(t)\rangle$ at any later time t.

(c) Determine the probability of finding the system at a time t in the state $|\phi_3\rangle$.

Exercise 3.8
The components of the initial state $|\psi_i\rangle$ of a quantum system are given in a complete and orthonormal basis of three states $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ by

$$\langle\phi_1|\psi_i\rangle = \frac{i}{\sqrt{3}}, \quad \langle\phi_2|\psi_i\rangle = \sqrt{\frac{2}{3}}, \quad \langle\phi_3|\psi_i\rangle = 0.$$

Calculate the probability of finding the system in a state $|\psi_f\rangle$ whose components are given in the same basis by

$$\langle\phi_1|\psi_f\rangle = \frac{1+i}{\sqrt{3}}, \quad \langle\phi_2|\psi_f\rangle = \frac{1}{\sqrt{6}}, \quad \langle\phi_3|\psi_f\rangle = \frac{1}{\sqrt{6}}.$$

Exercise 3.9

(a) Evaluate the Poisson bracket $\{x^2, p^2\}$.

(b) Express the commutator $\left[\hat{X}^2, \hat{P}^2\right]$ in terms of $\hat{X}\hat{P}$ plus a constant in $\hbar^2$.

(c) Find the classical limit of $\left[\hat{x}^2, \hat{p}^2\right]$ for this expression and then compare it with the result of part (a).

Exercise 3.10

A particle bound in a one-dimensional potential has a wave function

$$\psi(x) = \begin{cases} Ae^{5ikx}\cos(3\pi x/a), & -a/2 \le x \le a/2, \\ 0, & |x| > a/2. \end{cases}$$

(a) Calculate the constant A so that $\psi(x)$ is normalized.

(b) Calculate the probability of finding the particle between $x = 0$ and $x = a/4$.

Exercise 3.11

(a) Show that any component of the momentum operator of a particle is compatible with its kinetic energy operator.

(b) Show that the momentum operator is compatible with the Hamiltonian operator only if the potential operator is constant in space coordinates.

Exercise 3.12

Consider a physical system whose Hamiltonian H and an operator A are given by

$$H = \mathcal{E}_0 \begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad A = a_0 \begin{pmatrix} 5 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix},$$

where $\mathcal{E}_0$ has the dimensions of energy.

(a) Do H and A commute? If yes, give a basis of eigenvectors common to H and A.

(b) Which among the sets of operators $\{\hat{H}\}$, $\{\hat{A}\}$, $\{\hat{H}, \hat{A}\}$, $\{\hat{H}^2, \hat{A}\}$ form a complete set of commuting operators (CSCO)?

Exercise 3.13

Show that the momentum and the total energy can be measured simultaneously only when the potential is constant everywhere.

Exercise 3.14

The initial state of a system is given in terms of four orthonormal energy eigenfunctions $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$, and $|\phi_4\rangle$ as follows:

$$|\psi_0\rangle = |\psi(t=0)\rangle = \frac{1}{\sqrt{3}}|\phi_1\rangle + \frac{1}{2}|\phi_2\rangle + \frac{1}{\sqrt{6}}|\phi_3\rangle + \frac{1}{2}|\phi_4\rangle.$$

(a) If the four kets $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$, and $|\phi_4\rangle$ are eigenvectors to the Hamiltonian $\hat{H}$ with energies E_1, E_2, E_3, and E_4, respectively, find the state $|\psi(t)\rangle$ at any later time t.

(b) What are the possible results of measuring the energy of this system and with what probability will they occur?

(c) Find the expectation value of the system's Hamiltonian at $t = 0$ and $t = 10\,\text{s}$.

Exercise 3.15
The complete set expansion of an initial wave function $\psi(x,0)$ of a system in terms of orthonormal energy eigenfunctions $\phi_n(x)$ of the system has three terms, $n = 1, 2, 3$. The measurement of energy on the system represented by $\psi(x,0)$ gives three values, E_1 and E_2 with probability $1/4$ and E_3 with probability $1/2$.

(a) Write down $\psi(x,0)$ in terms of $\phi_1(x)$, $\phi_2(x)$, and $\phi_3(x)$.

(b) Find $\psi(x,0)$ at any later time t, i.e., find $\psi(x,t)$.

Exercise 3.16
Consider a system whose Hamiltonian H and an operator A are given by the matrices

$$H = \mathcal{E}_0 \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 2i \\ 0 & -2i & 0 \end{pmatrix}, \quad A = a_0 \begin{pmatrix} 0 & -i & 0 \\ i & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

(a) If we measure energy, what values will we obtain?

(b) Suppose that when we measure energy, we obtain a value of $\sqrt{5}\mathcal{E}_0$. Immediately afterwards, we measure A. What values will we obtain for A and what are the probabilities corresponding to each value?

(c) Calculate the expectation value $\langle A \rangle$.

Exercise 3.17
Consider a physical system whose Hamiltonian and initial state are given by

$$H = \mathcal{E}_0 \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad |\psi_0\rangle = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix},$$

where $\mathcal{E}_0$ has the dimensions of energy.

(a) What values will we obtain when measuring the energy and with what probabilities?

(b) Calculate the expectation value of the Hamiltonian $\langle \hat{H} \rangle$.

Exercise 3.18
Consider a system whose state $|\psi(t)\rangle$ and two observables A and B are given by

$$|\psi(t)\rangle = \begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix}, \qquad A = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

(a) We perform a measurement where A is measured first and then B immediately afterwards. Find the probability of obtaining a value of $\sqrt{2}$ for A and a value of -1 for B.

(b) Now we measure B first and then A immediately afterwards. Find the probability of obtaining a value of -1 for B and a value of $\sqrt{2}$ for A.

(c) Compare the results of (a) and (b). Explain.

Exercise 3.19
Consider a system whose state $|\psi(t)\rangle$ and two observables A and B are given by

$$|\psi(t)\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} -i \\ 2 \\ 0 \end{pmatrix}, \qquad A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i & 1 \\ -i & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & i \\ 0 & -i & 0 \end{pmatrix}.$$

(a) Are A and B compatible? Which among the sets of operators $\{\hat{A}\}$, $\{\hat{B}\}$, and $\{\hat{A}, \hat{B}\}$ form a complete set of commuting operators?

(b) Measuring A first and then B immediately afterwards, find the probability of obtaining a value of -1 for A and a value of 3 for B.

(c) Now, measuring B first then A immediately afterwards, find the probability of obtaining 3 for B and -1 for A. Compare this result with the probability obtained in (b).

Exercise 3.20

Consider a physical system which has a number of observables that are represented by the following matrices:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & i \\ -1 & -i & 4 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 3 & 1 \end{pmatrix}.$$

(a) Find the results of the measurements of the compatible observables.

(b) Which among these observables are compatible? Give a basis of eigenvectors common to these observables.

(c) Which among the sets of operators $\{\hat{A}\}$, $\{\hat{B}\}$, $\{\hat{C}\}$, $\{\hat{A}, \hat{B}\}$, $\{\hat{A}, \hat{C}\}$, $\{\hat{B}, \hat{C}\}$ form a complete set of commuting operators?

Exercise 3.21

Consider a system which is initially in a state $|\psi(0)\rangle$ and having a Hamiltonian $\hat{H}$, where

$$|\psi(0)\rangle = \begin{pmatrix} 4-i \\ -2+5i \\ 3+2i \end{pmatrix}, \qquad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 3 & 3 \\ 0 & 3 & 0 \end{pmatrix}.$$

(a) If a measurement of H is carried out, what values will we obtain and with what probabilities?

(b) Find the state of the system at a later time t; you may need to expand $|\psi(0)\rangle$ in terms of the eigenvectors of H.

(c) Find the total energy of the system at time $t = 0$ and any later time t; are these values different?

(d) Does $\hat{H}$ form a complete set of commuting operators?

Exercise 3.22

Consider a particle which moves in a scalar potential $V(\vec{r}) = V_x(x) + V_y(y) + V_z(z)$.

(a) Show that the Hamiltonian of this particle can be written as $\hat{H} = \hat{H}_x + \hat{H}_y + \hat{H}_z$, where $\hat{H}_x = p_x^2/(2m) + V_x(x)$, and so on.

(b) Do $\hat{H}_x$, $\hat{H}_y$, and $\hat{H}_z$ form a complete set of commuting operators?

Exercise 3.23

Consider a system whose Hamiltonian is $H = \mathcal{E} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, where $\mathcal{E}$ is a real constant with the dimensions of energy.

(a) Find the eigenenergies, E_1 and E_2, of H.

(b) If the system is initially (i.e., $t = 0$) in the state $|\psi_0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, find the probability so that a measurement of energy at $t = 0$ yields: (i) E_1, and (ii) E_2.

(c) Find the average value of the energy $\langle \hat{H} \rangle$ and the energy uncertainty $\sqrt{\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2}$.
(d) Find the state $|\psi(t)\rangle$.

Exercise 3.24
Prove the relation

$$\frac{d}{dt}\langle \hat{A}\hat{B} \rangle = \langle \frac{\partial \hat{A}}{\partial t}\hat{B} \rangle + \langle \hat{A}\frac{\partial \hat{B}}{\partial t} \rangle + \frac{1}{i\hbar}\langle [\hat{A}, \hat{H}]\hat{B} \rangle + \frac{1}{i\hbar}\langle \hat{A}[\hat{B}, \hat{H}] \rangle.$$

Exercise 3.25
Consider a particle of mass m which moves under the influence of gravity; the particle's Hamiltonian is $\hat{H} = \hat{P}_z^2/(2m) - mg\hat{Z}$, where g is the acceleration due to gravity, $g = 9.8\,\mathrm{m\,s^{-2}}$.

(a) Calculate $d\langle \hat{Z} \rangle/dt$, $d\langle \hat{P}_z \rangle/dt$, $d\langle \hat{H} \rangle/dt$.

(b) Solve the equation $d\langle \hat{Z} \rangle/dt$ and obtain $\langle \hat{Z} \rangle(t)$, such that $\langle \hat{Z} \rangle(0) = h$ and $\langle \hat{P}_z \rangle(0) = 0$. Compare the result with the classical relation $z(t) = -\frac{1}{2}gt^2 + h$.

Exercise 3.26
Calculate $d\langle \hat{X} \rangle/dt$, $d\langle \hat{P}_x \rangle/dt$, $d\langle \hat{H} \rangle/dt$ for a particle with $\hat{H} = \hat{P}_x^2/(2m) + \frac{1}{2}m\omega^2\hat{X}^2 + V_0\hat{X}^3$.

Exercise 3.27
Consider a system whose initial state at $t = 0$ is given in terms of a complete and orthonormal set of four vectors $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$, $|\phi_4\rangle$ as follows:

$$|\psi(0)\rangle = \frac{A}{\sqrt{12}}|\phi_1\rangle + \frac{1}{\sqrt{6}}|\phi_2\rangle + \frac{2}{\sqrt{12}}|\phi_3\rangle + \frac{1}{2}|\phi_4\rangle,$$

where A is a real constant.

(a) Find A so that $|\psi(0)\rangle$ is normalized.

(b) If the energies corresponding to $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$, $|\phi_4\rangle$ are given by E_1, E_2, E_3, and E_4, respectively, write down the state of the system $|\psi(t)\rangle$ at any later time t.

(c) Determine the probability of finding the system at a time t in the state $|\phi_2\rangle$.

Chapter 4

One-Dimensional Problems

4.1 Introduction

After presenting the formalism of quantum mechanics in the previous two chapters, we are now well equipped to apply it to the study of physical problems. Here we apply the Schrödinger equation to *one-dimensional* problems. These problems are interesting since there exist many physical phenomena whose motion is one-dimensional. The application of the Schrödinger equation to *one-dimensional* problems enables us to compare the predictions of classical and quantum mechanics in a simple setting. In addition to being simple to solve, one-dimensional problems will be used to illustrate some nonclassical effects.

The Schrödinger equation describing the dynamics of a microscopic particle of mass m in a one-dimensional time-independent potential $V(x)$ is given by

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x), \tag{4.1}$$

where E is the total energy of the particle. The solutions of this equation yield the allowed energy eigenvalues E_n and the corresponding wave functions $\psi_n(x)$. To solve this partial differential equation, we need to specify the potential $V(x)$ as well as the boundary conditions; the boundary conditions can be obtained from the physical requirements of the system.

We have seen in the previous chapter that the solutions of the Schrödinger equation for time-independent potentials are stationary,

$$\Psi(x,t) = \psi(x)e^{-iEt/\hbar}, \tag{4.2}$$

for the probability density does not depend on time. Recall that the state $\psi(x)$ has the physical dimensions of $1/\sqrt{L}$, where L is a length. Hence, the physical dimension of $|\psi(x)|^2$ is $1/L$: $\left[|\psi(x)|^2\right] = 1/L$.

We begin by examining some general properties of one-dimensional motion and discussing the symmetry character of the solutions. Then, in the rest of the chapter, we apply the Schrödinger equation to various one-dimensional potentials: the free particle, the potential step, the finite and infinite potential wells, and the harmonic oscillator. We conclude by showing how to solve the Schrödinger equation numerically.

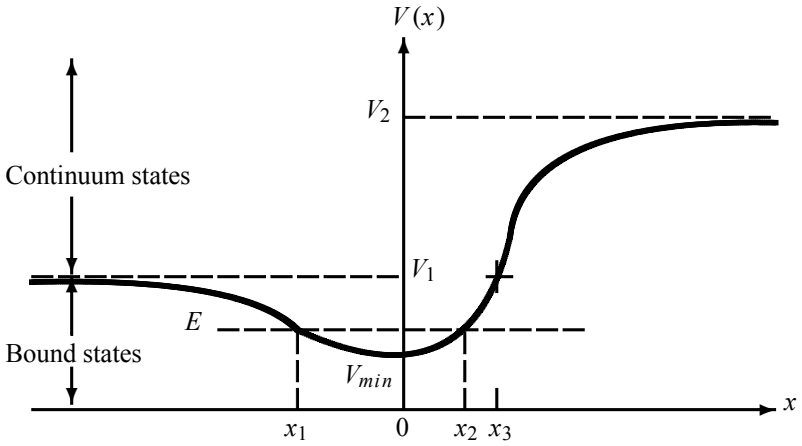


Figure 4.1 Shape of a general potential.

4.2 Properties of One-Dimensional Motion

To study the dynamic properties of a single particle moving in a one-dimensional potential, let us consider a potential $V(x)$ that is general enough to allow for the illustration of all the desired features. One such potential is displayed in Figure 4.1; it is finite at $x \to \pm\infty$, $V(-\infty) = V_1$ and $V(+\infty) = V_2$ with V_1 smaller than V_2, and it has a minimum, V_{min}. In particular, we want to study the conditions under which discrete and continuous spectra occur. As the character of the states is completely determined by the size of the system's energy, we will be considering separately the cases where the energy is smaller and larger than the potential.

4.2.1 Discrete Spectrum (Bound States)

Bound states occur whenever the particle cannot move to infinity. That is, the particle is *confined* or *bound* at all energies to move within a *finite* and limited region of space which is delimited by two classical turning points. The Schrödinger equation in this region admits only solutions that are *discrete*. The infinite square well potential and the harmonic oscillator are typical examples that display bound states.

In the potential of Figure 4.1, the motion of the particle is bounded between the classical turning points x_1 and x_2 when the particle's energy lies between V_{min} and V_1:

$$V_{min} < E < V_1. \tag{4.3}$$

The states corresponding to this energy range are called *bound* states. They are defined as states whose wave functions are finite (or zero) at $x \to \pm\infty$; usually the bound states have energies smaller than the potential $E < V$. For the bound states to exist, the potential $V(x)$ must have at least one minimum which is lower than V_1 (i.e., $V_{min} < V_1$). The energy spectra of bound states are discrete. We need to use the boundary conditions[1] to find the wave function and the energy.

Let us now list two theorems that are important to the study of bound states.

[1] Since the Schrödinger equation is a *second-order* differential equation, only *two* boundary conditions are required to solve it.

Theorem 4.1 *In a one-dimensional problem the energy levels of a bound state system are discrete and not degenerate.*

Theorem 4.2 *The wave function $\psi_n(x)$ of a one-dimensional bound state system has n nodes (i.e., $\psi_n(x)$ vanishes n times) if $n = 0$ corresponds to the ground state and $(n-1)$ nodes if $n = 1$ corresponds to the ground state.*

4.2.2 Continuous Spectrum (Unbound States)

Unbound states occur in those cases where the motion of the system is not confined; a typical example is the *free* particle. For the potential displayed in Figure 4.1 there are two energy ranges where the particle's motion is infinite: $V_1 < E < V_2$ and $E > V_2$.

- Case $V_1 < E < V_2$

  In this case the particle's motion is infinite only towards $x = -\infty$; that is, the particle can move between $x = x_3$ and $x \to -\infty$, x_3 being a classical turning point. The energy spectrum is continuous and none of the energy eigenvalues is degenerate. The nondegeneracy can be shown to result as follows. Since the Schrödinger equation (4.1) is a second-order differential equation, it has, for this case, two linearly independent solutions, but only one is physically acceptable. The solution is oscillatory for $x \leq x_3$ and rapidly decaying for $x > x_3$ so that it is finite (zero) at $x \to +\infty$, since divergent solutions are unphysical.

- Case $E > V_2$

  The energy spectrum is continuous and the particle's motion is infinite in both directions of x (i.e., towards $x \to \pm\infty$). All the energy levels of this spectrum are doubly degenerate. To see this, note that the general solution to (4.1) is a linear combination of two independent oscillatory solutions, one moving to the left and the other to the right. In the previous nondegenerate case only one solution is retained, since the other one diverges as $x \to +\infty$ and it has to be rejected.

In contrast to bound states, unbound states cannot be normalized and we cannot use boundary conditions.

4.2.3 Mixed Spectrum

Potentials that confine the particle for only some energies give rise to mixed spectra; the motion of the particle for such potentials is confined for some energy values only. For instance, for the potential displayed in Figure 4.1, if the energy of the particle is between $V_{min} < E < V_1$, the motion of the particle is confined (bound) and its spectrum is discrete, but if $E > V_2$, the particle's motion is unbound and its spectrum is continuous (if $V_1 < E < V_2$, the motion is unbound only along the $x = -\infty$ direction). Other typical examples where mixed spectra are encountered are the finite square well potential and the Coulomb or molecular potential.

4.2.4 Symmetric Potentials and Parity

Most of the potentials that are encountered at the microscopic level are symmetric (or even) with respect to space inversion, $\hat{V}(-x) = \hat{V}(x)$. This symmetry introduces considerable simplifications in the calculations.

When $\hat{V}(x)$ is even, the corresponding Hamiltonian, $\hat{H}(x) = -(\hbar^2/2m)d^2/dx^2 + \hat{V}(x)$, is also even. We saw in Chapter 2 that even operators commute with the parity operator; hence they can have a common eigenbasis.

Let us consider the following two cases pertaining to degenerate and nondegenerate spectra of this Hamiltonian:

- **Nondegenerate spectrum**

  First we consider the particular case where the eigenvalues of the Hamiltonian corresponding to this symmetric potential are not degenerate. According to Theorem 4.1, this Hamiltonian describes bound states. We saw in Chapter 2 that a nondegenerate, even operator has the same eigenstates as the parity operator. Since the eigenstates of the parity operator have definite parity, *the bound eigenstates of a particle moving in a one-dimensional symmetric potential have definite parity; they are either even or odd*:

$$\boxed{\hat{V}(-x) = \hat{V}(x) \quad \Longrightarrow \quad \psi(-x) = \pm\psi(x).} \tag{4.4}$$

- **Degenerate spectrum**

  If the spectrum of the Hamiltonian corresponding to a symmetric potential is degenerate, the eigenstates are expressed only in terms of even and odd states. That is, the eigenstates do not have definite parity.

Summary: The various properties of the one-dimensional motion discussed in this section can be summarized as follows:

- The energy spectrum of a bound state system is discrete and nondegenerate.
- The bound state wave function $\psi_n(x)$ has: (a) n nodes if $n = 0$ corresponds to the ground state and (b) $(n-1)$ nodes if $n = 1$ corresponds to the ground state.
- The bound state eigenfunctions in an even potential have definite parity.
- The eigenfunctions of a degenerate spectrum in an even potential do not have definite parity.

4.3 The Free Particle: Continuous States

This is the simplest one-dimensional problem; it corresponds to $V(x) = 0$ for any value of x. In this case the Schrödinger equation is given by

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x) \qquad \Longrightarrow \qquad \left(\frac{d^2}{dx^2} + k^2\right)\psi(x) = 0, \tag{4.5}$$

where $k^2 = 2mE/\hbar^2$, k being the wave number. The most general solution to (4.5) is a combination of two linearly independent *plane waves* $\psi_+(x) = e^{ikx}$ and $\psi_-(x) = e^{-ikx}$:

$$\psi_k(x) = A_+e^{ikx} + A_-e^{-ikx}, \tag{4.6}$$

where A_+ and A_- are two arbitrary constants. The complete wave function is thus given by the stationary state

$$\Psi_k(x,t) = A_+ e^{i(kx-\omega t)} + A_- e^{-i(kx+\omega t)} = A_+ e^{i(kx-\hbar k^2 t/2m)} + A_- e^{-i(kx+\hbar k^2 t/2m)}, \quad (4.7)$$

since $\omega = E/\hbar = \hbar k^2/2m$. The first term, $\Psi_+(x,t) = A_+ e^{i(kx-\omega t)}$, represents a wave traveling to the right, while the second term, $\Psi_-(x,t) = A_- e^{-i(kx+\omega t)}$, represents a wave traveling to the left. The intensities of these waves are given by $|A_+|^2$ and $|A_-|^2$, respectively. We should note that the waves $\Psi_+(x,t)$ and $\Psi_-(x,t)$ are associated, respectively, with a free particle traveling to the right and to the left with *well-defined* momenta and energy: $p_\pm = \pm\hbar k$, $E_\pm = \hbar^2 k^2/2m$. We will comment on the physical implications of this in a moment. Since there are no boundary conditions, there are no restrictions on k or on E; all values yield solutions to the equation.

The free particle problem is simple to solve mathematically, yet it presents a number of physical subtleties. Let us discuss briefly three of these subtleties. First, the probability densities corresponding to either solutions

$$P_\pm(x,t) = |\Psi_\pm(x,t)|^2 = |A_\pm|^2 \quad (4.8)$$

are constant, for they depend neither on x nor on t. This is due to the complete loss of information about the position and time for a state with definite values of momentum, $p_\pm = \pm\hbar k$, and energy, $E_\pm = \hbar^2 k^2/2m$. This is a consequence of Heisenberg's uncertainty principle: when the momentum and energy of a particle are known exactly, $\Delta p = 0$ and $\Delta E = 0$, there must be total uncertainty about its position and time: $\Delta x \longrightarrow \infty$ and $\Delta t \longrightarrow \infty$. The second subtlety pertains to an apparent discrepancy between the speed of the wave and the speed of the particle it is supposed to represent. The speed of the plane waves $\Psi_\pm(x,t)$ is given by

$$v_{wave} = \frac{\omega}{k} = \frac{E}{\hbar k} = \frac{\hbar^2 k^2/2m}{\hbar k} = \frac{\hbar k}{2m}. \quad (4.9)$$

On the other hand, the classical speed of the particle[2] is given by

$$v_{classical} = \frac{p}{m} = \frac{\hbar k}{m} = 2v_{wave}. \quad (4.10)$$

This means that the particle travels twice as fast as the wave that represents it! Third, the wave function is not normalizable:

$$\int_{-\infty}^{+\infty} \Psi_\pm^*(x,t)\Psi_\pm(x,t)\,dx = |A_\pm|^2 \int_{-\infty}^{+\infty} dx \to \infty. \quad (4.11)$$

The solutions $\Psi_\pm(x,t)$ are thus unphysical; physical wave functions must be square integrable. The problem can be traced to this: a free particle cannot have sharply defined momenta and energy.

In view of the three subtleties outlined above, the solutions of the Schrödinger equation (4.5) that are physically acceptable cannot be plane waves. Instead, we can construct physical

[2]The classical speed can be associated with the flux (or current density) which, as shown in Chapter 3, is $J_+ = i\hbar\frac{1}{2m}(\Psi_+\frac{\partial\Psi_+^*}{\partial x} - \Psi_+^*\frac{\partial\Psi_+}{\partial x}) = \frac{\hbar k}{m} = \frac{p}{m}$, where use was made of $A_+ = 1$.

solutions by means of a linear superposition of plane waves. The answer is provided by *wave packets*, which we have seen in Chapter 1:

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{i(kx-\omega t)} dk, \tag{4.12}$$

where $\phi(k)$, the amplitude of the wave packet, is given by the Fourier transform of $\psi(x,0)$ as

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \psi(x,0) e^{-ikx} dx. \tag{4.13}$$

The wave packet solution cures and avoids all the subtleties raised above. First, the momentum, the position and the energy of the particle are no longer known exactly; only probabilistic outcomes are possible. Second, as shown in Chapter 1, the wave packet (4.12) and the particle travel with the same speed $v_g = p/m$, called the *group* speed or the speed of the whole packet. Third, the wave packet (4.12) is normalizable.

To summarize, a free particle cannot be represented by a single (monochromatic) plane wave; it has to be represented by a wave packet. The physical solutions of the Schrödinger equation are thus given by wave packets, not by stationary solutions.

4.4 The Potential Step

Another simple problem consists of a particle that is free everywhere, but beyond a particular point, say $x = 0$, the potential increases sharply (i.e., it becomes repulsive or attractive). A potential of this type is called a potential step (see Figure 4.2):

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & x \geq 0. \end{cases} \tag{4.14}$$

In this problem we try to analyze the dynamics of a flux of particles (all having the same mass m and moving with the same velocity) moving from left to the right. We are going to consider two cases, depending on whether the energy of the particles is larger or smaller than V_0.

(a) Case $E > V_0$

The particles are free for $x < 0$ and feel a repulsive potential V_0 that starts at $x = 0$ and stays flat (constant) for $x > 0$. Let us analyze the dynamics of this flux of particles classically and then quantum mechanically.

Classically, the particles approach the potential step or barrier from the left with a constant momentum $\sqrt{2mE}$. As the particles enter the region $x \geq 0$, where the potential now is $V = V_0$, they slow down to a momentum $\sqrt{2m(E - V_0)}$; they will then conserve this momentum as they travel to the right. Since the particles have sufficient energy to penetrate into the region $x \geq 0$, there will be *total transmission*: all the particles will emerge to the right with a smaller kinetic energy $E - V_0$. This is then a simple *scattering* problem in one dimension.

Quantum mechanically, the dynamics of the particle is regulated by the Schrödinger equation, which is given in these two regions by

$$\left(\frac{d^2}{dx^2} + k_1^2\right) \psi_1(x) = 0 \qquad (x < 0), \tag{4.15}$$

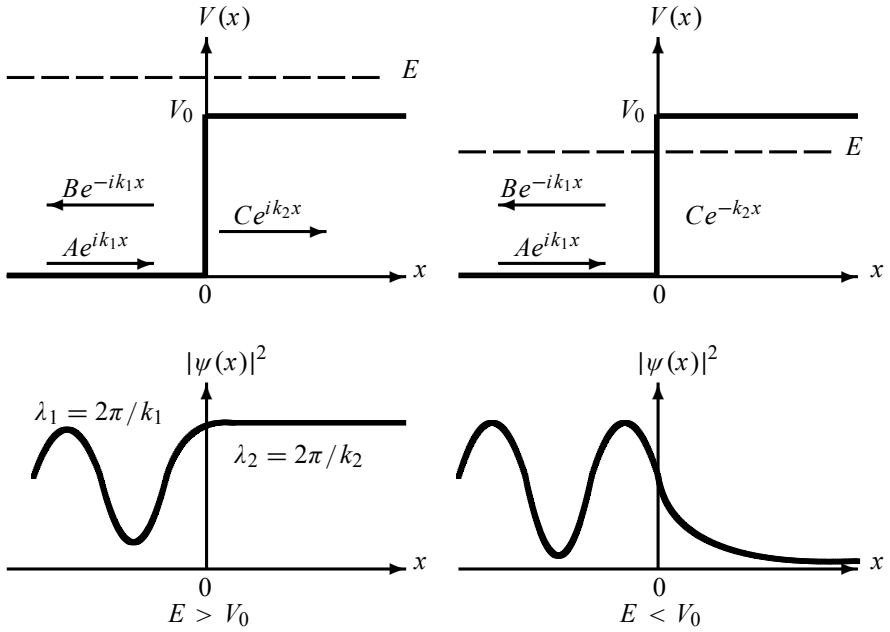


Figure 4.2 Potential step and propagation directions of the incident, reflected, and transmitted waves, plus their probability densities $|\psi(x)|^2$ when $E > V_0$ and $E < V_0$.

$$\left(\frac{d^2}{dx^2} + k_2^2\right)\psi_2(x) = 0 \qquad (x \geq 0), \tag{4.16}$$

where $k_1^2 = 2mE/\hbar^2$ and $k_2^2 = 2m(E - V_0)/\hbar^2$. The most general solutions to these two equations are plane waves:

$$\psi_1(x) = Ae^{ik_1x} + Be^{-ik_1x} \qquad (x < 0), \tag{4.17}$$

$$\psi_2(x) = Ce^{ik_2x} + De^{-ik_2x} \qquad (x \geq 0), \tag{4.18}$$

where Ae^{ik_1x} and Ce^{ik_2x} represent waves moving in the positive x-direction, but Be^{-ik_1x} and De^{-ik_2x} correspond to waves moving in the negative x-direction. We are interested in the case where the particles are initially incident on the potential step from the left: they can be reflected or transmitted at $x = 0$. Since no wave is reflected from the region $x > 0$ to the left, the constant D must vanish. Since we are dealing with stationary states, the complete wave function is thus given by

$$\Psi(x,t) = \begin{cases} \psi_1(x)e^{-i\omega t} = Ae^{i(k_1x-\omega t)} + Be^{-i(k_1x+\omega t)} & x < 0 \\ \psi_2(x)e^{-i\omega t} = Ce^{i(k_2x-\omega t)} & x \geq 0, \end{cases} \tag{4.19}$$

where $A\exp[i(k_1x-\omega t)]$, $B\exp[-i(k_1x+\omega t)]$, and $C\exp[i(k_2x-\omega t)]$ represent the *incident*, the *reflected*, and the *transmitted* waves, respectively; they travel to the right, the left, and the right (Figure 4.2). Note that the probability density $|\psi(x)|^2$ shown in the lower left plot of Figure 4.2 is a straight line for $x > 0$, since $|\psi_2(x)|^2 = |C\exp i(k_2x - \omega t)|^2 = |C|^2$.

Let us now evaluate the *reflection* and *transmission coefficients*, R and T, as defined by

$$R = \left|\frac{\text{reflected current density}}{\text{incident current density}}\right| = \left|\frac{J_{reflected}}{J_{incident}}\right|, \qquad T = \left|\frac{J_{transmitted}}{J_{incident}}\right|; \tag{4.20}$$

R represents the ratio of the reflected to the incident beams and T the ratio of the transmitted to the incident beams. To calculate R and T, we need to find $J_{incident}$, $J_{reflected}$, and $J_{transmitted}$. Since the incident wave is $\psi_i(x) = Ae^{ik_1x}$, the incident current density (or incident flux) is given by

$$J_{incident} = \frac{i\hbar}{2m}\left(\psi_i(x)\frac{d\psi_i^*(x)}{dx} - \psi_i^*(x)\frac{d\psi_i(x)}{dx}\right) = \frac{\hbar k_1}{m}|A|^2. \tag{4.21}$$

Similarly, since the reflected and transmitted waves are $\psi_r(x) = Be^{-ik_1x}$ and $\psi_t(x) = Ce^{ik_2x}$, we can verify that the reflected and transmitted fluxes are

$$J_{reflected} = -\frac{\hbar k_1}{m}|B|^2, \qquad J_{transmitted} = \frac{\hbar k_2}{m}|C|^2. \tag{4.22}$$

A combination of (4.20) to (4.22) yields

$$R = \frac{|B|^2}{|A|^2}, \qquad T = \frac{k_2}{k_1}\frac{|C|^2}{|A|^2}. \tag{4.23}$$

Thus, the calculation of R and T is reduced to determining the constants B and C. For this, we need to use the boundary conditions of the wave function at $x = 0$. Since both the wave function and its first derivative are continuous at $x = 0$,

$$\psi_1(0) = \psi_2(0), \qquad \frac{d\psi_1(0)}{dx} = \frac{d\psi_2(0)}{dx}, \tag{4.24}$$

equations (4.17) and (4.18) yield

$$A + B = C, \qquad k_1(A - B) = k_2C; \tag{4.25}$$

hence

$$B = \frac{k_1 - k_2}{k_1 + k_2}A, \qquad C = \frac{2k_1}{k_1 + k_2}A. \tag{4.26}$$

As for the constant A, it can be determined from the normalization condition of the wave function, but we don't need it here, since R and T are expressed in terms of ratios. A combination of (4.23) with (4.26) leads to

$$R = \frac{(k_1 - k_2)^2}{(k_1 + k_2)^2} = \frac{(1 - \mathcal{K})^2}{(1 + \mathcal{K})^2}, \qquad T = \frac{4k_1k_2}{(k_1 + k_2)^2} = \frac{4\mathcal{K}}{(1 + \mathcal{K})^2}, \tag{4.27}$$

where $\mathcal{K} = k_2/k_1 = \sqrt{1 - V_0/E}$. The sum of R and T is equal to 1, as it should be.

In contrast to classical mechanics, which states that none of the particles get reflected, equation (4.27) shows that the quantum mechanical reflection coefficient R is not zero: there are particles that get reflected in spite of their energies being higher than the step V_0. This effect must be attributed to the *wavelike behavior* of the particles.

From (4.27) we see that as E gets smaller and smaller, T also gets smaller and smaller so that when $E = V_0$ the transmission coefficient T becomes zero and $R = 1$. On the other hand, when $E \gg V_0$, we have $\mathcal{K} = \sqrt{1 - V_0/E} \simeq 1$; hence $R = 0$ and $T = 1$. This is expected since, when the incident particles have very high energies, the potential step is so weak that it produces no noticeable effect on their motion.

Remark: physical meaning of the boundary conditions

Throughout this chapter, we will encounter at numerous times the use of the boundary conditions of the wave function and its first derivative as in Eq (4.24). What is the underlying physics behind these continuity conditions? We can make two observations:

- Since the probability density $|\psi(x)|^2$ of finding the particle in any small region varies continuously from one point to another, the wave function $\psi(x)$ must, therefore, be a continuous function of x; thus, as shown in (4.24), we must have $\psi_1(0) = \psi_2(0)$.
- Since the linear momentum of the particle, $\hat{P}\psi(x) = -i\hbar d\psi(x)/dx$, must be a continuous function of x as the particle moves from left to right, the first derivative of the wave function, $d\psi(x)/dx$, must also be a continuous function of x, notably at $x = 0$. Hence, as shown in (4.24), we must have $d\psi_1(0)/dx = d\psi_2(0)/dx$.

(b) Case $E < V_0$

Classically, the particles arriving at the potential step from the left (with momenta $p = \sqrt{2mE}$) will come to a stop at $x = 0$ and then all will bounce back to the left with the magnitudes of their momenta unchanged. None of the particles will make it into the right side of the barrier $x = 0$; there is total reflection of the particles. So the motion of the particles is reversed by the potential barrier.

Quantum mechanically, the picture will be somewhat different. In this case, the Schrödinger equation and the wave function in the region $x < 0$ are given by (4.15) and (4.17), respectively. But for $x > 0$ the Schrödinger equation is given by

$$\left(\frac{d^2}{dx^2} - k_2'^2\right)\psi_2(x) = 0 \quad (x \geq 0), \tag{4.28}$$

where $k_2'^2 = 2m(V_0 - E)/\hbar^2$. This equation's solution is

$$\psi_2(x) = Ce^{-k_2'x} + De^{k_2'x} \quad (x \geq 0). \tag{4.29}$$

Since the wave function must be finite everywhere, and since the term $e^{k_2'x}$ diverges when $x \to \infty$, the constant D has to be zero. Thus, the complete wave function is

$$\Psi(x,t) = \begin{cases} Ae^{i(k_1x-\omega t)} + Be^{-i(k_1x+\omega t)}, & x < 0, \\ Ce^{-k_2'x}e^{-i\omega t}, & x \geq 0. \end{cases} \tag{4.30}$$

Let us now evaluate, as we did in the previous case, the reflected and the transmitted coefficients. First we should note that the transmitted coefficient, which corresponds to the transmitted wave function $\psi_t(x) = Ce^{-k_2'x}$, is zero since $\psi_t(x)$ is a *purely real* function ($\psi_t^*(x) = \psi_t(x)$) and therefore

$$J_{transmitted} = \frac{\hbar}{2im}\left(\psi_t(x)\frac{d\psi_t(x)}{dx} - \psi_t(x)\frac{d\psi_t(x)}{dx}\right) = 0. \tag{4.31}$$

Hence, the reflected coefficient R must be equal to 1. We can obtain this result by applying the continuity conditions at $x = 0$ for (4.17) and (4.29):

$$B = \frac{k_1 - ik_2'}{k_1 + ik_2'}A, \qquad C = \frac{2k_1}{k_1 + ik_2'}A. \tag{4.32}$$

Thus, the reflected coefficient is given by

$$R = \frac{|B|^2}{|A|^2} = \frac{k_1^2 + k_2'^2}{k_1^2 + k_2'^2} = 1. \tag{4.33}$$

We therefore have total reflection, as in the classical case.

There is, however, a difference with the classical case: while none of the particles can be found classically in the region $x > 0$, quantum mechanically there is a *nonzero probability* that the wave function penetrates this *classically forbidden* region. To see this, note that the relative probability density

$$P(x) = |\psi_t(x)|^2 = |C|^2 e^{-2k_2' x} = \frac{4k_1^2 |A|^2}{k_1^2 + k_2'^2} e^{-2k_2' x} \tag{4.34}$$

is appreciable near $x = 0$ and falls exponentially to small values as x becomes large; the behavior of the probability density is shown in Figure 4.2.

4.5 The Potential Barrier and Well

Consider a beam of particles of mass m that are sent from the left on a potential barrier

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & 0 \le x \le a, \\ 0, & x > a. \end{cases} \tag{4.35}$$

This potential, which is repulsive, supports no bound states (Figure 4.3). We are dealing here, as in the case of the potential step, with a one-dimensional *scattering* problem.

Again, let us consider the following two cases which correspond to the particle energies being respectively larger and smaller than the potential barrier.

4.5.1 The Case $E > V_0$

Classically, the particles that approach the barrier from the left at constant momentum, $p_1 = \sqrt{2mE}$, as they enter the region $0 \le x \le a$ will slow down to a momentum $p_2 = \sqrt{2m(E - V_0)}$. They will maintain the momentum p_2 until they reach the point $x = a$. Then, as soon as they pass beyond the point $x = a$, they will accelerate to a momentum $p_3 = \sqrt{2mE}$ and maintain this value in the entire region $x > a$. Since the particles have enough energy to cross the barrier, none of the particles will be reflected back; all the particles will emerge on the right side of $x = a$: *total transmission*.

It is easy to infer the quantum mechanical study from the treatment of the potential step presented in the previous section. We need only to mention that the wave function will display an oscillatory pattern in all three regions; its amplitude reduces every time the particle enters a new region (see Figure 4.3):

$$\psi(x) = \begin{cases} \psi_1(x) = Ae^{ik_1 x} + Be^{-ik_1 x}, & x \le 0, \\ \psi_2(x) = Ce^{ik_2 x} + De^{-ik_2 x}, & 0 < x < a, \\ \psi_3(x) = Ee^{ik_1 x}, & x \ge a, \end{cases} \tag{4.36}$$

where $k_1 = \sqrt{2mE/\hbar^2}$ and $k_2 = \sqrt{2m(E - V_0)/\hbar^2}$. The constants B, C, D, and E can be obtained in terms of A from the boundary conditions: $\psi(x)$ and $d\psi/dx$ must be continuous at $x = 0$ and $x = a$, respectively:

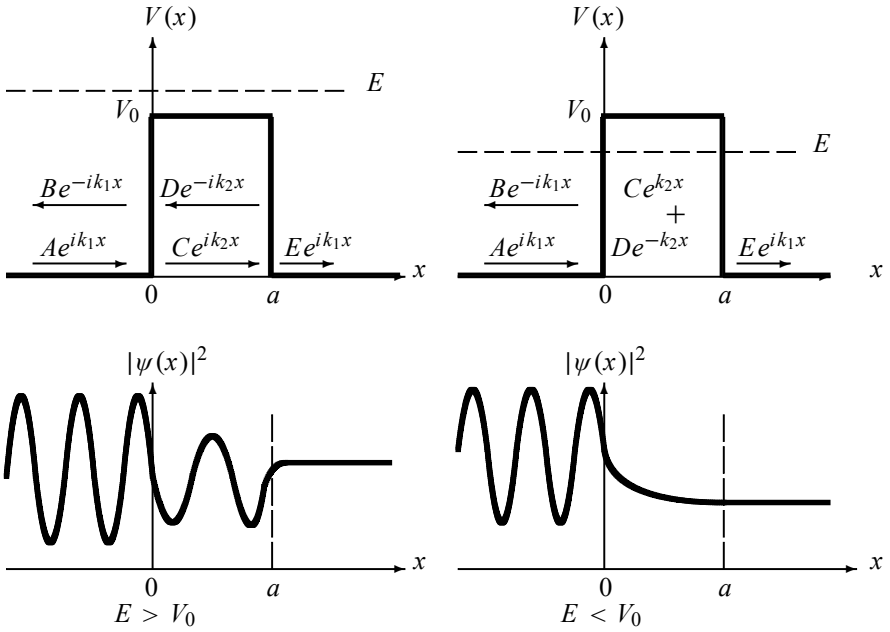


Figure 4.3 Potential barrier and propagation directions of the incident, reflected, and transmitted waves, plus their probability densities $|\psi(x)|^2$ when $E > V_0$ and $E < V_0$.

$$\psi_1(0) = \psi_2(0), \qquad \frac{d\psi_1(0)}{dx} = \frac{d\psi_2(0)}{dx}, \tag{4.37}$$

$$\psi_2(a) = \psi_3(a), \qquad \frac{d\psi_2(a)}{dx} = \frac{d\psi_3(a)}{dx}. \tag{4.38}$$

These equations yield

$$A + B = C + D, \qquad ik_1(A - B) = ik_2(C - D), \tag{4.39}$$

$$Ce^{ik_2a} + De^{-ik_2a} = Ee^{ik_1a}, \qquad ik_2\left(Ce^{ik_2a} - De^{-ik_2a}\right) = ik_1Ee^{ik_1a}. \tag{4.40}$$

Solving for E, we obtain

$$\begin{aligned} E &= 4k_1k_2Ae^{-ik_1a}[(k_1 + k_2)^2 e^{-ik_2a} - (k_1 - k_2)^2 e^{ik_2a}]^{-1} \\ &= 4k_1k_2Ae^{-ik_1a}\left[4k_1k_2\cos(k_2a) - 2i\left(k_1^2 + k_2^2\right)\sin(k_2a)\right]^{-1}. \end{aligned} \tag{4.41}$$

The transmission coefficient is thus given by

$$\begin{aligned} T &= \frac{k_1|E|^2}{k_1|A|^2} = \left[1 + \frac{1}{4}\left(\frac{k_1^2 - k_2^2}{k_1k_2}\right)^2 \sin^2(k_2a)\right]^{-1} \\ &= \left[1 + \frac{V_0^2}{4E(E - V_0)}\sin^2\left(a\sqrt{2mV_0/\hbar^2}\sqrt{E/V_0 - 1}\right)\right]^{-1}, \end{aligned} \tag{4.42}$$

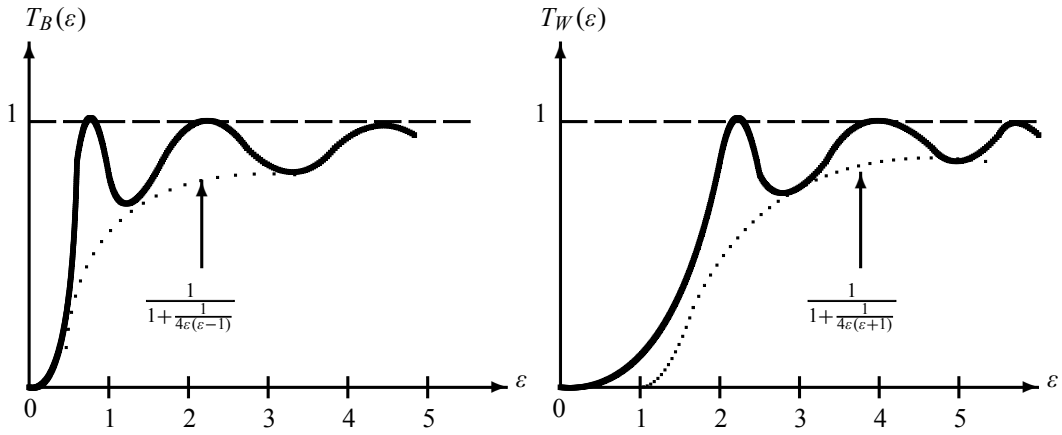


Figure 4.4 Transmission coefficients for a potential barrier, $T_B(\varepsilon) = \frac{4\varepsilon(\varepsilon-1)}{4\varepsilon(\varepsilon-1)+\sin^2(\lambda\sqrt{\varepsilon-1})}$, and for a potential well, $T_W(\varepsilon) = \frac{4\varepsilon(\varepsilon+1)}{4\varepsilon(\varepsilon+1)+\sin^2(\lambda\sqrt{\varepsilon+1})}$.

because

$$\left(\frac{k_1^2 - k_2^2}{k_1 k_2}\right)^2 = \frac{V_0^2}{E(E - V_0)}. \tag{4.43}$$

Using the notation $\lambda = a\sqrt{2mV_0/\hbar^2}$ and $\varepsilon = E/V_0$, we can rewrite T as

$$T = \left[1 + \frac{1}{4\varepsilon(\varepsilon - 1)} \sin^2(\lambda\sqrt{\varepsilon - 1})\right]^{-1}. \tag{4.44}$$

Similarly, we can show that

$$R = \frac{\sin^2(\lambda\sqrt{\varepsilon - 1})}{4\varepsilon(\varepsilon - 1) + \sin^2(\lambda\sqrt{\varepsilon - 1})} = \left[1 + \frac{4\varepsilon(\varepsilon - 1)}{\sin^2(\lambda\sqrt{\varepsilon - 1})}\right]^{-1}. \tag{4.45}$$

Special cases

- If $E \gg V_0$, and hence $\varepsilon \gg 1$, the transmission coefficient T becomes asymptotically equal to unity, $T \simeq 1$, and $R \simeq 0$. So, at very high energies and weak potential barrier, the particles would not feel the effect of the barrier; we have total transmission.
- We also have total transmission when $\sin(\lambda\sqrt{\varepsilon - 1}) = 0$ or $\lambda\sqrt{\varepsilon - 1} = n\pi$. As shown in Figure 4.4, the total transmission, $T(\varepsilon_n) = 1$, occurs whenever $\varepsilon_n = E_n/V_0 = n^2\pi^2\hbar^2/(2ma^2V_0) + 1$ or whenever the incident energy of the particle is $E_n = V_0 + n^2\pi^2\hbar^2/(2ma^2)$ with $n = 1, 2, 3, \ldots$. The maxima of the transmission coefficient coincide with the energy eigenvalues of the infinite square well potential; these are known as resonances. This resonance phenomenon, which does not occur in classical physics, results from a constructive interference between the incident and the reflected waves. This phenomenon is observed experimentally in a number of cases such as when scattering low-energy ($E \sim 0.1$ eV) electrons off noble atoms (known as the *Ramsauer–Townsend effect*, a consequence of symmetry of noble atoms) and neutrons off nuclei.

- In the limit $\varepsilon \to 1$ we have $\sin(\lambda\sqrt{\varepsilon-1}) \sim \lambda\sqrt{\varepsilon-1}$, hence (4.44) and (4.45) become

$$T = \left(1 + \frac{ma^2V_0}{2\hbar^2}\right)^{-1}, \qquad R = \left(1 + \frac{2\hbar^2}{ma^2V_0}\right)^{-1}. \tag{4.46}$$

The potential well ($V_0 < 0$)
The transmission coefficient (4.44) was derived for the case where $V_0 > 0$, i.e., for a *barrier potential*. Following the same procedure that led to (4.44), we can show that the transmission coefficient for a finite *potential well*, $V_0 < 0$, is given by

$$T_W = \left[1 + \frac{1}{4\varepsilon(\varepsilon+1)}\sin^2(\lambda\sqrt{\varepsilon+1})\right]^{-1}, \tag{4.47}$$

where $\varepsilon = E/|V_0|$ and $\lambda = a\sqrt{2m|V_0|/\hbar^2}$. Notice that there is total transmission whenever $\sin(\lambda\sqrt{\varepsilon+1}) = 0$ or $\lambda\sqrt{\varepsilon+1} = n\pi$. As shown in Figure 4.4, the total transmission, $T_W(\varepsilon_n) = 1$, occurs whenever $\varepsilon_n = E_n/|V_0| = n^2\pi^2\hbar^2/(2ma^2V_0) - 1$ or whenever the incident energy of the particle is $E_n = n^2\pi^2\hbar^2/(2ma^2) - |V_0|$ with $n = 1, 2, 3, \ldots$. We will study in more detail the *symmetric* potential well in Section 4.7.

4.5.2 The Case $E < V_0$: Tunneling

Classically, we would expect total reflection: every particle that arrives at the barrier ($x = 0$) will be reflected back; no particle can penetrate the barrier, where it would have a negative kinetic energy.

We are now going to show that the quantum mechanical predictions differ sharply from their classical counterparts, for the wave function is not zero beyond the barrier. The solutions of the Schrödinger equation in the three regions yield expressions that are similar to (4.36) except that $\psi_2(x) = Ce^{ik_2x} + De^{-ik_2x}$ should be replaced with $\psi_2(x) = Ce^{k_2x} + De^{-k_2x}$:

$$\psi(x) = \begin{cases} \psi_1(x) = Ae^{ik_1x} + Be^{-ik_1x}, & x \le 0, \\ \psi_2(x) = Ce^{k_2x} + De^{-k_2x}, & 0 < x < a, \\ \psi_3(x) = Ee^{ik_1x}, & x \ge a, \end{cases} \tag{4.48}$$

where $k_1^2 = 2mE/\hbar^2$ and $k_2^2 = 2m(V_0 - E)/\hbar^2$. The behavior of the probability density corresponding to this wave function is expected, as displayed in Figure 4.3, to be oscillatory in the regions $x < 0$ and $x > a$, and exponentially decaying for $0 \le x \le a$.

To find the reflection and transmission coefficients,

$$R = \frac{|B|^2}{|A|^2}, \qquad T = \frac{|E|^2}{|A|^2}, \tag{4.49}$$

we need only to calculate B and E in terms of A. The continuity conditions of the wave function and its derivative at $x = 0$ and $x = a$ yield

$$A + B = C + D, \tag{4.50}$$

$$ik_1(A - B) = k_2(C - D), \tag{4.51}$$

$$Ce^{k_2a} + De^{-k_2a} = Ee^{ik_1a}, \tag{4.52}$$

$$k_2\left(Ce^{k_2a} - De^{-k_2a}\right) = ik_1Ee^{ik_1a}. \tag{4.53}$$

The last two equations lead to the following expressions for C and D:

$$C = \frac{E}{2}\left(1 + i\frac{k_1}{k_2}\right)e^{(ik_1-k_2)a}, \qquad D = \frac{E}{2}\left(1 - i\frac{k_1}{k_2}\right)e^{(ik_1+k_2)a}. \tag{4.54}$$

Inserting these two expressions into the two equations (4.50) and (4.51) and dividing by A, we can show that these two equations reduce, respectively, to

$$1 + \frac{B}{A} = \frac{E}{A}e^{ik_1a}\left[\cosh(k_2a) - i\frac{k_1}{k_2}\sinh(k_2a)\right], \tag{4.55}$$

$$1 - \frac{B}{A} = \frac{E}{A}e^{ik_1a}\left[\cosh(k_2a) + i\frac{k_2}{k_1}\sinh(k_2a)\right]. \tag{4.56}$$

Solving these two equations for B/A and E/A, we obtain

$$\frac{B}{A} = -i\frac{k_1^2+k_2^2}{k_1k_2}\sinh(k_2a)\left[2\cosh(k_2a) + i\frac{k_2^2-k_1^2}{k_1k_2}\sinh(k_2a)\right]^{-1}, \tag{4.57}$$

$$\frac{E}{A} = 2e^{-ik_1a}\left[2\cosh(k_2a) + i\frac{k_2^2-k_1^2}{k_1k_2}\sinh(k_2a)\right]^{-1}. \tag{4.58}$$

Thus, the coefficients R and T become

$$R = \left(\frac{k_1^2+k_2^2}{k_1k_2}\right)^2\sinh^2(k_2a)\left[4\cosh^2(k_2a) + \left(\frac{k_2^2-k_1^2}{k_1k_2}\right)^2\sinh^2(k_2a)\right]^{-1}, \tag{4.59}$$

$$T = \frac{|E|^2}{|A|^2} = 4\left[4\cosh^2(k_2a) + \left(\frac{k_2^2-k_1^2}{k_1k_2}\right)^2\sinh^2(k_2a)\right]^{-1}. \tag{4.60}$$

We can rewrite R in terms of T as

$$R = \frac{1}{4}T\left(\frac{k_1^2+k_2^2}{k_1k_2}\right)^2\sinh^2(k_2a). \tag{4.61}$$

Since $\cosh^2(k_2a) = 1 + \sinh^2(k_2a)$ we can reduce (4.60) to

$$T = \left[1 + \frac{1}{4}\left(\frac{k_1^2+k_2^2}{k_1k_2}\right)^2\sinh^2(k_2a)\right]^{-1}. \tag{4.62}$$

Note that T is *finite*. This means that the probability for the transmission of the particles into the region $x \geq a$ is *not zero* (in classical physics, however, the particle can in no way make it into the $x \geq 0$ region). This is a purely quantum mechanical effect which is due to the *wave aspect* of microscopic objects; it is known as the *tunneling effect*: *quantum mechanical objects can tunnel through classically impenetrable barriers*. This *barrier penetration* effect has important applications in various branches of modern physics ranging from particle and nuclear physics

to semiconductor devices. For instance, radioactive decays and charge transport in electronic devices are typical examples of the tunneling effect.

Now since

$$\left(\frac{k_1^2+k_2^2}{k_1k_2}\right)^2=\left(\frac{V_0}{\sqrt{E(V_0-E)}}\right)^2=\frac{V_0^2}{E(V_0-E)}, \tag{4.63}$$

we can rewrite (4.61) and (4.62) as follows:

$$R=\frac{1}{4}\frac{V_0^2T}{E(V_0-E)}\sinh^2\left(\frac{a}{\hbar}\sqrt{2m(V_0-E)}\right), \tag{4.64}$$

$$T=\left[1+\frac{1}{4}\frac{V_0^2}{E(V_0-E)}\sinh^2\left(\frac{a}{\hbar}\sqrt{2m(V_0-E)}\right)\right]^{-1}, \tag{4.65}$$

or

$$R=\frac{T}{4\varepsilon(1-\varepsilon)}\sinh^2\left(\lambda\sqrt{1-\varepsilon}\right), \tag{4.66}$$

$$T=\left[1+\frac{1}{4\varepsilon(1-\varepsilon)}\sinh^2\left(\lambda\sqrt{1-\varepsilon}\right)\right]^{-1}, \tag{4.67}$$

where $\lambda=a\sqrt{2mV_0/\hbar^2}$ and $\varepsilon=E/V_0$.

Special cases

- If $E\ll V_0$, hence $\varepsilon\ll 1$ or $\lambda\sqrt{1-\varepsilon}\gg 1$, we may approximate $\sinh\left(\lambda\sqrt{1-\varepsilon}\right)\simeq\frac{1}{2}\exp\left(\lambda\sqrt{1-\varepsilon}\right)$. We can thus show that the transmission coefficient (4.67) becomes asymptotically equal to

$$\begin{aligned}T &\simeq \left\{\frac{1}{4\varepsilon(1-\varepsilon)}\left[\frac{1}{2}e^{\lambda\sqrt{1-\varepsilon}}\right]^2\right\}^{-1}=16\varepsilon(1-\varepsilon)e^{-2\lambda\sqrt{1-\varepsilon}}\\ &= \frac{16E}{V_0}\left(1-\frac{E}{V_0}\right)e^{-(2a/\hbar)\sqrt{2m(V_0-E)}}.\end{aligned} \tag{4.68}$$

  This shows that the transmission coefficient is not zero, as it would be classically, but has a finite value. So, quantum mechanically, there is a finite tunneling beyond the barrier, $x>a$.

- When $E\simeq V_0$, hence $\varepsilon\simeq 1$, we can verify that (4.66) and (4.67) lead to the relations (4.46).

- Taking the classical limit $\hbar\to 0$, the coefficients (4.66) and (4.67) reduce to the classical result: $R\to 1$ and $T\to 0$.

4.5.3 The Tunneling Effect

In general, the tunneling effect consists of the propagation of a particle through a region where the particle's energy is smaller than the potential energy $E<V(x)$. Classically this region, defined by $x_1<x<x_2$ (Figure 4.5a), is forbidden to the particle where its kinetic energy

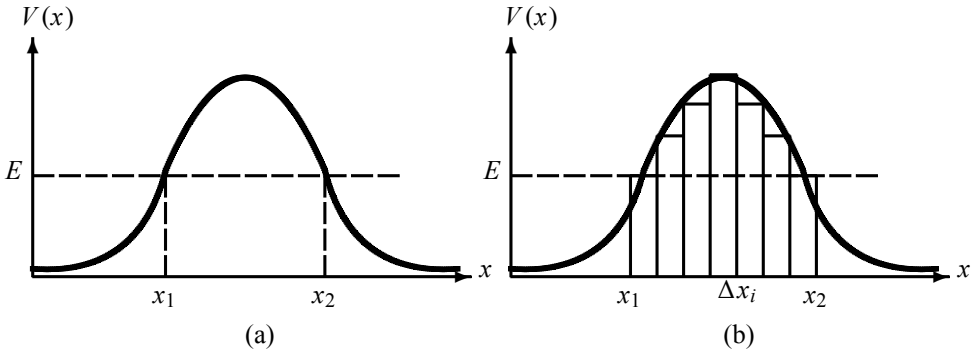


Figure 4.5 (a) Tunneling though a potential barrier. (b) Approximation of a smoothly varying potential $V(x)$ by square barriers.

would be negative; the points $x = x_1$ and $x = x_2$ are known as the *classical turning points*. Quantum mechanically, however, since particles display wave features, the quantum waves can tunnel through the barrier.

As shown in the square barrier example, the particle has a finite probability of tunneling through the barrier. In this case we managed to find an analytical expression (4.67) for the tunneling probability only because we dealt with a simple square potential. Analytic expressions cannot be obtained for potentials with arbitrary spatial dependence. In such cases one needs approximations. The Wentzel–Kramers–Brillouin (WKB) method (Chapter 9) provides one of the most useful approximation methods. We will show that the transmission coefficient for a barrier potential $V(x)$ is given by

$$T \sim \exp\left\{-\frac{2}{\hbar}\int_{x_1}^{x_2} dx\sqrt{2m\,[V(x)-E]}\right\}. \tag{4.69}$$

We can obtain this relation by means of a crude approximation. For this, we need simply to take the classically forbidden region $x_1 < x < x_2$ (Figure 4.5b) and divide it into a series of small intervals Δx_i. If Δx_i is small enough, we may approximate the potential $V(x_i)$ at each point x_i by a square potential barrier. Thus, we can use (4.68) to calculate the transmission probability corresponding to $V(x_i)$:

$$T_i \sim \exp\left[-\frac{2\Delta x_i}{\hbar}\sqrt{2m(V(x_i)-E)}\right]. \tag{4.70}$$

The transmission probability for the general potential of Figure 4.5, where we divided the region $x_1 < x < x_2$ into a very large number of small intervals Δx_i, is given by

$$\begin{aligned} T \quad &\sim \quad \lim_{N\to\infty}\prod_{i=1}^{N}\exp\left[-\frac{2\Delta x_i}{\hbar}\sqrt{2m(V(x_i)-E)}\right] \\ &= \quad \exp\left[-\frac{2}{\hbar}\lim_{\Delta x_i\to 0}\sum_i \Delta x_i\sqrt{2m(V(x_i)-E)}\right] \\ &\longrightarrow \quad \exp\left[-\frac{2}{\hbar}\int_{x_1}^{x_2} dx\sqrt{2m\,[V(x)-E]}\right]. \end{aligned} \tag{4.71}$$

The approximation leading to this relation is valid, as will be shown in Chapter 9, only if the potential $V(x)$ is a smooth, slowly varying function of x.

4.6 The Infinite Square Well Potential

4.6.1 The Asymmetric Square Well

Consider a particle of mass m confined to move inside an infinitely deep asymmetric potential well

$$V(x) = \begin{cases} +\infty, & x < 0, \\ 0, & 0 \le x \le a, \\ +\infty, & x > a. \end{cases} \tag{4.72}$$

Classically, the particle remains confined inside the well, moving at constant momentum $p = \pm\sqrt{2mE}$ back and forth as a result of repeated reflections from the walls of the well.

Quantum mechanically, we expect this particle to have only bound state solutions and a discrete nondegenerate energy spectrum. Since $V(x)$ is infinite outside the region $0 \le x \le a$, the wave function of the particle must be zero outside the boundary. Hence we can look for solutions only inside the well

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0, \tag{4.73}$$

with $k^2 = 2mE/\hbar^2$; the solutions are

$$\psi(x) = A'e^{ikx} + B'e^{-ikx} \quad \Longrightarrow \quad \psi(x) = A\sin(kx) + B\cos(kx). \tag{4.74}$$

The wave function vanishes at the walls, $\psi(0) = \psi(a) = 0$: the condition $\psi(0) = 0$ gives $B = 0$, while $\psi(a) = A\sin(ka) = 0$ gives

$$k_n a = n\pi \qquad (n = 1, 2, 3, \cdots). \tag{4.75}$$

This condition determines the energy

$$\boxed{E_n = \frac{\hbar^2}{2m}k_n^2 = \frac{\hbar^2\pi^2}{2ma^2}n^2 \qquad (n = 1, 2, 3, \cdots).} \tag{4.76}$$

The energy is *quantized*; only certain values are permitted. This is expected since the states of a particle which is confined to a limited region of space are *bound states* and the energy spectrum is *discrete*. This is in sharp contrast to classical physics where the energy of the particle, given by $E = p^2/(2m)$, takes any value; the classical energy evolves *continuously*.

As it can be inferred from (4.76), we should note that the energy between adjacent levels is not constant:

$$E_{n+1} - E_n = 2n + 1, \tag{4.77}$$

which leads to

$$\frac{E_{n+1} - E_n}{E_n} = \frac{(n+1)^2 - n^2}{n^2} = \frac{2n+1}{n^2}. \tag{4.78}$$

In the classical limit $n \to \infty$,

$$\lim_{n\to\infty} \frac{E_{n+1} - E_n}{E_n} = \lim_{n\to\infty} \frac{2n+1}{n^2} = 0, \tag{4.79}$$

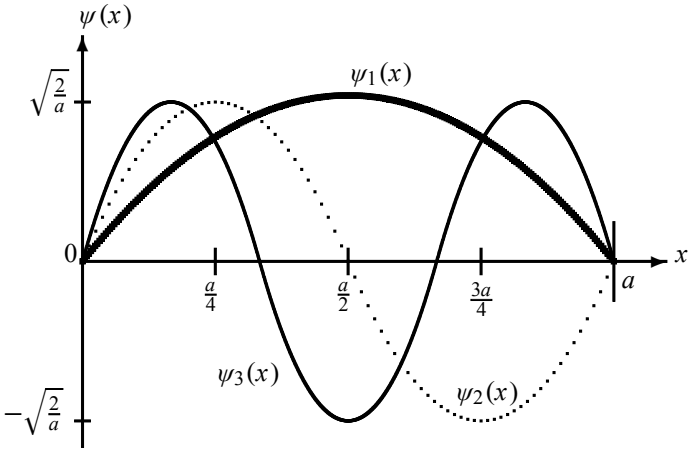


Figure 4.6 Three lowest states of an infinite potential well, $\psi_n(x) = \sqrt{2/a}\sin(n\pi x/a)$; the states $\psi_{2n+1}(x)$ and $\psi_{2n}(x)$ are even and odd, respectively, with respect to $x = a/2$.

the levels become so close together as to be practically indistinguishable.

Since $B = 0$ and $k_n = n\pi/a$, (4.74) yields $\psi_n(x) = A\sin(n\pi x/a)$. We can choose the constant A so that $\psi_n(x)$ is normalized:

$$1 = \int_0^a |\psi_n(x)|^2 dx = |A|^2 \int_0^a \sin^2\left(\frac{n\pi}{a}x\right) dx \quad \Longrightarrow \quad A = \sqrt{\frac{2}{a}}, \tag{4.80}$$

hence

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right) \qquad (n = 1, 2, 3, \cdots).} \tag{4.81}$$

The first few functions are plotted in Figure 4.6.

The solution of the time-independent Schrödinger equation has thus given us the energy (4.76) and the wave function (4.81). There is then an infinite sequence of discrete energy levels corresponding to the positive integer values of the *quantum number* n. It is clear that $n = 0$ yields an uninteresting result: $\psi_0(x) = 0$ and $E_0 = 0$; later, we will examine in more detail the physical implications of this. So, the lowest energy, or *ground state* energy, corresponds to $n = 1$; it is $E_1 = \hbar^2\pi^2/(2ma^2)$. As will be explained later, this is called the *zero-point energy*, for there exists no state with zero energy. The states corresponding to $n = 2, 3, 4, \ldots$ are called *excited states*; their energies are given by $E_n = n^2 E_1$. As mentioned in Theorem 4.2, each function $\psi_n(x)$ has $(n-1)$ nodes. Figure 4.6 shows that the functions $\psi_{2n+1}(x)$ are even and the functions $\psi_{2n}(x)$ are odd with respect to the center of the well; we will study this in Section 4.6.2 when we consider the symmetric potential well. Note that none of the energy levels is degenerate (there is only one eigenfunction for each energy level) and that the wave functions corresponding to different energy levels are orthogonal:

$$\int_0^a \psi_m^*(x)\psi_n(x)\,dx = \delta_{mn}. \tag{4.82}$$

Since we are dealing with stationary states and since $E_n = n^2 E_1$, the most general solutions of

the time-dependent Schrödinger equation are given by

$$\Psi(x,t) = \sum_{n=1}^{\infty} \psi_n(x) e^{-iE_n t/\hbar} = \sqrt{\frac{2}{a}} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{a}\right) e^{-in^2 E_1 t/\hbar}. \tag{4.83}$$

Zero-point energy

Let us examine why there is no state with zero energy for a square well potential. If the particle has zero energy, it will be at rest inside the well, and this violates Heisenberg's uncertainty principle. By *localizing* or confining the particle to a limited region in space, it will acquire a *finite* momentum leading to a minimum kinetic energy. That is, the localization of the particle's motion to $0 \leq x \leq a$ implies a position uncertainty of order $\Delta x \sim a$ which, according to the uncertainty principle, leads to a minimum momentum uncertainty $\Delta p \sim \hbar/a$ and this in turn leads to a minimum kinetic energy of order $\hbar^2/(2ma^2)$. This is in qualitative agreement with the exact value $E_1 = \pi^2\hbar^2/(2ma^2)$. In fact, as will be shown in (4.216), an accurate evaluation of Δp_1 leads to a zero-point energy which is equal to E_1.

Note that, as the momentum uncertainty is inversely proportional to the width of the well, $\Delta p \sim \hbar/a$, if the width decreases (i.e., the particle's position is confined further and further), the uncertainty on $\hat{P}$ will increase. This makes the particle move faster and faster, so the zero-point energy will also increase. Conversely, if the width of the well increases, the zero-point energy decreases, but it will never vanish.

The zero-point energy therefore reflects the necessity of a *minimum motion* of a particle due to localization. The zero-point energy occurs in all bound state potentials. In the case of binding potentials, the lowest energy state has an energy which is higher than the minimum of the potential energy. This is in sharp contrast to classical mechanics, where the lowest possible energy is equal to the minimum value of the potential energy, with zero kinetic energy. In quantum mechanics, however, the lowest state does not minimize the potential alone, but applies to the sum of the kinetic and potential energies, and this leads to a finite ground state or zero-point energy. This concept has far-reaching physical consequences in the realm of the microscopic world. For instance, without the zero-point motion, atoms would not be stable, for the electrons would fall into the nuclei. Also, it is the zero-point energy which prevents helium from freezing at very low temperatures.

The following example shows that the zero-point energy is also present in macroscopic systems, but it is infinitesimally small. In the case of microscopic systems, however, it has a nonnegligible size.

Example 4.1 (Zero-point energy)

To illustrate the idea that the zero-point energy gets larger by going from macroscopic to microscopic systems, calculate the zero-point energy for a particle in an infinite potential well for the following three cases:

(a) a 100 g ball confined on a 5 m long line,

(b) an oxygen atom confined to a 2×10^{-10} m lattice, and

(c) an electron confined to a 10^{-10} m atom.

Solution

(a) The zero-point energy of a 100 g ball that is confined to a 5 m long line is

$$E = \frac{\hbar^2\pi^2}{2ma^2} \simeq \frac{10 \times 10^{-68}\,\text{J}}{2 \times 0.1 \times 25} \simeq 2 \times 10^{-68}\,\text{J} = 1.25 \times 10^{-49}\,\text{eV}. \tag{4.84}$$

This energy is too small to be detected, much less measured, by any known experimental technique.

(b) For the zero-point energy of an oxygen atom confined to a 2×10^{-10} m lattice, since the oxygen atom has 16 nucleons, its mass is of the order of $m \simeq 16 \times 1.6 \times 10^{-27}$ kg $\simeq 26 \times 10^{-27}$ kg, so we have

$$E = \frac{10^{-67}\,\text{J}}{2 \times 26 \times 10^{-27} \times 4 \times 10^{-20}} \simeq 0.5 \times 10^{-22}\,\text{J} \simeq 3 \times 10^{-4}\,\text{eV}. \tag{4.85}$$

(c) The zero-point energy of an electron ($m \sim 10^{-30}$ kg) that is confined to an atom ($a \sim 1 \times 10^{-10}$ m) is

$$E = \frac{10^{-67}\,\text{J}}{2 \times 10^{-30} \times 10^{-20}} \simeq 5 \times 10^{-18}\,\text{J} \simeq 30\,\text{eV}. \tag{4.86}$$

This energy is important at the atomic scale, for the binding energy of a hydrogen electron is about 14 eV. So the zero-point energy is negligible for macroscopic objects, but important for microscopic systems.

4.6.2 The Symmetric Potential Well

What happens if the potential (4.72) is translated to the left by a distance of $a/2$ to become symmetric?

$$V(x) = \begin{cases} +\infty, & x < -a/2, \\ 0, & -a/2 \le x \le a/2, \\ +\infty, & x > a/2. \end{cases} \tag{4.87}$$

First, we would expect the energy spectrum (4.76) to remain unaffected by this translation, since the Hamiltonian is invariant under spatial translations; as it contains only a kinetic part, it commutes with the particle's momentum, $[\hat{H}, \hat{P}] = 0$. The energy spectrum is discrete and nondegenerate.

Second, earlier in this chapter we saw that for symmetric potentials, $V(-x) = V(x)$, the wave function of bound states must be either even or odd. The wave function corresponding to the potential (4.87) can be written as follows:

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left[\frac{n\pi}{a}\left(x + \frac{a}{2}\right)\right] = \begin{cases} \sqrt{\frac{2}{a}} \cos(\frac{n\pi}{a}x) & (n = 1, 3, 5, 7, \cdots), \\ \sqrt{\frac{2}{a}} \sin(\frac{n\pi}{a}x) & (n = 2, 4, 6, 8, \cdots). \end{cases} \tag{4.88}$$

That is, the wave functions corresponding to odd quantum numbers $n = 1, 3, 5, \ldots$ are symmetric, $\psi(-x) = \psi(x)$, and those corresponding to even numbers $n = 2, 4, 6, \ldots$ are antisymmetric, $\psi(-x) = -\psi(x)$.

4.7 The Finite Square Well Potential

Consider a particle of mass m moving in the following symmetric potential:

$$V(x) = \begin{cases} V_0, & x < -a/2, \\ 0, & -a/2 \le x \le a/2, \\ V_0, & x > a/2. \end{cases} \tag{4.89}$$

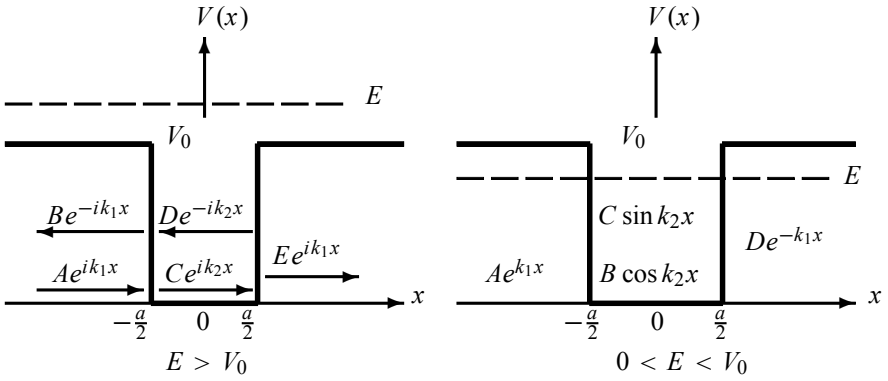


Figure 4.7 Finite square well potential and propagation directions of the incident, reflected and transmitted waves when $E > V_0$ and $0 < E < V_0$.

The two physically interesting cases are $E > V_0$ and $E < V_0$ (see Figure 4.7). We expect the solutions to yield a *continuous* doubly-degenerate energy spectrum for $E > V_0$ and a *discrete* nondegenerate spectrum for $0 < E < V_0$.

4.7.1 The Scattering Solutions ($E > V_0$)

Classically, if the particle is initially incident from left with constant momentum $\sqrt{2m(E - V_0)}$, it will speed up to $\sqrt{2mE}$ between $-a/2 \leq x \leq a/2$ and then slow down to its initial momentum in the region $x > a$. All the particles that come from the left will be transmitted, none will be reflected back; therefore $T = 1$ and $R = 0$.

Quantum mechanically, and as we did for the step and barrier potentials, we can verify that we get a *finite* reflection coefficient. The solution is straightforward to obtain; just follow the procedure outlined in the previous two sections. The wave function has an oscillating pattern in all three regions (see Figure 4.7).

4.7.2 The Bound State Solutions ($0 < E < V_0$)

Classically, when $E < V_0$ the particle is completely confined to the region $-a/2 \leq x \leq a/2$; it will bounce back and forth between $x = -a/2$ and $x = a/2$ with constant momentum $p = \sqrt{2mE}$.

Quantum mechanically, the solutions are particularly interesting for they are expected to yield a *discrete* energy spectrum and wave functions that decay in the two regions $x < -a/2$ and $x > a/2$, but oscillate in $-a/2 \leq x \leq a/2$. In these three regions, the Schrödinger equation can be written as

$$\left(\frac{d^2}{dx^2} - k_1^2\right)\psi_1(x) = 0 \qquad \left(x < -\frac{1}{2}a\right), \tag{4.90}$$

$$\left(\frac{d^2}{dx^2} + k_2^2\right)\psi_2(x) = 0 \qquad \left(-\frac{1}{2}a \leq x \leq \frac{1}{2}a\right), \tag{4.91}$$

$$\left(\frac{d^2}{dx^2} - k_1^2\right)\psi_3(x) = 0 \qquad \left(x > \frac{1}{2}a\right), \tag{4.92}$$

where $k_1^2 = 2m(V_0 - E)/\hbar^2$ and $k_2^2 = 2mE/\hbar^2$. Eliminating the physically unacceptable solutions which grow exponentially for large values of $|x|$, we can write the solution to this Schrödinger equation in the regions $x < -a/2$ and $x > a/2$ as follows:

$$\psi_1(x) = Ae^{k_1 x} \qquad \left(x < -\frac{1}{2}a\right), \tag{4.93}$$

$$\psi_3(x) = De^{-k_1 x} \qquad \left(x > \frac{1}{2}a\right). \tag{4.94}$$

As mentioned in (4.4), since the bound state eigenfunctions of symmetric one-dimensional Hamiltonians are either even or odd under space inversion, the solutions of (4.90) to (4.92) are then either antisymmetric (odd)

$$\psi_a(x) = \begin{cases} Ae^{k_1 x}, & x < -a/2, \\ C\sin(k_2 x), & -a/2 \le x \le a/2, \\ De^{-k_1 x}, & x > a/2, \end{cases} \tag{4.95}$$

or symmetric (even)

$$\psi_s(x) = \begin{cases} Ae^{k_1 x}, & x < -a/2, \\ B\cos(k_2 x), & -a/2 \le x \le a/2, \\ De^{-k_1 x}, & x > a/2. \end{cases} \tag{4.96}$$

To determine the eigenvalues, we need to use the continuity conditions at $x = \pm a/2$. The continuity of the logarithmic derivative, $(1/\psi_a(x))d\psi_a(x)/dx$, of $\psi_a(x)$ at $x = \pm a/2$ yields

$$k_2 \cot\left(\frac{k_2 a}{2}\right) = -k_1. \tag{4.97}$$

Similarly, the continuity of $(1/\psi_s(x))d\psi_s(x)/dx$ at $x = \pm a/2$ gives

$$k_2 \tan\left(\frac{k_2 a}{2}\right) = k_1. \tag{4.98}$$

The transcendental equations (4.97) and (4.98) cannot be solved directly; we can solve them either graphically or numerically. To solve these equations graphically, we need only to rewrite them in the following suggestive forms:

$$-\alpha_n \cot\alpha_n = \sqrt{R^2 - \alpha_n^2} \qquad \text{(for odd states)}, \tag{4.99}$$

$$\alpha_n \tan\alpha_n = \sqrt{R^2 - \alpha_n^2} \qquad \text{(for even states)}, \tag{4.100}$$

where $\alpha_n^2 = (k_2 a/2)^2 = ma^2 E_n/(2\hbar^2)$ and $R^2 = ma^2 V_0/(2\hbar^2)$; these equations are obtained by inserting $k_1 = \sqrt{2m(V_0 - E)/\hbar^2}$ and $k_2 = \sqrt{2mE/\hbar^2}$ into (4.97) and (4.98). The left-hand sides of (4.99) and (4.100) consist of trigonometric functions; the right-hand sides consist of a circle of radius R. The solutions are given by the points where the circle $\sqrt{R^2 - \alpha_n^2}$ intersects the functions $-\alpha_n \cot\alpha_n$ and $\alpha_n \tan\alpha_n$ (Figure 4.8). The solutions form a *discrete* set. As illustrated in Figure 4.8, the intersection of the small circle with the curve $\alpha_n \tan\alpha_n$ yields only one bound state, $n = 0$, whereas the intersection of the larger circle with $\alpha_n \tan\alpha_n$ yields two

bound states, $n = 0, 2$, and its intersection with $-\alpha_n \cot \alpha_n$ yields two other bound states, $n = 1,\ 3$.

The number of solutions depends on the size of R, which in turn depends on the depth V_0 and the width a of the well, since $R = \sqrt{ma^2 V_0/(2\hbar^2)}$. The deeper and broader the well, the larger the value of R, and hence the greater the number of bound states. Note that there is always at least one bound state (i.e., one intersection) no matter how small V_0 is. When

$$0 < R < \frac{\pi}{2} \qquad \text{or} \qquad 0 < V_0 < \left(\frac{\pi}{2}\right)^2 \frac{2\hbar^2}{ma^2}, \tag{4.101}$$

there is only one bound state corresponding to $n = 0$ (see Figure 4.8); this state—the ground state—is even. Then, and when

$$\frac{\pi}{2} < R < \pi \qquad \text{or} \qquad \left(\frac{\pi}{2}\right)^2 \frac{2\hbar^2}{ma^2} < V_0 < \pi^2 \frac{2\hbar^2}{ma^2}, \tag{4.102}$$

there are two bound states: an even state (the ground state) corresponding to $n = 0$ and the first odd state corresponding to $n = 1$. Now, if

$$\pi < R < \frac{3\pi}{2} \qquad \text{or} \qquad \pi^2 \frac{2\hbar^2}{ma^2} < V_0 < \left(\frac{3\pi}{2}\right)^2 \frac{2\hbar^2}{ma^2}, \tag{4.103}$$

there exist three bound states: the ground state (even state), $n = 0$, the first excited state (odd state), corresponding to $n = 1$, and the second excited state (even state), which corresponds to $n = 2$. In general, the well width at which n states are allowed is given by

$$R = \frac{n\pi}{2} \qquad \text{or} \qquad V_0 = \left(\frac{\pi}{2}\right)^2 \frac{2\hbar^2}{ma^2} n^2. \tag{4.104}$$

The spectrum, therefore, consists of a set of alternating even and odd states: the lowest state, the ground state, is even, the next state (first excited sate) is odd, and so on.

In the limiting case $V_0 \to \infty$, the circle's radius R also becomes infinite, and hence the function $\sqrt{R^2 - \alpha_n^2}$ will cross $-\alpha_n \cot \alpha_n$ and $\alpha_n \tan \alpha_n$ at the asymptotes $\alpha_n = n\pi/2$, because when $V_0 \to \infty$ both $\tan \alpha_n$ and $\cot \alpha_n$ become infinite:

$$\tan \alpha_n \to \infty \quad \Longrightarrow \quad \alpha_n = \frac{2n+1}{2}\pi \qquad (n = 0, 1, 2, 3, \cdots), \tag{4.105}$$

$$\cot \alpha_n \to \infty \quad \Longrightarrow \quad \alpha_n = n\pi \qquad (n = 1, 2, 3, \cdots). \tag{4.106}$$

Combining these two cases, we obtain

$$\alpha_n = \frac{n\pi}{2} \qquad (1, 2, 3, \ldots). \tag{4.107}$$

Since $\alpha_n^2 = ma^2 E_n/(2\hbar^2)$ we see that we recover the energy expression for the infinite well:

$$\alpha_n = \frac{n\pi}{2} \longrightarrow E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2. \tag{4.108}$$

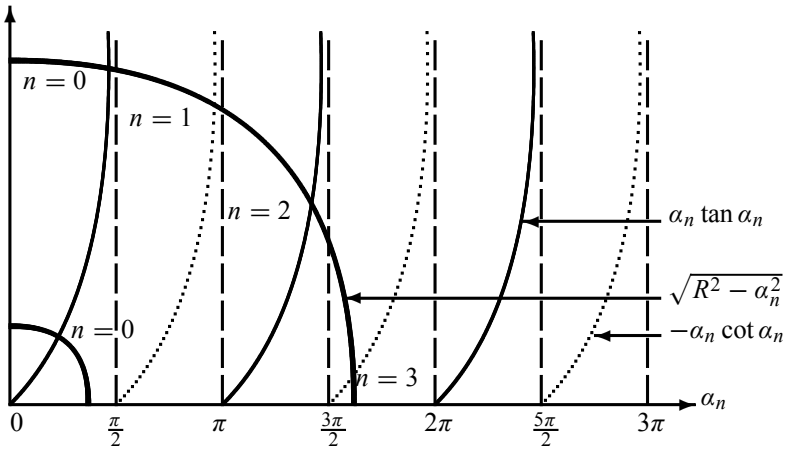


Figure 4.8 Graphical solutions for the finite square well potential: they are given by the intersections of $\sqrt{R^2-\alpha_n^2}$ with $\alpha_n \tan\alpha_n$ and $-\alpha_n \cot\alpha_n$, where $\alpha_n^2 = ma^2E_n/(2\hbar^2)$ and $R^2 = ma^2V_0/(2\hbar^2)$.

Example 4.2
Find the number of bound states and the corresponding energies for the finite square well potential when: (a) $R = 1$ (i.e., $\sqrt{ma^2V_0/(2\hbar^2)} = 1$), and (b) $R = 2$.

Solution

(a) From Figure 4.8, when $R = \sqrt{ma^2V_0/(2\hbar^2)} = 1$, there is only one bound state since $\alpha_n \leq R$. This bound state corresponds to $n = 0$. The corresponding energy is given by the intersection of $\alpha_0 \tan\alpha_0$ with $\sqrt{1-\alpha_0^2}$:

$$\alpha_0 \tan\alpha_0 = \sqrt{1-\alpha_0^2} \quad \Longrightarrow \quad \alpha_0^2(1+\tan^2\alpha_0) = 1 \quad \Longrightarrow \quad \cos^2\alpha_0 = \alpha_0^2. \tag{4.109}$$

The solution of $\cos^2\alpha_0 = \alpha_0^2$ is given numerically by $\alpha_0 = 0.739\,09$. Thus, the corresponding energy is determined by the relation $\sqrt{ma^2E_0/(2\hbar^2)} = 0.739\,09$, which yields $E_0 \simeq 1.1\hbar^2/(ma^2)$.

(b) When $R = 2$ there are two bound states resulting from the intersections of $\sqrt{4-\alpha_0^2}$ with $\alpha_0 \tan\alpha_0$ and $-\alpha_1 \cot\alpha_1$; they correspond to $n = 0$ and $n = 1$, respectively. The numerical solutions of the corresponding equations

$$\alpha_0 \tan\alpha_0 = \sqrt{4-\alpha_0^2} \quad \Longrightarrow \quad 4\cos^2\alpha_0 = \alpha_0^2, \tag{4.110}$$

$$-\alpha_1 \cot\alpha_1 = \sqrt{4-\alpha_1^2} \quad \Longrightarrow \quad 4\sin^2\alpha_1 = \alpha_1^2, \tag{4.111}$$

yield $\alpha_0 \simeq 1.03$ and $\alpha_1 \simeq 1.9$, respectively. The corresponding energies are

$$\alpha_0 = \sqrt{\frac{ma^2E_0}{2\hbar^2}} \simeq 1.03 \quad \Longrightarrow \quad E_0 \simeq \frac{2.12\hbar^2}{ma^2}, \tag{4.112}$$

$$\alpha_1 = \sqrt{\frac{ma^2E_1}{2\hbar^2}} \simeq 1.9 \quad \Longrightarrow \quad E_1 \simeq \frac{7.22\hbar^2}{ma^2}. \tag{4.113}$$

4.8 The Harmonic Oscillator

The harmonic oscillator is one of those few problems that are important to all branches of physics. It provides a useful model for a variety of vibrational phenomena that are encountered, for instance, in classical mechanics, electrodynamics, statistical mechanics, solid state, atomic, nuclear, and particle physics. In quantum mechanics, it serves as an invaluable tool to illustrate the basic concepts and the formalism.

The Hamiltonian of a particle of mass m which oscillates with an angular frequency ω under the influence of a one-dimensional harmonic potential is

$$\hat{H} = \frac{\hat{P}^2}{2m} + \frac{1}{2}m\omega^2\hat{X}^2. \tag{4.114}$$

The problem is how to find the energy eigenvalues and eigenstates of this Hamiltonian. This problem can be studied by means of two separate methods. The first method, called the *analytic method*, consists in solving the time-independent Schrödinger equation (TISE) for the Hamiltonian (4.114). The second method, called the *ladder* or *algebraic method*, does not deal with solving the Schrödinger equation, but deals instead with operator algebra involving operators known as the *creation* and *annihilation* or *ladder* operators; this method is in essence a matrix formulation, because it expresses the various quantities in terms of matrices. In our presentation, we are going to adopt the second method, for it is more straightforward, more elegant and much simpler than solving the Schrödinger equation. Unlike the examples seen up to now, solving the Schrödinger equation for the potential $V(x) = \frac{1}{2}m\omega x^2$ is no easy job. Before embarking on the second method, let us highlight the main steps involved in the first method.

Brief outline of the analytic method

This approach consists in using the power series method to solve the following differential (Schrödinger) equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(x) = E\psi(x), \tag{4.115}$$

which can be reduced to

$$\frac{d^2\psi(x)}{dx^2} + \left(\frac{2mE}{\hbar^2} - \frac{x^2}{x_0^4}\right)\psi(x) = 0, \tag{4.116}$$

where $x_0 = \sqrt{\hbar/(m\omega)}$ is a constant that has the dimensions of length; it sets the length scale of the oscillator, as will be seen later. The solutions of differential equations like (4.116) have been worked out by our mathematician colleagues well before the arrival of quantum mechanics (the solutions are expressed in terms of some special functions, the Hermite polynomials). The occurrence of the term $x^2\psi(x)$ in (4.116) suggests trying a Gaussian type solution[3]: $\psi(x) =$

[3] Solutions of the type $\psi(x) = f(x)\exp(x^2/2x_0^2)$ are physically unacceptable, for they diverge when $x \longrightarrow \pm\infty$.

$f(x)\exp(-x^2/2x_0^2)$, where $f(x)$ is a function of x. Inserting this trial function into (4.116), we obtain a differential equation for $f(x)$. This new differential equation can be solved by expanding $f(x)$ out in a power series (i.e., $f(x) = \sum_{n=0}^{\infty} a_n x^n$, where a_n is just a coefficient), which when inserted into the differential equation leads to a recursion relation. By demanding the power series of $f(x)$ to terminate at some finite value of n (because the wave function $\psi(x)$ has to be finite everywhere, notably when $x \longrightarrow \pm\infty$), the recursion relation yields an expression for the energy eigenvalues which are discrete or quantized:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \qquad (n = 0, 1, 2, 3, \ \ldots). \tag{4.117}$$

After some calculations, we can show that the wave functions that are physically acceptable and that satisfy (4.116) are given by

$$\psi_n(x) = \frac{1}{\sqrt{\sqrt{\pi}2^n n! x_0}} e^{-x^2/2x_0^2} H_n\left(\frac{x}{x_0}\right), \tag{4.118}$$

where $H_n(y)$ are nth order polynomials called *Hermite polynomials*:

$$H_n(y) = (-1)^n e^{y^2} \frac{d^n}{dy^n} e^{-y^2}. \tag{4.119}$$

From this relation it is easy to calculate the first few polynomials:

$$\begin{aligned} &H_0(y) = 1, && H_1(y) = 2y, \\ &H_2(y) = 4y^2 - 2, && H_3(y) = 8y^3 - 12y, \\ &H_4(y) = 16y^4 - 48y^2 + 12, && H_5(y) = 32y^5 - 160y^3 + 120y. \end{aligned} \tag{4.120}$$

We will deal with the physical interpretations of the harmonic oscillator results when we study the second method.

Algebraic method

Let us now show how to solve the harmonic oscillator eigenvalue problem using the algebraic method. For this, we need to rewrite the Hamiltonian (4.114) in terms of the two Hermitian, dimensionless operators $\hat{p} = \hat{P}/\sqrt{m\hbar\omega}$ and $\hat{q} = \hat{X}\sqrt{m\omega/\hbar}$:

$$\hat{H} = \frac{\hbar\omega}{2}(\hat{p}^2 + \hat{q}^2), \tag{4.121}$$

and then introduce two non-Hermitian, dimensionless operators:

$$\hat{a} = \frac{1}{\sqrt{2}}(\hat{q} + i\,\hat{p}), \qquad \hat{a}^\dagger = \frac{1}{\sqrt{2}}(\hat{q} - i\,\hat{p}). \tag{4.122}$$

The physical meaning of the operators $\hat{a}$ and $\hat{a}^\dagger$ will be examined later. Note that

$$\hat{a}^\dagger\hat{a} = \frac{1}{2}(\hat{q} - i\,\hat{p})(\hat{q} + i\,\hat{p}) = \frac{1}{2}(\hat{q}^2 + \hat{p}^2 + i\hat{q}\,\hat{p} - i\,\hat{p}\hat{q}) = \frac{1}{2}(\hat{q}^2 + \hat{p}^2) + \frac{i}{2}[\hat{q},\ \hat{p}], \tag{4.123}$$

where, using $[\hat{X}, \hat{P}] = i\hbar$, we can verify that the commutator between $\hat{q}$ and $\hat{p}$ is

$$[\hat{q},\ \hat{p}] = \left[\sqrt{\frac{m\omega}{\hbar}}\hat{X}, \frac{1}{\sqrt{\hbar m\omega}}\hat{P}\right] = \frac{1}{\hbar}\left[\hat{X}, \hat{P}\right] = i; \tag{4.124}$$

hence

$$\hat{a}^\dagger \hat{a} = \frac{1}{2}(\hat{q}^2 + \hat{p}^2) - \frac{1}{2} \tag{4.125}$$

or

$$\frac{1}{2}(\hat{q}^2 + \hat{p}^2) = \hat{a}^\dagger \hat{a} + \frac{1}{2}. \tag{4.126}$$

Inserting (4.126) into (4.121) we obtain

$$\boxed{\hat{H} = \hbar\omega\left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right) = \hbar\omega\left(\hat{N} + \frac{1}{2}\right) \qquad \text{with} \quad \hat{N} = \hat{a}^\dagger \hat{a},} \tag{4.127}$$

where $\hat{N}$ is known as the *number* operator or *occupation number* operator, which is clearly Hermitian.

Let us now derive the commutator $[\hat{a}, \hat{a}^\dagger]$. Since $[\hat{X}, \hat{P}] = i\hbar$ we have $[\hat{q}, \hat{p}] = \frac{1}{\hbar}[\hat{X}, \hat{P}] = i$; hence

$$[\hat{a}, \hat{a}^\dagger] = \frac{1}{2}\left[\hat{q} + i\hat{p}, \hat{q} - i\hat{p}\right] = -i\left[\hat{q}, \hat{p}\right] = 1 \tag{4.128}$$

or

$$\boxed{[\hat{a}, \hat{a}^\dagger] = 1.} \tag{4.129}$$

4.8.1 Energy Eigenvalues

Note that $\hat{H}$ as given by (4.127) commutes with $\hat{N}$, since $\hat{H}$ is linear in $\hat{N}$. Thus, $\hat{H}$ and $\hat{N}$ can have a set of joint eigenstates, to be denoted by $\mid n\rangle$:

$$\hat{N} \mid n\rangle = n \mid n\rangle \tag{4.130}$$

and

$$\hat{H} \mid n\rangle = E_n \mid n\rangle; \tag{4.131}$$

the states $\mid n\rangle$ are called energy eigenstates. Combining (4.127) and (4.131), we obtain the energy eigenvalues at once:

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega.} \tag{4.132}$$

We will show later that n is a *positive integer*; it cannot have negative values.

The physical meaning of the operators $\hat{a}$, $\hat{a}^\dagger$, and $\hat{N}$ can now be clarified. First, we need the following two commutators that can be extracted from (4.129) and (4.127):

$$\boxed{[\hat{a}, \hat{H}] = \hbar\omega\hat{a}, \qquad [\hat{a}^\dagger, \hat{H}] = -\hbar\omega\hat{a}^\dagger.} \tag{4.133}$$

These commutation relations along with (4.131) lead to

$$\hat{H}\left(\hat{a} \mid n\rangle\right) \quad = (\hat{a}\hat{H} - \hbar\omega\hat{a}) \mid n\rangle \quad = (E_n - \hbar\omega)\left(\hat{a} \mid n\rangle\right), \tag{4.134}$$

$$\hat{H}\left(\hat{a}^\dagger \mid n\rangle\right) \quad = (\hat{a}^\dagger\hat{H} + \hbar\omega\hat{a}^\dagger) \mid n\rangle \; = (E_n + \hbar\omega)\left(\hat{a}^\dagger \mid n\rangle\right). \tag{4.135}$$

Thus, $\hat{a} \mid n\rangle$ and $\hat{a}^\dagger \mid n\rangle$ are eigenstates of $\hat{H}$ with eigenvalues $(E_n - \hbar\omega)$ and $(E_n + \hbar\omega)$, respectively. So the actions of $\hat{a}$ and $\hat{a}^\dagger$ on $\mid n\rangle$ generate new energy states that are lower and

higher by one unit of $\hbar\omega$, respectively. As a result, $\hat{a}$ and $\hat{a}^\dagger$ are respectively known as the *lowering* and *raising* operators, or the *annihilation* and *creation* operators; they are also known as the *ladder* operators.

Let us now find out how the operators $\hat{a}$ and $\hat{a}^\dagger$ act on the energy eigenstates $|\, n\rangle$. Since $\hat{a}$ and $\hat{a}^\dagger$ do not commute with $\hat{N}$, the states $|\, n\rangle$ are eigenstates neither to $\hat{a}$ nor to $\hat{a}^\dagger$. Using (4.129) along with $[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$, we can show that

$$\boxed{[\hat{N}, \hat{a}] = -\hat{a}, \qquad [\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger;} \tag{4.136}$$

hence $\hat{N}\hat{a} = \hat{a}(\hat{N} - 1)$ and $\hat{N}\hat{a}^\dagger = \hat{a}^\dagger(\hat{N} + 1)$. Combining these relations with (4.130), we obtain

$$\hat{N}\left(\hat{a} \mid n\rangle\right) = \hat{a}(\hat{N} - 1) \mid n\rangle = (n - 1)\left(\hat{a} \mid n\rangle\right), \tag{4.137}$$

$$\hat{N}\left(\hat{a}^\dagger \mid n\rangle\right) = \hat{a}^\dagger(\hat{N} + 1) \mid n\rangle = (n + 1)(\hat{a}^\dagger \mid n\rangle). \tag{4.138}$$

These relations reveal that $\hat{a} \mid n\rangle$ and $\hat{a}^\dagger \mid n\rangle$ are eigenstates of $\hat{N}$ with eigenvalues $(n - 1)$ and $(n + 1)$, respectively. This implies that when $\hat{a}$ and $\hat{a}^\dagger$ operate on $|\, n\rangle$, respectively, they decrease and increase n by one unit. That is, while the action of $\hat{a}$ on $|\, n\rangle$ generates a new state $|\, n - 1\rangle$ (i.e., $\hat{a} \mid n\rangle \sim |\, n - 1\rangle$), the action of $\hat{a}^\dagger$ on $|\, n\rangle$ generates $|\, n + 1\rangle$. Hence from (4.137) we can write

$$\hat{a} \mid n\rangle = c_n \mid n - 1\rangle, \tag{4.139}$$

where c_n is a constant to be determined from the requirement that the states $|\, n\rangle$ be normalized for all values of n. On the one hand, (4.139) yields

$$\left(\langle n \mid \hat{a}^\dagger\right) \cdot \left(\hat{a} \mid n\rangle\right) = \langle n \mid \hat{a}^\dagger\hat{a} \mid n\rangle = |c_n|^2 \langle n - 1 \mid n - 1\rangle = |c_n|^2 \tag{4.140}$$

and, on the other hand, (4.130) gives

$$\left(\langle n \mid \hat{a}^\dagger\right) \cdot \left(\hat{a} \mid n\rangle\right) = \langle n \mid \hat{a}^\dagger\hat{a} \mid n\rangle = n\langle n \mid n\rangle = n. \tag{4.141}$$

When combined, the last two equations yield

$$|c_n|^2 = n. \tag{4.142}$$

This implies that n, which is equal to the norm of $\hat{a} \mid n\rangle$ (see (4.141)), *cannot be negative*, $n \geq 0$, since the norm is a positive quantity. Substituting (4.142) into (4.139) we end up with

$$\boxed{\hat{a} \mid n\rangle = \sqrt{n} \mid n - 1\rangle.} \tag{4.143}$$

This equation shows that repeated applications of the operator $\hat{a}$ on $|\, n\rangle$ generate a sequence of eigenvectors $|\, n - 1\rangle$, $|\, n - 2\rangle$, $|\, n - 3\rangle$, ... Since $n \geq 0$ and since $\hat{a} \mid 0\rangle = 0$, this sequence has to terminate at $n = 0$; this is true if we start with an integer value of n. But if we start with a noninteger n, the sequence will not terminate; hence it leads to eigenvectors with negative values of n. But as shown above, since n cannot be negative, we conclude that n has to be a *nonnegative integer*.

Now, we can easily show, as we did for (4.143), that

$$\boxed{\hat{a}^\dagger \mid n\rangle = \sqrt{n+1} \mid n+1\rangle.} \tag{4.144}$$

This implies that repeated applications of $\hat{a}^\dagger$ on $\mid n\rangle$ generate an infinite sequence of eigenvectors $\mid n+1\rangle$, $\mid n+2\rangle$, $\mid n+3\rangle$, ... Since n is a positive integer, the energy spectrum of a harmonic oscillator as specified by (4.132) is therefore *discrete*:

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega \qquad (n = 0, 1, 2, 3, \ldots).} \tag{4.145}$$

This expression is similar to the one obtained from the first method (see Eq. (4.117)). The energy spectrum of the harmonic oscillator consists of energy levels that are equally spaced: $E_{n+1} - E_n = \hbar\omega$. This is Planck's famous equidistant energy idea—the energy of the radiation emitted by the oscillating charges (from the inside walls of the cavity) must come only in bundles (quanta) that are integral multiples of $\hbar\omega$—which, as mentioned in Chapter 1, led to the birth of quantum mechanics.

As expected for bound states of one-dimensional potentials, the energy spectrum is both discrete and nondegenerate. Once again, as in the case of the infinite square well potential, we encounter the zero-point energy phenomenon: the lowest energy eigenvalue of the oscillator is not zero but is instead equal to $E_0 = \hbar\omega/2$. It is called the *zero-point energy* of the oscillator, for it corresponds to $n = 0$. The zero-point energy of bound state systems cannot be zero, otherwise it would violate the uncertainty principle. For the harmonic oscillator, for instance, the classical minimum energy corresponds to $x = 0$ and $p = 0$; there would be no oscillations in this case. This would imply that we know simultaneously and with absolute precision both the position and the momentum of the system. This would contradict the uncertainty principle.

4.8.2 Energy Eigenstates

The algebraic or operator method can also be used to determine the energy eigenvectors. First, using (4.144), we see that the various eigenvectors can be written in terms of the ground state $\mid 0\rangle$ as follows:

$$\mid 1\rangle = \hat{a}^\dagger \mid 0\rangle, \tag{4.146}$$

$$\mid 2\rangle = \frac{1}{\sqrt{2}}\hat{a}^\dagger \mid 1\rangle = \frac{1}{\sqrt{2!}}\left(\hat{a}^\dagger\right)^2 \mid 0\rangle, \tag{4.147}$$

$$\mid 3\rangle = \frac{1}{\sqrt{3}}\hat{a}^\dagger \mid 2\rangle = \frac{1}{\sqrt{3!}}\left(\hat{a}^\dagger\right)^3 \mid 0\rangle, \tag{4.148}$$

$$\vdots$$

$$\mid n\rangle = \frac{1}{\sqrt{n}}\hat{a}^\dagger \mid n-1\rangle = \frac{1}{\sqrt{n!}}\left(\hat{a}^\dagger\right)^n \mid 0\rangle. \tag{4.149}$$

So, to find any excited eigenstate $\mid n\rangle$, we need simply to operate $\hat{a}^\dagger$ on $\mid 0\rangle$ n successive times.

Note that any set of kets $\mid n\rangle$ and $\mid n'\rangle$, corresponding to different eigenvalues, must be orthogonal, $\langle n' \mid n\rangle \sim \delta_{n',n}$, since $\hat{H}$ is Hermitian and none of its eigenstates is degenerate.

Moreover, the states $\mid 0\rangle, \mid 1\rangle, \mid 2\rangle, \mid 3\rangle, \ldots, \mid n\rangle, \ldots$ are *simultaneous eigenstates* of $\hat{H}$ and $\hat{N}$; the set $\{\mid n\rangle\}$ constitutes an orthonormal and complete basis:

$$\langle n' \mid n\rangle = \delta_{n',n}, \qquad \sum_{n=0}^{+\infty} \mid n\rangle\langle n \mid = 1. \tag{4.150}$$

4.8.3 Energy Eigenstates in Position Space

Let us now determine the harmonic oscillator wave function in the position representation.

Equations (4.146) to (4.149) show that, knowing the ground state wave function, we can determine any other eigenstate by successive applications of the operator $a^\dagger$ on the ground state. So let us first determine the ground state wave function in the position representation.

The operator $\hat{p}$, defined by $\hat{p} = \hat{P}/\sqrt{m\hbar\omega}$, is given in the position space by

$$\hat{p} = -\frac{i\hbar}{\sqrt{m\hbar\omega}}\frac{d}{dx} = -ix_0\frac{d}{dx}, \tag{4.151}$$

where, as mentioned above, $x_0 = \sqrt{\hbar/(m\omega)}$ is a constant that has the dimensions of length; it sets the length scale of the oscillator. We can easily show that the annihilation and creation operators $\hat{a}$ and $\hat{a}^\dagger$, defined in (4.122), can be written in the position representation as

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\frac{\hat{X}}{x_0} + x_0\frac{d}{dx}\right) = \frac{1}{\sqrt{2}x_0}\left(\hat{X} + x_0^2\frac{d}{dx}\right), \tag{4.152}$$

$$\hat{a}^\dagger = \frac{1}{\sqrt{2}}\left(\frac{\hat{X}}{x_0} - x_0\frac{d}{dx}\right) = \frac{1}{\sqrt{2}x_0}\left(\hat{X} - x_0^2\frac{d}{dx}\right). \tag{4.153}$$

Using (4.152) we can write the equation $\hat{a} \mid 0\rangle = 0$ in the position space as

$$\langle x|\hat{a} \mid 0\rangle = \frac{1}{\sqrt{2}x_0}\langle x|\hat{X} + x_0^2\frac{d}{dx} \mid 0\rangle = \frac{1}{\sqrt{2}x_0}\left(x\,\psi_0(x) + x_0^2\frac{d\psi_0(x)}{dx}\right) = 0; \tag{4.154}$$

hence

$$\frac{d\psi_0(x)}{dx} = -\frac{x}{x_0^2}\psi_0(x), \tag{4.155}$$

where $\psi_0(x) = \langle x \mid 0\rangle$ represents the ground state wave function. The solution of this differential equation is

$$\psi_0(x) = A\exp\left(-\frac{x^2}{2x_0^2}\right), \tag{4.156}$$

where A is a constant that can be determined from the normalization condition

$$1 = \int_{-\infty}^{+\infty} dx\,|\psi_0(x)|^2 = A^2\int_{-\infty}^{+\infty} dx\exp\left(-\frac{x^2}{x_0^2}\right) = A^2\sqrt{\pi}x_0; \tag{4.157}$$

hence $A = (m\omega/(\pi\hbar))^{1/4} = 1/\sqrt{\sqrt{\pi}x_0}$. The normalized ground state wave function is then given by

$$\boxed{\psi_0(x) = \frac{1}{\sqrt{\sqrt{\pi}x_0}} \exp\left(-\frac{x^2}{2x_0^2}\right).} \tag{4.158}$$

This is a Gaussian function.

We can then obtain the wave function of any excited state by a series of applications of $\hat{a}^\dagger$ on the ground state. For instance, the first excited state is obtained by one single application of the operator $\hat{a}^\dagger$ of (4.153) on the ground state:

$$\begin{aligned} \langle x \mid 1\rangle &= \langle x|\hat{a}^\dagger \mid 0\rangle = \frac{1}{\sqrt{2}x_0}\left(x - x_0^2\frac{d}{dx}\right)\langle x \mid 0\rangle \\ &= \frac{1}{\sqrt{2}x_0}\left(x - x_0^2\left(-\frac{x}{x_0^2}\right)\right)\psi_0(x) = \frac{\sqrt{2}}{x_0}x\,\psi_0(x) \end{aligned} \tag{4.159}$$

or

$$\psi_1(x) = \frac{\sqrt{2}}{x_0}x\,\psi_0(x) = \sqrt{\frac{2}{\sqrt{\pi}x_0^3}}x\exp\left(-\frac{x^2}{2x_0^2}\right). \tag{4.160}$$

As for the eigenstates of the second and third excited states, we can obtain them by applying $\hat{a}^\dagger$ on the ground state twice and three times, respectively:

$$\langle x \mid 2\rangle = \frac{1}{\sqrt{2!}}\langle x|\left(a^\dagger\right)^2 \mid 0\rangle = \frac{1}{\sqrt{2!}}\left(\frac{1}{\sqrt{2}x_0}\right)^2\left(x - x_0^2\frac{d}{dx}\right)^2\psi_0(x), \tag{4.161}$$

$$\langle x \mid 3\rangle = \frac{1}{\sqrt{3!}}\langle x|\left(a^\dagger\right)^3 \mid 0\rangle = \frac{1}{\sqrt{3!}}\left(\frac{1}{\sqrt{2}x_0}\right)^3\left(x - x_0^2\frac{d}{dx}\right)^3\psi_0(x) \tag{4.162}$$

or

$$\psi_2(x) = \frac{1}{\sqrt{2\sqrt{\pi}x_0}}\left(\frac{2x^2}{x_0^2} - 1\right)\exp\left(-\frac{x^2}{2x_0^2}\right), \quad \psi_3(x) = \frac{1}{\sqrt{3\sqrt{\pi}x_0}}\left(\frac{2x^3}{x_0^3} - \frac{3x}{x_0}\right)\exp\left(-\frac{x^2}{2x_0^2}\right). \tag{4.163}$$

Similarly, using (4.149), (4.153), and (4.158), we can easily infer the energy eigenstate for the nth excited state:

$$\langle x \mid n\rangle = \frac{1}{\sqrt{n!}}\langle x|\left(a^\dagger\right)^n \mid 0\rangle = \frac{1}{\sqrt{n!}}\left(\frac{1}{\sqrt{2}x_0}\right)^n\left(x - x_0^2\frac{d}{dx}\right)^n\psi_0(x), \tag{4.164}$$

which in turn can be rewritten as

$$\boxed{\psi_n(x) = \frac{1}{\sqrt{\sqrt{\pi}2^n n!}}\frac{1}{x_0^{n+1/2}}\left(x - x_0^2\frac{d}{dx}\right)^n\exp\left(-\frac{x^2}{2x_0^2}\right).} \tag{4.165}$$

In summary, by successive applications of $\hat{a}^\dagger = (\hat{X} - x_0^2 d/dx)/(\sqrt{2}x_0)$ on $\psi_0(x)$, we can find the wave function of any excited state $\psi_n(x)$.

Oscillator wave functions and the Hermite polynomials

At this level, we can show that the wave function (4.165) derived from the algebraic method is similar to the one obtained from the first method (4.118). To see this, we simply need to use the following operator identity:

$$e^{-x^2/2}\left(x-\frac{d}{dx}\right)e^{x^2/2}=-\frac{d}{dx}\quad\text{or}\quad e^{-x^2/2x_0^2}\left(x-x_0^2\frac{d}{dx}\right)e^{x^2/2x_0^2}=-x_0^2\frac{d}{dx}. \tag{4.166}$$

An application of this operator n times leads at once to

$$e^{-x^2/2x_0^2}\left(x-x_0^2\frac{d}{dx}\right)^n e^{x^2/2x_0^2}=(-1)^n(x_0^2)^n\frac{d^n}{dx^n}, \tag{4.167}$$

which can be shown to yield

$$\left(x-x_0^2\frac{d}{dx}\right)^n e^{-x^2/2x_0^2}=(-1)^n(x_0^2)^n e^{x^2/2x_0^2}\frac{d^n}{dx^n}e^{-x^2/x_0^2}. \tag{4.168}$$

We can now rewrite the right-hand side of this equation as follows:

$$\begin{aligned}(-1)^n(x_0^2)^n e^{x^2/2x_0^2}\frac{d^n}{dx^n}e^{-x^2/x_0^2} &= x_0^n e^{-x^2/2x_0^2}\left[(-1)^n e^{x^2/x_0^2}\frac{d^n}{d(x/x_0)^n}e^{-x^2/x_0^2}\right]\\ &= x_0^n e^{-x^2/2x_0^2}\left[(-1)^n e^{y^2}\frac{d^n}{dy^n}e^{-y^2}\right]\\ &= x_0^n e^{-x^2/2x_0^2}H_n(y),\end{aligned} \tag{4.169}$$

where $y=x/x_0$ and where $H_n(y)$ are the Hermite polynomials listed in (4.119):

$$H_n(y)=(-1)^n e^{y^2}\frac{d^n}{dy^n}e^{-y^2}. \tag{4.170}$$

Note that the polynomials $H_{2n}(y)$ are even and $H_{2n+1}(y)$ are odd, since $H_n(-y)=(-1)^n H_n(y)$.

Inserting (4.169) into (4.168), we obtain

$$\left(x-x_0^2\frac{d}{dx}\right)^n e^{-x^2/2x_0^2}=x_0^n e^{-x^2/2x_0^2}H_n\left(\frac{x}{x_0}\right); \tag{4.171}$$

substituting this equation into (4.165), we can write the oscillator wave function in terms of the Hermite polynomials as follows:

$$\boxed{\psi_n(x)=\frac{1}{\sqrt{\sqrt{\pi}2^n n!x_0}}e^{-x^2/2x_0^2}H_n\left(\frac{x}{x_0}\right).} \tag{4.172}$$

This wave function is identical with the one obtained from the first method (see Eq. (4.118)).

Remark

This wave function is either even or odd depending on n; in fact, the functions $\psi_{2n}(x)$ are even (i.e., $\psi_{2n}(-x)=\psi_{2n}(x)$) and $\psi_{2n+1}(x)$ are odd (i.e., $\psi_{2n}(-x)=-\psi_{2n}(x)$) since, as can be inferred from Eq (4.120), the Hermite polynomials $H_{2n}(x)$ are even and $H_{2n+1}(x)$ are odd. This is expected because, as mentioned in Section 4.2.4, the wave functions of even one-dimensional potentials have definite parity. Figure 4.9 displays the shapes of the first few wave functions.

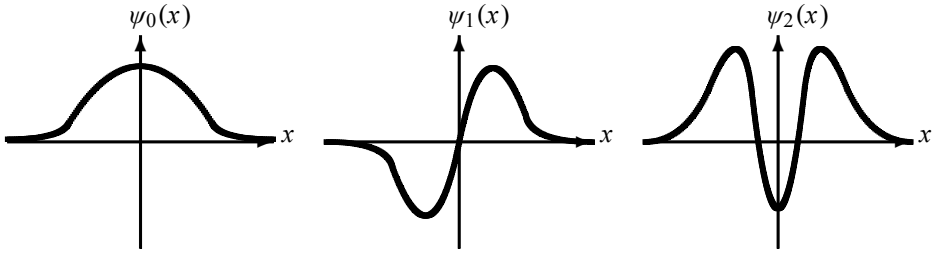


Figure 4.9 Shapes of the first three wave functions of the harmonic oscillator.

4.8.4 The Matrix Representation of Various Operators

Here we look at the matrix representation of several operators in the N-space. In particular, we focus on the representation of the operators $\hat{a}$, $\hat{a}^\dagger$, $\hat{X}$, and $\hat{P}$. First, since the states $|\, n\rangle$ are joint eigenstates of $\hat{H}$ and $\hat{N}$, it is easy to see from (4.130) and (4.132) that $\hat{H}$ and $\hat{N}$ are represented within the $\{|\, n\rangle\}$ basis by infinite diagonal matrices:

$$\langle n'|\hat{N} \mid n\rangle = n\delta_{n',n}, \qquad \langle n'|\hat{H} \mid n\rangle = \hbar\omega\left(n + \frac{1}{2}\right)\delta_{n',n}; \tag{4.173}$$

that is,

$$\hat{N} = \begin{pmatrix} 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & 2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad \hat{H} = \frac{\hbar\omega}{2}\begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 3 & 0 & \cdots \\ 0 & 0 & 5 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{4.174}$$

As for the operators $\hat{a}$, $\hat{a}^\dagger$, $\hat{X}$, $\hat{P}$, none of them are diagonal in the N-representation, since they do not commute with $\hat{N}$. The matrix elements of $\hat{a}$ and $\hat{a}^\dagger$ can be obtained from (4.143) and (4.144):

$$\langle n'|\hat{a} \mid n\rangle = \sqrt{n}\delta_{n',n-1}, \qquad \langle n'|\hat{a}^\dagger \mid n\rangle = \sqrt{n+1}\delta_{n',n+1}; \tag{4.175}$$

that is,

$$\hat{a} = \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{2} & 0 & \cdots \\ 0 & 0 & 0 & \sqrt{3} & \cdots \\ 0 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad \hat{a}^\dagger = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots \\ \sqrt{1} & 0 & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{4.176}$$

Now, let us find the N-representation of the position and momentum operators, $\hat{X}$ and $\hat{P}$. From (4.122) we can show that $\hat{X}$ and $\hat{P}$ are given in terms of $\hat{a}$ and $\hat{a}^\dagger$ as follows:

$$\hat{X} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} + \hat{a}^\dagger\right), \qquad \hat{P} = i\sqrt{\frac{m\hbar\omega}{2}}\left(\hat{a}^\dagger - \hat{a}\right). \tag{4.177}$$

Their matrix elements are given by

$$\langle n'|\hat{X} \mid n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n}\delta_{n',n-1} + \sqrt{n+1}\delta_{n',n+1}\right), \tag{4.178}$$

$$\langle n'|\hat{P} \mid n\rangle = i\sqrt{\frac{m\hbar\omega}{2}}\left(-\sqrt{n}\delta_{n',n-1} + \sqrt{n+1}\delta_{n',n+1}\right), \tag{4.179}$$

in particular

$$\langle n \mid \hat{X} \mid n\rangle = \langle n \mid \hat{P} \mid n\rangle = 0. \tag{4.180}$$

The matrices corresponding to $\hat{X}$ and $\hat{P}$ are thus given by

$$\hat{X} = \sqrt{\frac{\hbar}{2m\omega}}\begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & \cdots \\ \sqrt{1} & 0 & \sqrt{2} & 0 & \cdots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \tag{4.181}$$

$$\hat{P} = i\sqrt{\frac{m\hbar\omega}{2}}\begin{pmatrix} 0 & -\sqrt{1} & 0 & 0 & \cdots \\ \sqrt{1} & 0 & -\sqrt{2} & 0 & \cdots \\ 0 & \sqrt{2} & 0 & -\sqrt{3} & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{4.182}$$

As mentioned in Chapter 2, the momentum operator is Hermitian, but not equal to its own complex conjugate: (4.182) shows that $\hat{P}^\dagger = \hat{P}$ and $\hat{P}^* = -\hat{P}$. As for $\hat{X}$, however, it is both Hermitian and equal to its complex conjugate: from (4.181) we have that $\hat{X}^\dagger = \hat{X}^* = \hat{X}$.

Finally, we should mention that the eigenstates $|n\rangle$ are represented by infinite column matrices; the first few states can be written as

$$|\,0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \qquad |\,1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \qquad |\,2\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \qquad |\,3\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ \vdots \end{pmatrix}, \ldots. \tag{4.183}$$

The set of states $\{|\,n\rangle\}$ forms indeed a complete and orthonormal basis.

4.8.5 Expectation Values of Various Operators

Let us evaluate the expectation values for $\hat{X}^2$ and $\hat{P}^2$ in the N-representation:

$$\hat{X}^2 = \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}\right) = \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}^{\dagger 2} + 2\hat{a}^\dagger\hat{a} + 1\right), \tag{4.184}$$

$$\hat{P}^2 = -\frac{m\hbar\omega}{2}\left(\hat{a}^2 + \hat{a}^{\dagger 2} - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}\right) = -\frac{m\hbar\omega}{2}\left(\hat{a}^2 + \hat{a}^{\dagger 2} - 2\hat{a}^\dagger\hat{a} - 1\right), \tag{4.185}$$

where we have used the fact that $\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} = 2\hat{a}^\dagger\hat{a} + 1$. Since the expectation values of $\hat{a}^2$ and $\hat{a}^{\dagger 2}$ are zero, $\langle n \mid \hat{a}^2 \mid n\rangle = \langle n \mid \hat{a}^{\dagger 2} \mid n\rangle = 0$, and $\langle n \mid \hat{a}^\dagger\hat{a} \mid n\rangle = n$, we have

$$\langle n \mid \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} \mid n\rangle = \langle n \mid 2\hat{a}^\dagger\hat{a} + 1 \mid n\rangle = 2n + 1; \tag{4.186}$$

hence

$$\langle n \mid \hat{X}^2 \mid n\rangle = \frac{\hbar}{2m\omega}\langle n \mid \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} \mid n\rangle = \frac{\hbar}{2m\omega}(2n+1), \tag{4.187}$$

$$\langle n \mid \hat{P}^2 \mid n\rangle = \frac{m\hbar\omega}{2}\langle n \mid \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} \mid n\rangle = \frac{m\hbar\omega}{2}(2n+1). \tag{4.188}$$

Comparing (4.187) and (4.188) we see that the expectation values of the potential and kinetic energies are equal and are also equal to half the total energy:

$$\boxed{\frac{m\omega^2}{2}\langle n \mid \hat{X}^2 \mid n\rangle = \frac{1}{2m}\langle n \mid \hat{P}^2 \mid n\rangle = \frac{1}{2}\langle n \mid \hat{H} \mid n\rangle.} \tag{4.189}$$

This result is known as the *Virial theorem*.

We can now easily calculate the product $\Delta x\,\Delta p$ from (4.187) and (4.188). Since $\langle\hat{X}\rangle = \langle\hat{P}\rangle = 0$ we have

$$\Delta x = \sqrt{\langle\hat{X}^2\rangle - \langle\hat{X}\rangle^2} = \sqrt{\langle\hat{X}^2\rangle} = \sqrt{\frac{\hbar}{2m\omega}(2n+1)}, \tag{4.190}$$

$$\Delta p = \sqrt{\langle\hat{P}^2\rangle - \langle\hat{P}\rangle^2} = \sqrt{\langle\hat{P}^2\rangle} = \sqrt{\frac{m\hbar\omega}{2}(2n+1)}; \tag{4.191}$$

hence

$$\Delta x\,\Delta p = \left(n + \frac{1}{2}\right)\hbar \quad\Longrightarrow\quad \Delta x\,\Delta p \geq \frac{\hbar}{2}, \tag{4.192}$$

since $n \geq 0$; this is the Heisenberg uncertainty principle.

4.9 Numerical Solution of the Schrödinger Equation

In this section we are going to show how to solve a one-dimensional Schrödinger equation numerically. The numerical solutions provide an idea about the properties of stationary states.

4.9.1 Numerical Procedure

We want to solve the following equation numerically:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x) = E\psi(x) \Longrightarrow \frac{d^2\psi}{dx^2} + k^2\psi(x) = 0, \tag{4.193}$$

where $k^2 = 2m[E - V(x)]/\hbar^2$.

First, divide the x-axis into a set of equidistant points with a spacing of $h_0 = \Delta x$, as shown in Figure 4.10a. The wave function $\psi(x)$ can be approximately described by its values at the

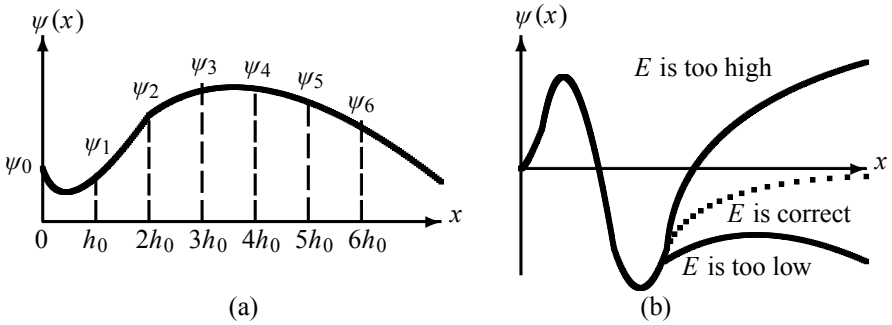


Figure 4.10 (a) Discretization of the wave function. (b) If the energy E used in the computation is too high (too low), the wave function will diverge as $x \to \pm\infty$; but at the appropriate value of E, the wave function converges to the correct values.

points of the grid (i.e., $\psi_0 = \psi(x = 0)$, $\psi_1 = \psi(h_0)$, $\psi_2 = \psi(2h_0)$, $\psi_3 = \psi(3h_0)$, and so on). The first derivative of ψ can then be approximated by

$$\frac{d\psi}{dx} \approx \frac{\psi_{n+1} - \psi_n}{h_0}. \tag{4.194}$$

An analogous approximation for the second derivative is actually a bit tricky. There are several methods to calculate it, but a very efficient procedure is called the *Numerov algorithm* (which is described in standard numerical analysis textbooks). In short, the second derivative is approximated by the so-called three-point difference formula:

$$\frac{\psi_{n+1} - 2\psi_n + \psi_{n-1}}{h_0^2} = \psi_n'' + \frac{h_0^2}{12}\psi_n'''' + 0(h_0^4). \tag{4.195}$$

From (4.193) we have

$$\psi_n'''' = \frac{d^2}{dx^2}(-k^2\psi)\bigg|_{x=x_n} = -\frac{(k^2\psi)_{n+1} - 2(k^2\psi)_n + (k^2\psi)_{n-1}}{h_0^2}. \tag{4.196}$$

Using $\psi_n'' = -k_n^2\psi_n$ and substituting (4.196) into (4.195) we can show that

$$\psi_{n+1} = \frac{2\left(1 - \frac{5}{12}h_0^2k_n^2\right)\psi_n - \left(1 + \frac{1}{12}h_0^2k_{n-1}^2\right)\psi_{n-1}}{1 + \frac{1}{12}h_0^2k_{n+1}^2}. \tag{4.197}$$

We can thus assign arbitrary values for ψ_0 and ψ_1; this is equivalent to providing the starting (or initial) values for $\psi(x)$ and $\psi'(x)$. Knowing ψ_0 and ψ_1, we can use (4.197) to calculate ψ_2, then ψ_3, then ψ_4, and so on. The solution of a linear equation, equation (4.197), for either ψ_{n+1} or ψ_{n-1} yields a recursion relation for integrating either *forward* or *backward* in x with a local error $O(h_0^6)$. In this way, the solution depends on two arbitrary constants, ψ_0 and ψ_1, as it should for any second-order differential equation (i.e., there are two linearly independent solutions).

The *boundary conditions* play a crucial role in solving any Schrödinger equation. Every boundary condition gives a linear homogeneous equation satisfied by the wave function or its

derivative. For example, in the case of the infinite square well potential and the harmonic oscillator, the conditions $\psi(x_{min}) = 0$, $\psi(x_{max}) = 0$ are satisfied as follows:

- Infinite square well: $\psi(-a/2) = \psi(a/2) = 0$
- Harmonic oscillator: $\psi(-\infty) = \psi(+\infty) = 0$

4.9.2 Algorithm

To solve the Schrödinger equation with the boundary conditions $\psi(x_{min}) = \psi(x_{max}) = 0$, you may proceed as follows. Suppose you want to find the wave function, $\psi^{(n)}(x)$, and the energy E_n for the nth excited[4] state of a system:

- Take $\psi_0 = 0$ and choose ψ_1 (any small number you like), because the value of ψ_1 must be very close to that of ψ_0.
- Choose a *trial* energy E_n.
- With this value of the energy, E_n, together with ψ_0 and ψ_1, you can calculate iteratively the wave function at different values of x; that is, you can calculate ψ_2, ψ_3, ψ_4, ... How? You need simply to inject $\psi_0 = 0$, ψ_1, and E_n into (4.197) and proceed incrementally to calculate ψ_2; then use ψ_1 and ψ_2 to calculate ψ_3; then use ψ_2 and ψ_3 to calculate ψ_4; and so on till you end up with the value of the wave function at $x_n = nh_0$, $\psi_n = \psi(nh_0)$.
- Next, you need to check whether the ψ_n you obtained is zero or not. If ψ_n is zero, this means that you have made the right choice for the trial energy. This value E_n can then be taken as a possible eigenenergy for the system; at this value of E_n, the wave function converges to the correct value (dotted curve in Figure 4.10b). Of course, it is highly unlikely to have chosen the correct energy from a first trial. In this case you need to proceed as follows. If the value of ψ_n obtained is a nonzero positive number or if it diverges, this means that the trial E_n you started with is larger than the correct eigenvalue (Figure 4.10b); on the other hand, if ψ_n is a negative nonzero number, this means that the E_n you started with is less than the true energy. If the ψ_n you end up with is a positive nonzero number, you need to start all over again with a smaller value of the energy. But if the ψ_n you end up with is negative, you need to start again with a larger value of E. You can continue in this way, improving every time, till you end up with a zero value for ψ_n. Note that in practice there is no way to get ψ_n exactly equal to zero. You may stop the procedure the moment ψ_n is sufficiently small; that is, you stop the iteration at the desired accuracy, say at 10^{-8} of its maximum value.

Example 4.3 (Numerical solution of the Schrödinger equation)
A proton is subject to a harmonic oscillator potential $V(x) = m\omega^2 x^2/2$, $\omega = 5.34 \times 10^{21} s^{-1}$.

(a) Find the exact energies of the five lowest states (express them in MeV).

(b) Solve the Schrödinger equation *numerically* and find the energies of the five lowest states and compare them with the exact results obtained in (a). Note: You may use these quantities: rest mass energy of the proton $mc^2 \simeq 10^3$ MeV, $\hbar c \simeq 200\,\text{MeV fm}$, and $\hbar\omega \simeq 3.5\,\text{MeV}$.

[4] We have denoted here the wave function of the nth excited state by $\psi^{(n)}(x)$ to distinguish it from the value of the wave function at $x_n = nh_0$, $\psi_n = \psi(nh_0)$.

Table 4.1 Exact and numerical energies for the five lowest states of the harmonic oscillator.

n	E_n^{Exact} (MeV)	$E_n^{Numeric}$ (MeV)
0.0	1.750 000	1.749 999 999 795
1.0	5.250 000	5.249 999 998 112
2.0	8.750 000	8.749 999 992 829
3.0	12.250 000	12.249 999 982 320
4.0	15.750 000	15.749 999 967 590

Solution

(a) The exact energies can be calculated at once from $E_n = \hbar\omega(n + \frac{1}{2}) \simeq 3.5(n + \frac{1}{2})$ MeV. The results for the five lowest states are listed in Table 4.1.

(b) To obtain the numerical values, we need simply to make use of the Numerov relation (4.197), where $k_n^2(x) = 2m(E_n - \frac{1}{2}m\omega^2x^2)/\hbar^2$. The numerical values involved here can be calculated as follows:

$$\frac{m^2\omega^2}{\hbar^2} = \frac{(mc^2)^2(\hbar\omega)^2}{(\hbar c)^4} \simeq \frac{(10^3\,\text{MeV})^2(3.5\,\text{MeV})^2}{(200\,\text{MeV fm})^4} = 7.66 \times 10^{-4}\,\text{fm}^{-3}, \quad (4.198)$$

$$\frac{2m}{\hbar^2} = \frac{2mc^2}{(\hbar c)^2} \simeq \frac{2 \times 10^3\,\text{MeV}}{(200\,\text{MeV fm})^2} = 0.05\,\text{MeV}^{-1}\,\text{fm}^{-2}. \quad (4.199)$$

The boundary conditions for the harmonic oscillator imply that the wave function vanishes at $x = \pm\infty$, i.e., at $x_{min} = -\infty$ and $x_{max} = \infty$. How does one deal with infinities within a computer program? For this, we need to choose the numerical values of x_{min} and x_{max} in a way that the wave function would not feel the "edge" effects. That is, we simply need to assign numerical values to x_{min} and x_{max} so that they are far away from the turning points $x_{Left} = -\sqrt{2E_n/(m\omega^2)}$ and $x_{Right} = \sqrt{2E_n/(m\omega^2)}$, respectively. For instance, in the case of the ground state, where $E_0 = 1.75$ MeV, we have $x_{Left} = -3.38$ fm and $x_{Right} = 3.38$ fm; we may then take $x_{min} = -20$ fm and $x_{max} = 20$ fm. The wave function should be practically zero at $x = \pm 20$ fm.

To calculate the energies numerically for the five lowest states, a C++ computer code has been prepared (see Appendix C). The numerical results generated by this code are listed in Table 4.1; they are in excellent agreement with the exact results. Figure 4.11 displays the wave functions obtained from this code for the five lowest states for the proton moving in a harmonic oscillator potential (these plotted wave functions are normalized).

4.10 Solved Problems

Problem 4.1

A particle moving in one dimension is in a stationary state whose wave function

$$\psi(x) = \begin{cases} 0, & x < -a, \\ A(1 + \cos\frac{\pi x}{a}), & -a \le x \le a, \\ 0, & x > a, \end{cases}$$

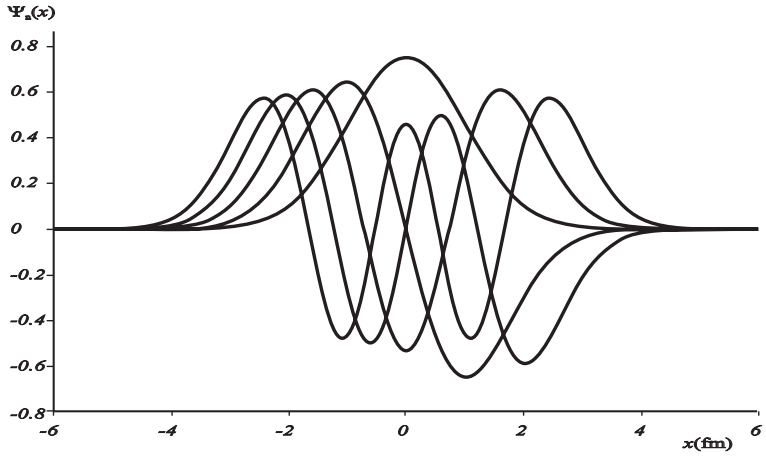


Figure 4.11 Wave functions $\psi_n(x)$ of the five lowest states of a harmonic oscillator potential in terms of x, where the x-axis values are in fm (obtained from the C++ code of Appendix C).

where A and a are real constants.

(a) Is this a physically acceptable wave function? Explain.

(b) Find the magnitude of A so that $\psi(x)$ is normalized.

(c) Evaluate Δx and Δp. Verify that $\Delta x\,\Delta p \geq \hbar/2$.

(d) Find the classically allowed region.

Solution

(a) Since $\psi(x)$ is square integrable, single-valued, continuous, and has a continuous first derivative, it is indeed physically acceptable.

(b) Normalization of $\psi(x)$: using the relation $\cos^2 y = (1+\cos 2y)/2$, we have

$$\begin{aligned} 1 &= \int_{-\infty}^{+\infty} |\psi(x)|^2 dx = A^2 \int_{-a}^{a} dx \left[1 + 2\cos\frac{\pi x}{a} + \cos^2\left(\frac{\pi x}{a}\right)\right] \\ &= A^2 \int_{-a}^{a} dx \left[\frac{3}{2} + 2\cos\frac{\pi x}{a} + \frac{1}{2}\cos\frac{2\pi x}{a}\right] \\ &= \frac{3}{2}A^2 \int_{-a}^{a} dx = 3aA^2; \end{aligned} \tag{4.200}$$

hence $A = 1/\sqrt{3a}$.

(c) As $\psi(x)$ is even, we have $\langle \hat{X}\rangle = \int_{-a}^{+a} \psi^*(x) x \psi(x)\, dx = 0$, since the symmetric integral of an odd function (i.e., $\psi^*(x)x\psi(x)$ is odd) is zero. On the other hand, we also have $\langle \hat{P}\rangle = 0$ because $\psi(x)$ is real and even. We can thus write

$$\Delta x = \sqrt{\langle \hat{X}^2\rangle}, \qquad \Delta p = \sqrt{\langle \hat{P}^2\rangle}, \tag{4.201}$$

since $\Delta A = \sqrt{\langle \hat{A}^2\rangle - \langle \hat{A}\rangle^2}$. The calculations of $\langle \hat{X}^2\rangle$ and $\langle \hat{P}^2\rangle$ are straightforward:

$$\langle \hat{X}^2\rangle = -\hbar^2 \int_{-a}^{a} \psi^*(x) x^2 \psi(x)\, dx = \frac{1}{3a}\int_{-a}^{a} \left[x^2 + 2x^2\cos\left(\frac{\pi x}{a}\right) + x^2\cos^2\left(\frac{\pi x}{a}\right)\right] dx$$

$$= \frac{a^2}{6\pi^2}\left(2\pi^2 - 15\right), \tag{4.202}$$

$$\begin{aligned}\langle \hat{P}^2 \rangle &= -\hbar^2 \int_{-a}^{+a} \psi(x) \frac{d^2\psi(x)}{dx^2}\, dx = \frac{\pi^2\hbar^2}{a^2} A^2 \int_{-a}^{a} \left[\cos\frac{\pi x}{a} + cos^2\left(\frac{\pi x}{a}\right)\right] dx \\ &= \frac{\pi^2\hbar^2}{3a^3} \int_{-a}^{a} \left[\frac{1}{2} + \cos\frac{\pi x}{a} + \frac{1}{2}\cos\frac{2\pi x}{a}\right] dx = \frac{\pi^2\hbar^2}{3a^2}; \end{aligned} \tag{4.203}$$

hence $\Delta x = a\sqrt{1/3 - 5/(2\pi^2)}$ and $\Delta p = \pi\hbar/(\sqrt{3}a)$. We see that the uncertainties product

$$\Delta x\, \Delta p = \frac{\pi\hbar}{3}\sqrt{1 - \frac{15}{2\pi^2}} \tag{4.204}$$

satisfies Heisenberg's uncertainty principle, $\Delta x\, \Delta p > \hbar/2$.

(d) Since $d\psi^2/dx^2$ is zero at the inflection points, we have

$$\frac{d^2\psi}{dx^2} = -\frac{\pi^2}{a^2} A \cos\frac{\pi x}{a} = 0. \tag{4.205}$$

This relation holds when $x = \pm a/2$; hence the classically allowed region is defined by the interval between the inflection points $-a/2 \leq x \leq a/2$. That is, since $\psi(x)$ decays exponentially for $x > a/2$ and for $x < -a/2$, the energy of the system must be smaller than the potential. Classically, the system cannot be found in this region.

Problem 4.2

Consider a particle of mass m moving freely between $x = 0$ and $x = a$ inside an infinite square well potential.

(a) Calculate the expectation values $\langle \hat{X} \rangle_n$, $\langle \hat{P} \rangle_n$, $\langle \hat{X}^2 \rangle_n$, and $\langle \hat{P}^2 \rangle_n$, and compare them with their classical counterparts.

(b) Calculate the uncertainties product $\Delta x_n\, \Delta p_n$.

(c) Use the result of (b) to estimate the zero-point energy.

Solution

(a) Since $\psi_n(x) = \sqrt{2/a}\sin(n\pi x/a)$ and since it is a real function, we have $\langle \psi_n | \hat{P} | \psi_n \rangle = 0$ because for any real function $\phi(x)$ the integral $\langle \hat{P} \rangle = -i\hbar \int \phi^*(x)(d\phi(x)/dx)\, dx$ is imaginary and this contradicts the fact that $\langle \hat{P} \rangle$ has to be real. On the other hand, the expectation values of $\hat{X}$, $\hat{X}^2$, and $\hat{P}^2$ are

$$\begin{aligned}\langle \psi_n | \hat{X} | \psi_n \rangle &= \int_0^a \psi_n^*(x)\, x\, \psi_n(x)\, dx = \frac{2}{a}\int_0^a x \sin^2\left(\frac{n\pi x}{a}\right) dx \\ &= \frac{1}{a}\int_0^a x\left[1 - \cos\left(\frac{2n\pi x}{a}\right)\right] dx = \frac{a}{2}, \end{aligned} \tag{4.206}$$

$$\begin{aligned}\langle \psi_n | \hat{X}^2 | \psi_n \rangle &= \frac{2}{a}\int_0^a x^2 \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{1}{a}\int_0^a x^2\left[1 - \cos\left(\frac{2n\pi x}{a}\right)\right] dx \\ &= \frac{a^2}{3} - \frac{1}{a}\int_0^a x^2 \cos\left(\frac{2n\pi x}{a}\right) dx \end{aligned}$$

$$= \frac{a^2}{3} - \frac{1}{2n\pi}x^2 \sin\left(\frac{2n\pi x}{a}\right)\Big|_{x=0}^{x=a} + \frac{1}{n\pi}\int_0^a x \sin\left(\frac{2n\pi x}{a}\right) dx$$

$$= \frac{a^2}{3} - \frac{a^2}{2n^2\pi^2}, \tag{4.207}$$

$$\langle\psi_n|\hat{P}^2|\psi_n\rangle = -\hbar^2\int_0^a \psi_n^*(x)\frac{d^2\psi_n(x)}{dx^2}dx = \frac{n^2\pi^2\hbar^2}{a^2}\int_0^a |\psi_n(x)|^2 dx = \frac{n^2\pi^2\hbar^2}{a^2}. \tag{4.208}$$

In deriving the previous three expressions, we have used integrations by parts. Since $E_n = n^2\pi^2\hbar^2/(2ma^2)$, we may write

$$\langle\psi_n|\hat{P}^2|\psi_n\rangle = \frac{n^2\pi^2\hbar^2}{a^2} = 2mE_n. \tag{4.209}$$

To calculate the classical average values x_{av}, p_{av}, x_{av}^2, p_{av}^2, it is easy first to infer that $p_{av} = 0$ and $p_{av}^2 = 2mE$, since the particle moves to the right with *constant* momentum $p = m\upsilon$ and to the left with $p = -m\upsilon$. As the particle moves at constant speed, we have $x = \upsilon t$, hence

$$x_{av} = \frac{1}{T}\int_0^T x(t)\,dt = \frac{\upsilon}{T}\int_0^T t\,dt = \upsilon\frac{T}{2} = \frac{a}{2}, \tag{4.210}$$

$$x_{av}^2 = \frac{1}{T}\int_0^T x^2(t)\,dt = \frac{\upsilon^2}{T}\int_0^T t^2 dt = \frac{1}{3}\upsilon^2 T^2 = \frac{a^2}{3}, \tag{4.211}$$

where T is half[5] of the period of the motion, with $a = \upsilon T$.

We conclude that, while the average classical and quantum expressions for x, p and p^2 are identical, a comparison of (4.207) and (4.211) yields

$$\langle\psi_n|\hat{X}^2|\psi_n\rangle = \frac{a^2}{3} - \frac{a^2}{2n^2\pi^2} = x_{av}^2 - \frac{a^2}{2n^2\pi^2}, \tag{4.212}$$

so that in the limit of large quantum numbers, the quantum expression $\langle\psi_n|\hat{X}^2|\psi_n\rangle$ matches with its classical counterpart x_{av}^2: $\lim_{n\longrightarrow\infty}\langle\psi_n|\hat{X}^2|\psi_n\rangle = a^2/3 = x_{av}^2$.

(b) The position and the momentum uncertainties can be calculated from (4.206) to (4.208):

$$\Delta x_n = \sqrt{\langle\psi_n|\hat{X}^2|\psi_n\rangle - \langle\psi_n|\hat{X}|\psi_n\rangle^2} = \sqrt{\frac{a^2}{3} - \frac{a^2}{2n^2\pi^2} - \frac{a^2}{4}} = a\sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}, \tag{4.213}$$

$$\Delta p_n = \sqrt{\langle\psi_n|\hat{P}^2|\psi_n\rangle - \langle\psi_n|\hat{P}|\psi_n\rangle^2} = \sqrt{\langle\psi_n|\hat{P}^2|\psi_n\rangle} = \frac{n\pi\hbar}{a}, \tag{4.214}$$

hence

$$\Delta x_n\,\Delta p_n = n\pi\hbar\sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}. \tag{4.215}$$

(c) Equation (4.214) shows that the momentum uncertainty for the ground state is not zero, but

$$\Delta p_1 = \frac{\pi\hbar}{a}. \tag{4.216}$$

[5] We may parameterize the other half of the motion by $x = -\upsilon t$, which when inserted in (4.210) and (4.211), where the variable t varies between $-T$ and 0, the integrals would yield the same results, namely $x_{av} = a/2$ and $x_{av}^2 = a^2/3$, respectively.

This leads to a nonzero kinetic energy. Therefore, the lowest value of the particle's kinetic energy is of the order of $E_{min} \sim (\Delta p_1)^2/(2m) \sim \pi^2\hbar^2/(2ma^2)$. This value, which is in full agreement with the ground state energy, $E_1 = \pi^2\hbar^2/(2ma^2)$, is the zero-point energy of the particle.

Problem 4.3
An electron is moving freely inside a one-dimensional infinite potential box with walls at $x = 0$ and $x = a$. If the electron is initially in the ground state ($n = 1$) of the box and if we *suddenly* quadruple the size of the box (i.e., the right-hand side wall is moved instantaneously from $x = a$ to $x = 4a$), calculate the probability of finding the electron in:

(a) the ground state of the new box and

(b) the first excited state of the new box.

Solution
Initially, the electron is in the ground state of the box $x = 0$ and $x = a$; its energy and wave function are

$$E_1 = \frac{\pi^2\hbar^2}{2ma^2}, \qquad \phi_1(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{\pi x}{a}\right). \tag{4.217}$$

(a) Once in the new box, $x = 0$ and $x = 4a$, the ground state energy and wave function of the electron are

$$E_1' = \frac{\pi^2\hbar^2}{2m(4a)^2} = \frac{\pi^2\hbar^2}{32ma^2}, \qquad \psi_1(x) = \frac{1}{\sqrt{2a}}\sin\left(\frac{\pi x}{4a}\right). \tag{4.218}$$

The probability of finding the electron in $\psi_1(x)$ is

$$P(E_1') = |\langle\psi_1|\phi_1\rangle|^2 = \left|\int_0^a \psi_1^*(x)\phi_1(x)dx\right|^2 = \frac{1}{a^2}\left|\int_0^a \sin\left(\frac{\pi x}{4a}\right)\sin\left(\frac{\pi x}{a}\right)dx\right|^2; \tag{4.219}$$

the upper limit of the integral sign is a (and not $4a$) because $\phi_1(x)$ is limited to the region between 0 and a. Using the relation $\sin a \sin b = \frac{1}{2}\cos(a-b) - \frac{1}{2}\cos(a+b)$, we have $\sin(\pi x/4a)\sin(\pi x/a) = \frac{1}{2}\cos(3\pi x/4a) - \frac{1}{2}\cos(5\pi x/4a)$; hence

$$\begin{aligned} P(E_1') &= \frac{1}{a^2}\left|\frac{1}{2}\int_0^a \cos\left(\frac{3\pi x}{4a}\right)dx - \frac{1}{2}\int_0^a \cos\left(\frac{5\pi x}{4a}\right)dx\right|^2 \\ &= \frac{128}{15^2\pi^2} = 0.058 = 5.8\%. \end{aligned} \tag{4.220}$$

(b) If the electron is in the first excited state of the new box, its energy and wave function are

$$E_2' = \frac{\pi^2\hbar^2}{8ma^2}, \qquad \psi_2(x) = \frac{1}{\sqrt{2a}}\sin\left(\frac{\pi x}{2a}\right). \tag{4.221}$$

The corresponding probability is

$$\begin{aligned} P(E_2') &= |\langle\psi_2|\phi_1\rangle|^2 = \left|\int_0^a \psi_2^*(x)\phi_1(x)dx\right|^2 = \frac{1}{a^2}\left|\int_0^a \sin\left(\frac{\pi x}{2a}\right)\sin\left(\frac{\pi x}{a}\right)dx\right|^2 \\ &= \frac{16}{9\pi^2} = 0.18 = 18\%. \end{aligned} \tag{4.222}$$

Problem 4.4

Consider a particle of mass m subject to an attractive delta potential $V(x) = -V_0\,\delta(x)$, where $V_0 > 0$ (V_0 has the dimensions of Energy×Distance).

(a) In the case of negative energies, show that this particle has only one bound state; find the binding energy and the wave function.

(b) Calculate the probability of finding the particle in the interval $-a \leq x \leq a$.

(c) What is the probability that the particle remains bound when V_0 is (i) halved suddenly, (ii) quadrupled suddenly?

(d) Study the scattering case (i.e., $E > 0$) and calculate the reflection and transmission coefficients as a function of the wave number k.

Solution

(a) Let us consider first the bound state case $E < 0$. We can write the Schrödinger equation as follows:

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mV_0}{\hbar^2}\delta(x)\psi(x) + \frac{2mE}{\hbar^2}\psi(x) = 0. \tag{4.223}$$

Since $\delta(x)$ vanishes for $x \neq 0$, this equation becomes

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0. \tag{4.224}$$

The bound solutions require that $\psi(x)$ vanishes at $x = \pm\infty$; these bound solutions are given by

$$\psi(x) = \begin{cases} \psi_-(x) = Ae^{kx}, & x < 0, \\ \psi_+(x) = Be^{-kx}, & x > 0, \end{cases} \tag{4.225}$$

where $k = \sqrt{2m|E|}/\hbar$. Since $\psi(x)$ is continuous at $x = 0$, $\psi_-(0) = \psi_+(0)$, we have $A = B$. Thus, the wave function is given by $\psi(x) = Ae^{-k|x|}$; note that $\psi(x)$ is even.

The energy can be obtained from the discontinuity condition of the first derivative of the wave function, which in turn can be obtained by integrating (4.223) from $-\varepsilon$ to $+\varepsilon$,

$$\int_{-\varepsilon}^{+\varepsilon} dx \frac{d^2\psi(x)}{dx^2} + \frac{2mV_0}{\hbar^2}\int_{-\varepsilon}^{+\varepsilon}\delta(x)\psi(x)dx + \frac{2mE}{\hbar^2}\int_{-\varepsilon}^{+\varepsilon}\psi(x)dx = 0, \tag{4.226}$$

and then letting $\varepsilon \to 0$. Using the facts that

$$\int_{-\varepsilon}^{+\varepsilon} dx \frac{d^2\psi(x)}{dx^2} = \left.\frac{d\psi(x)}{dx}\right|_{x=+\varepsilon} - \left.\frac{d\psi(x)}{dx}\right|_{x=-\varepsilon} = \left.\frac{d\psi_+(x)}{dx}\right|_{x=+\varepsilon} - \left.\frac{d\psi_-(x)}{dx}\right|_{x=-\varepsilon} \tag{4.227}$$

and that $\int_{-\varepsilon}^{+\varepsilon}\psi(x)dx = 0$ (because $\psi(x)$ is even), we can rewrite (4.226) as follows:

$$\lim_{\varepsilon\to 0}\left(\left.\frac{d\psi_+(x)}{dx}\right|_{x=+\varepsilon} - \left.\frac{d\psi_-(x)}{dx}\right|_{x=-\varepsilon}\right)_{\varepsilon=0} + \frac{2mV_0}{\hbar^2}\psi(0) = 0, \tag{4.228}$$

since the wave function is continuous at $x = 0$, but its first derivative is not. Substituting (4.225) into (4.228) and using $A = B$, we obtain

$$(-2kA) + \frac{2mV_0}{\hbar^2}A = 0 \tag{4.229}$$

or $k = mV_0/\hbar^2$. But since $k = \sqrt{2m|E|/\hbar^2}$, we have $mV_0/\hbar^2 = \sqrt{2m|E|/\hbar^2}$, and since the energy is negative, we conclude that $E = -mV_0^2/(2\hbar^2)$. There is, therefore, only one bound state solution. As for the excited states, all of them are unbound. We may normalize $\psi(x)$,

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} \psi^*(x)\psi(x)\,dx = A^2 \int_{-\infty}^{0} \exp(2kx)\,dx + A^2 \int_{0}^{\infty} \exp(-2kx)\,dx \\ &= 2A^2 \int_0^{\infty} \exp(-2kx)\,dx = \frac{A^2}{k}, \end{aligned} \tag{4.230}$$

hence $A = \sqrt{k}$. The normalized wave function is thus given by $\psi(x) = \sqrt{k}e^{-k|x|}$. So the energy and normalized wave function of the bound state are given by

$$E = -\frac{mV_0^2}{2\hbar^2}, \qquad \psi(x) = \sqrt{\frac{mV_0}{\hbar^2}} \exp\left(-\frac{mV_0}{\hbar^2}|x|\right). \tag{4.231}$$

(b) Since the wave function $\psi(x) = \sqrt{k}e^{-k|x|}$ is normalized, the probability of finding the particle in the interval $-a \le x \le a$ is given by

$$\begin{aligned} P &= \frac{\int_{-a}^{a} |\psi(x)|^2\,dx}{\int_{-\infty}^{\infty} |\psi(x)|^2\,dx} = \int_{-a}^{a} |\psi(x)|^2\,dx = k\int_{-a}^{a} e^{-2k|x|}dx \\ &= k\int_{-a}^{0} e^{2kx}dx + k\int_{0}^{a} e^{-2kx}dx = 2k\int_{0}^{a} e^{-2kx}dx \\ &= 1 - e^{-2ka} = 1 - e^{-2mV_0a/\hbar^2}. \end{aligned} \tag{4.232}$$

(c) If the strength of the potential changed suddenly from V_0 to V_1, the wave function will be given by $\psi_1(x) = \sqrt{mV_1/\hbar^2}\exp(-mV_1|x|/\hbar^2)$. The probability that the particle remains in the bound state $\psi_1(x)$ is

$$\begin{aligned} P &= |\langle\psi_1|\psi\rangle|^2 = \left|\int_{-\infty}^{\infty} {\psi_1}^*(x)\psi(x)\,dx\right|^2 \\ &= \left|\frac{m}{\hbar^2}\sqrt{V_0V_1}\int_{-\infty}^{\infty} \exp\left(-\frac{m(V_0+V_1)}{\hbar^2}|x|\right)dx\right|^2 \\ &= \left|2\frac{m}{\hbar^2}\sqrt{V_0V_1}\int_{0}^{\infty} \exp\left(-\frac{m(V_0+V_1)}{\hbar^2}x\right)dx\right|^2 = \frac{4V_0V_1}{(V_0+V_1)^2}. \end{aligned} \tag{4.233}$$

(i) In the case where the strength of the potential is halved, $V_1 = \frac{1}{2}V_0$, the probability that the particle remains bound is

$$P = \frac{2V_0^2}{(V_0 + \frac{1}{2}V_0)^2} = \frac{8}{9} = 89\%. \tag{4.234}$$

(ii) When the strength is quadrupled, $V_1 = 4V_0$, the probability is given by

$$P = \frac{16V_0^2}{(5V_0)^2} = \frac{16}{25} = 64\%. \tag{4.235}$$

(d) The case $E > 0$ corresponds to a free motion and the energy levels represent a continuum. The solution of the Schrödinger equation for $E > 0$ is given by

$$\psi(x) = \begin{cases} \psi_-(x) = Ae^{ikx} + Be^{-ikx}, & x < 0, \\ \psi_+(x) = Ce^{ikx}, & x > 0, \end{cases} \tag{4.236}$$

where $k = \sqrt{2mE}/\hbar$; this corresponds to a plane wave incident from the left together with a reflected wave in the region $x < 0$, and only a transmitted wave for $x > 0$.

The values of the constants A and B are to be found from the continuity relations. On the one hand, the continuity of $\psi(x)$ at $x = 0$ yields

$$A + B = C \tag{4.237}$$

and, on the other hand, substituting (4.236) into (4.228), we end up with

$$ik(C - A + B) + \frac{2mV_0}{\hbar^2}C = 0. \tag{4.238}$$

Solving (4.237) and (4.238) for B/A and C/A, we find

$$\frac{B}{A} = \frac{-1}{1 + \frac{ik\hbar^2}{mV_0}}, \qquad \frac{C}{A} = \frac{1}{1 - \frac{imV_0}{\hbar^2 k}}. \tag{4.239}$$

Thus, the reflection and transmission coefficients are

$$R = \left|\frac{B}{A}\right|^2 = \frac{1}{1 + \frac{\hbar^4 k^2}{m^2 V_0^2}} = \frac{1}{1 + \frac{2\hbar^2 E}{mV_0^2}}, \quad T = \left|\frac{C}{A}\right|^2 = \frac{1}{1 + \frac{m^2 V_0^2}{\hbar^4 k^2}} = \frac{1}{1 + \frac{mV_0^2}{2\hbar^2 E}}, \tag{4.240}$$

with $R + T = 1$.

Problem 4.5

A particle of mass m is subject to an attractive double-delta potential $V(x) = -V_0\delta(x - a) - V_0\delta(x + a)$, where $V_0 > 0$. Consider only the case of negative energies.

(a) Obtain the wave functions of the bound states.

(b) Derive the eigenvalue equations.

(c) Specify the number of bound states and the limit on their energies. Is the ground state an even state or an odd state?

(d) Estimate the ground state energy for the limits $a \to 0$ and $a \to \infty$.

Solution

(a) The Schrödinger equation for this problem is

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mV_0}{\hbar^2}\left[\delta(x - a) + \delta(x + a)\right]\psi(x) + \frac{2mE}{\hbar^2}\psi(x) = 0. \tag{4.241}$$

For $x \neq \pm a$ this equation becomes

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \qquad \text{or} \qquad \frac{d^2\psi(x)}{dx^2} - k^2\psi(x) = 0, \tag{4.242}$$

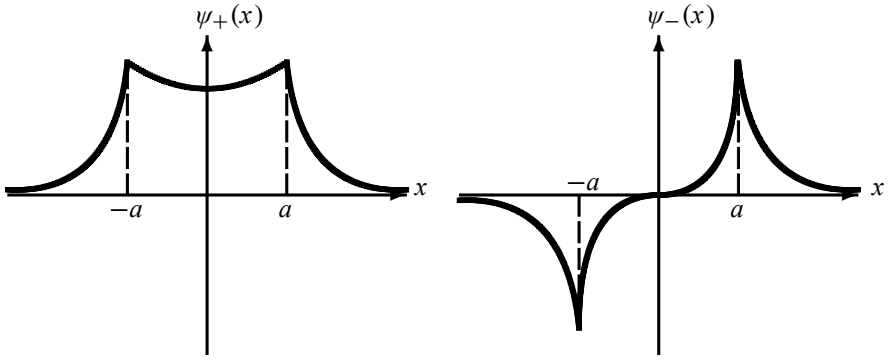


Figure 4.12 Shapes of the even and odd wave functions for $V(x) = -V_0\delta(x-a) - V_0\delta(x+a)$.

where $k^2 = -2mE/\hbar^2 = 2m|E|/\hbar^2$, since this problem deals only with the bound states $E < 0$.

Since the potential is symmetric, $V(-x) = V(x)$, the wave function is either even or odd; we will denote the even states by $\psi_+(x)$ and the odd states by $\psi_-(x)$. The bound state solutions for $E < 0$ require that $\psi_\pm(x)$ vanish at $x = \pm\infty$:

$$\psi_\pm(x) = \begin{cases} Ae^{-kx}, & x > a, \\ \frac{B}{2}\left(e^{kx} \pm e^{-kx}\right), & -a < x < a, \\ \pm Ae^{kx}, & x < -a; \end{cases} \tag{4.243}$$

hence

$$\psi_+(x) = \begin{cases} Ae^{-kx}, \\ B\cosh kx, \\ Ae^{kx}, \end{cases} \qquad \psi_-(x) = \begin{cases} Ae^{-kx}, & x > a, \\ B\sinh kx, & -a < x < a, \\ -Ae^{kx}, & x < -a. \end{cases} \tag{4.244}$$

The shapes of $\psi_\pm(x)$ are displayed in Figure 4.12.

(b) As for the energy eigenvalues, they can be obtained from the boundary conditions. The continuity condition at $x = a$ of $\psi_+(x)$ leads to

$$Ae^{-ka} = B\cosh ka \tag{4.245}$$

and that of $\psi_-(x)$ leads to

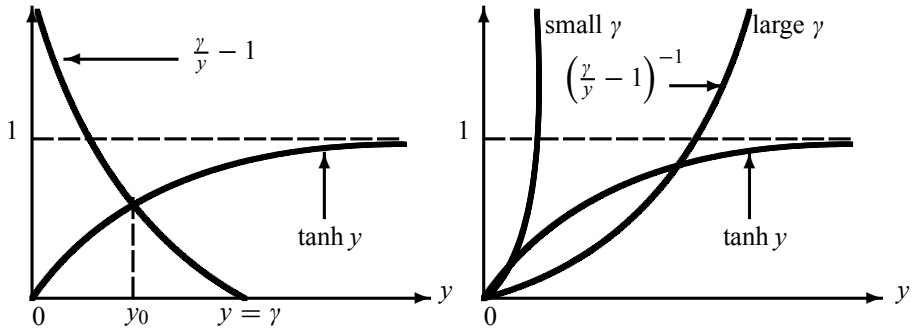
$$Ae^{-ka} = B\sinh ka. \tag{4.246}$$

To obtain the discontinuity condition for the first derivative of $\psi_+(x)$ at $x = a$, we need to integrate (4.241):

$$\lim_{\varepsilon\to 0}\left[\psi'_+(a+\varepsilon) - \psi'_+(a-\varepsilon)\right] + \frac{2mV_0}{\hbar^2}\psi_+(a) = 0; \tag{4.247}$$

hence

$$-kAe^{-ka} - kB\sinh ka + \frac{2mV_0}{\hbar^2}Ae^{-ka} = 0 \implies A\left(\frac{2mV_0}{k\hbar^2} - 1\right)e^{-ka} = B\sinh ka. \tag{4.248}$$



(a) Eigenvalues for even states (b) Eigenvalues for odd states

Figure 4.13 Graphical solutions of the eigenvalue equations for the even states and the odd states for the double-delta potential $V(x) = -V_0\delta(x-a) - V_0\delta(x+a)$.

Similarly, the continuity of the first derivative of $\psi_-(x)$ at $x = a$ yields

$$-kAe^{-ka} - kB\cosh ka + \frac{2mV_0}{\hbar^2}Ae^{-ka} = 0 \implies A\left(\frac{2mV_0}{k\hbar^2} - 1\right)e^{-ka} = B\cosh ka. \tag{4.249}$$

Dividing (4.248) by (4.245) we obtain the eigenvalue equation for the even solutions:

$$\frac{2mV_0}{k\hbar^2} - 1 = \tanh ka \quad \implies \quad \tanh y = \frac{\gamma}{y} - 1, \tag{4.250}$$

where $y = ka$ and $\gamma = 2maV_0/\hbar^2$. The eigenvalue equation for the odd solutions can be obtained by dividing (4.249) by (4.246):

$$\frac{2mV_0}{k\hbar^2} - 1 = \coth ka \quad \implies \quad \coth y = \frac{\gamma}{y} - 1 \quad \implies \quad \tanh y = \left(\frac{\gamma}{y} - 1\right)^{-1}, \tag{4.251}$$

because $\coth y = 1/\tanh y$.

To obtain the energy eigenvalues for the even and odd solutions, we need to solve the transcendental equations (4.250) and (4.251). These equations can be solved graphically. In what follows, let us determine the upper and lower limits of the energy for both the even and odd solutions.

(c) To find the number of bound states and the limits on the energy, let us consider the even and odd states separately.

Energies corresponding to the even solutions

There is only one bound state, since the curves $\tanh y$ and $\gamma/y - 1$ intersect only once (Figure 4.13a); we call this point $y = y_0$. When $y = \gamma$ we have $\gamma/y - 1 = 0$, while $\tanh\gamma > 0$. Therefore $y_0 < \gamma$. On the other hand, since $\tanh y_0 < 1$ we have $\gamma/y_0 - 1 < 1$ or $y_0 > \gamma/2$. We conclude then that $\gamma/2 < y_0 < \gamma$ or

$$\frac{\gamma}{2} < y_0 < \gamma \quad \implies \quad -\frac{2mV_0^2}{\hbar^2} < E_{even} < -\frac{mV_0^2}{2\hbar^2}. \tag{4.252}$$

In deriving this relation, we have used the fact that $\gamma^2/4 < y_0^2 < \gamma^2$ where $\gamma = 2maV_0/\hbar^2$ and $y_0^2 = k_0^2a^2 = -2ma^2E_{even}/\hbar^2$. So there is always one even bound state, the ground state, whose energy lies within the range specified by (4.252).

Energies corresponding to the odd solutions

As shown in Figure 4.13b, if the slope of $(\gamma/y-1)^{-1}$ at $y=0$ is smaller than the slope of $\tanh y$, i.e.,

$$\left.\frac{d}{dy}\left(\frac{\gamma}{y}-1\right)^{-1}\right|_{y=0} < \left.\frac{d\tanh y}{dy}\right|_{y=0} \quad\Longrightarrow\quad \frac{1}{\gamma}<1 \tag{4.253}$$

or

$$\gamma>1 \qquad\Longrightarrow\qquad V_0>\frac{\hbar^2}{2ma}, \tag{4.254}$$

there would be only one bound state because the curves $\tanh y$ and $(\gamma/y-1)^{-1}$ would intersect once. But if $\gamma<1$ or $V_0<\hbar^2/(2ma)$, there would be no odd bound states, for the curves of $\tanh y$ and $(\gamma/y-1)^{-1}$ would never intersect.

Note that if $y=\gamma/2$ we have $(\gamma/y-1)^{-1}=1$. Thus the intersection of $\tanh y$ and $(\gamma/y-1)^{-1}$, if it takes place at all, has to take place for $y<\gamma/2$. That is, the odd bound states occur only when

$$y<\frac{\gamma}{2} \qquad\Longrightarrow\qquad E_{odd}>-\frac{mV_0^2}{2\hbar^2}. \tag{4.255}$$

A comparison of (4.252) and (4.255) shows that the energies corresponding to even states are smaller than those of odd states:

$$E_{even}<E_{odd}. \tag{4.256}$$

Thus, the even bound state is the ground state. Using this result, we may infer (a) if $\gamma<1$ there are no odd bound states, but there is always one even bound state, the ground state; (b) if $\gamma>1$ there are two bound states: the ground state (even) and the first excited state (odd).

We may summarize these results as follows:

$$\boxed{\text{If}\quad \gamma<1 \quad\text{or}\quad V_0<\frac{\hbar^2}{2ma} \longrightarrow \text{there is only one bound state.}} \tag{4.257}$$

$$\boxed{\text{If}\quad \gamma>1 \quad\text{or}\quad V_0>\frac{\hbar^2}{2ma} \longrightarrow \text{there are two bound states .}} \tag{4.258}$$

(d) In the limit $a\longrightarrow 0$ we have $y\longrightarrow 0$ and $\gamma\longrightarrow 0$; hence the even transcendental equation $\tanh y=\gamma/y-1$ reduces to $y\simeq\gamma/y-1$ or $y=\gamma$, which in turn leads to $y^2=(ka)^2=\gamma^2$ or $-2ma^2E_{even}/\hbar^2=(2maV_0/\hbar^2)^2$:

$$E_{even}=-\frac{2mV_0^2}{\hbar^2}. \tag{4.259}$$

Note that in the limit $a\longrightarrow 0$, the potential $V(x)=-V_0\delta(x-a)-V_0\delta(x+a)$ reduces to $V(x)=-2V_0\delta(x)$. We can see that the ground state energy (4.231) of the single-delta potential is identical with (4.259) provided we replace V_0 in (4.231) by $2V_0$.

In the limit $a\longrightarrow\infty$, we have $y\longrightarrow\infty$ and $\gamma\longrightarrow\infty$; hence $\tanh y=\gamma/y-1$ reduces to $1\simeq\gamma/y-1$ or $y=\gamma/2$. This leads to $y^2=(ka)^2=\gamma^2/4$ or $-2ma^2E_{even}/\hbar^2=(maV_0/\hbar^2)^2$:

$$E_{even}=-\frac{mV_0^2}{2\hbar^2}. \tag{4.260}$$

This relation is identical with that of the single-delta potential (4.231).

Problem 4.6
Consider a particle of mass m subject to the potential

$$V(x) = \begin{cases} \infty, & x \leq 0, \\ -V_0\delta(x-a), & x > 0, \end{cases}$$

where $V_0 > 0$. Discuss the existence of bound states in terms of the size of a.

Solution
The Schrödinger equation for $x > 0$ is

$$\frac{d^2\psi(x)}{dx^2} + \left[\frac{2mV_0}{\hbar^2}\delta(x-a) - k^2\right]\psi(x) = 0, \tag{4.261}$$

where $k^2 = -2mE/\hbar^2$, since we are looking here at the bound states only, $E < 0$. The solutions of this equation are

$$\psi(x) = \begin{cases} \psi_1(x) = Ae^{kx} + Be^{-kx}, & 0 < x < a, \\ \psi_2(x) = Ce^{-kx}, & x > a. \end{cases} \tag{4.262}$$

The energy eigenvalues can be obtained from the boundary conditions. As the wave function vanishes at $x = 0$, we have

$$\psi_1(0) = 0 \quad \Longrightarrow \quad A + B = 0 \quad \Longrightarrow \quad B = -A. \tag{4.263}$$

The continuity condition at $x = a$ of $\psi(x)$, $\psi_1(a) = \psi_2(a)$, leads to

$$Ae^{ka} - Ae^{-ka} = Ce^{-ka}. \tag{4.264}$$

To obtain the discontinuity condition for the first derivative of $\psi(x)$ at $x = a$, we need to integrate (4.261):

$$\lim_{\varepsilon\to a}\left[\psi_2'(a+\varepsilon) - \psi_1'(a-\varepsilon)\right] + \frac{2mV_0}{\hbar^2}\psi_2(a) = 0 \tag{4.265}$$

or

$$-kCe^{-ka} - kAe^{ka} - kAe^{-ka} + \frac{2mV_0}{\hbar^2}Ce^{-ka} = 0 \tag{4.266}$$

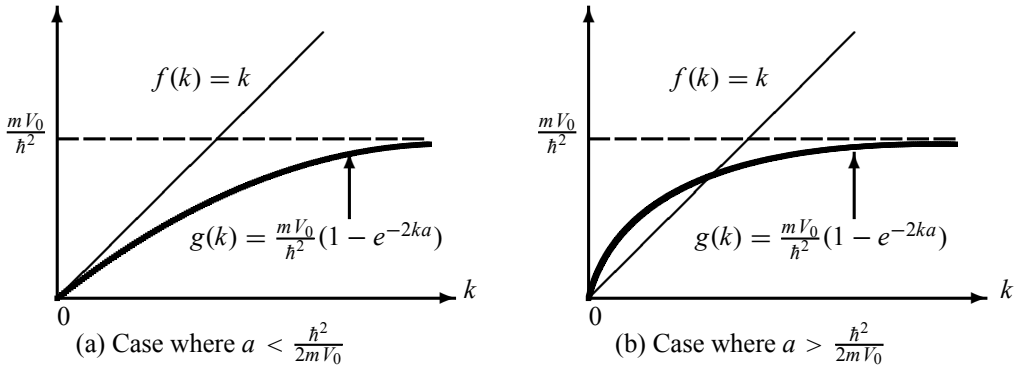
Substituting $Ce^{-ka} = Ae^{ka} - Ae^{-ka}$ or (4.264) into (4.266) we have

$$-kAe^{ka} + kAe^{-ka} - kAe^{ka} - kAe^{-ka} + \frac{2mV_0}{\hbar^2}\left(Ae^{ka} - Ae^{-ka}\right) = 0. \tag{4.267}$$

From this point on, we can proceed in two different, yet equivalent, ways. These two methods differ merely in the way we exploit (4.267). For completeness of the presentation, let us discuss both methods.

First method
The second and fourth terms of (4.267) cancel each other, so we can reduce it to

$$-kAe^{ka} - kAe^{ka} + \frac{2mV_0}{\hbar^2}\left(Ae^{ka} - Ae^{-ka}\right) = 0, \tag{4.268}$$



(a) Case where $a < \frac{\hbar^2}{2mV_0}$ (b) Case where $a > \frac{\hbar^2}{2mV_0}$

Figure 4.14 Graphical solutions of $f(k) = g(k)$ or $k = (mV_0/\hbar^2)(1 - e^{-2ka})$. If the slope of $g(k)$ is smaller than 1, i.e., $a < \hbar^2/(2mV_0)$, no bound state will exist, but if the slope of $g(k)$ is greater than 1, i.e., $a > \frac{\hbar^2}{2mV_0}$, there will be only one bound state.

which in turn leads to the following transcendental equation:

$$k = \frac{mV_0}{\hbar^2}\left(1 - e^{-2ka}\right). \tag{4.269}$$

The energy eigenvalues are given by the intersection of the curves $f(k) = k$ and $g(k) = mV_0(1 - e^{-2ka})/\hbar^2$. As the slope of $f(k)$ is equal to 1, if the slope of $g(k)$ at $k = 0$ is smaller than 1 (i.e., $a < \hbar^2/(2mV_0)$), there will be no bound states (Figure 4.14a). But if the slope of $g(k)$ is greater than 1 (i.e., $a > \hbar^2/(2mV_0)$),

$$\left.\frac{dg(k)}{dk}\right|_{k=0} > 1 \qquad \text{or} \qquad a > \frac{\hbar^2}{2mV_0}, \tag{4.270}$$

and there will be one bound state (Figure 4.14b).

Second method

We simply combine the first and second terms of (4.267) to generate $-2kA\sinh(ka)$; the third and fourth terms yield $-2kA\cosh(ka)$; and the fifth and sixth terms lead to $2A(2mV_0/\hbar^2)\sinh ka$. Hence

$$-2kA\sinh ka - 2kA\cosh ka + (2A)\frac{2mV_0}{\hbar^2}\sinh ka = 0, \tag{4.271}$$

which leads to

$$\gamma\coth\gamma = \frac{2mV_0}{\hbar^2}a - \gamma, \tag{4.272}$$

where $\gamma = ka$. The energy eigenvalues are given by the intersection of the curves $h(\gamma) = \gamma\coth\gamma$ and $u(\gamma) = 2mV_0a/\hbar^2 - \gamma$. As displayed in Figure 4.15a, if $a < 2mV_0/\hbar^2$, no bound state solution will exist, since the curves of $h(\gamma)$ and $u(\gamma)$ do not intersect. But if $a > 2mV_0/\hbar^2$, the curves intersect only once; hence there will be one bound state (Figure 4.15b).

We may summarize the results as follows:

$$a < \frac{\hbar^2}{2mV_0} \quad \Longrightarrow \quad \text{no bound states,} \tag{4.273}$$

$$a > \frac{\hbar^2}{2mV_0} \quad \Longrightarrow \quad \text{one bound state.} \tag{4.274}$$

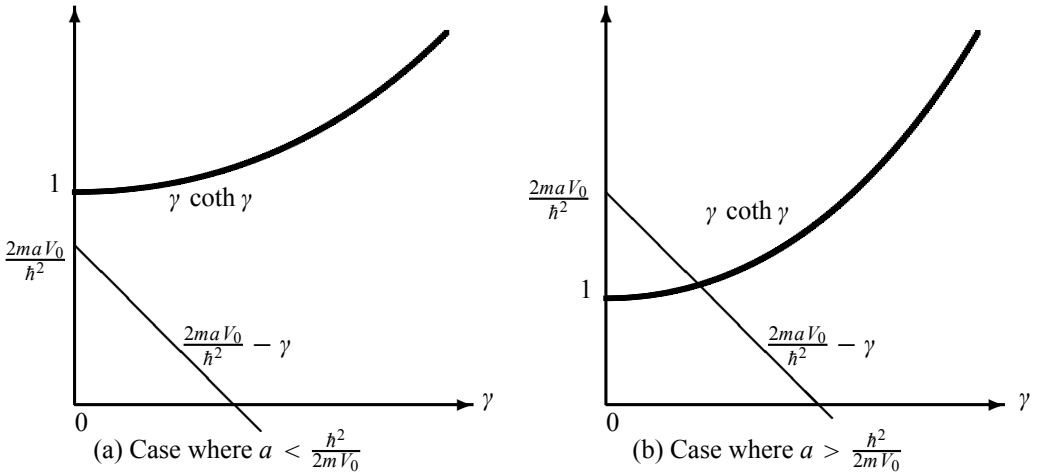


Figure 4.15 Graphical solutions of $h(\gamma) = u(\gamma)$, with $\gamma = ka$, $h(\gamma) = \gamma \coth \gamma$, and $u(\gamma) = 2mV_0a/\hbar^2 - \gamma$. If $a < 2mV_0/\hbar^2$ there is no bound state. If $a > 2mV_0/\hbar^2$ there is one bound state.

Problem 4.7
A particle of mass m, besides being confined to move inside an infinite square well potential of size a with walls at $x = 0$ and $x = a$, is subject to a delta potential of strength V_0

$$V(x) = \begin{cases} V_0\delta(x - a/2), & 0 < x < a, \\ \infty, & \text{elsewhere,} \end{cases}$$

where $V_0 > 0$. Show how to calculate the energy levels of the system in terms of V_0 and a.

Solution
The Schrödinger equation

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mV_0}{\hbar^2}\delta\left(x - \frac{a}{2}\right)\psi(x) + \frac{2mE}{\hbar^2}\psi(x) = 0 \tag{4.275}$$

can be written for $x \neq a/2$ as

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0. \tag{4.276}$$

The solutions of this equation must vanish at $x = 0$ and $x = a$:

$$\psi(x) = \begin{cases} \psi_L(x) = A \sin kx, & 0 \leq x < a/2, \\ \psi_R(x) = B \sin k(x - a), & a/2 < x \leq a, \end{cases} \tag{4.277}$$

where $k = \sqrt{2m|E|}/\hbar$. The continuity of $\psi(x)$ at $x = a/2$, $\psi_L(a/2) = \psi_R(a/2)$, leads to $A \sin(a/2) = -B \sin(a/2)$; hence $B = -A$. The wave function is thus given by

$$\psi(x) = \begin{cases} \psi_L(x) = A \sin kx, & 0 \leq x < a/2, \\ \psi_R(x) = -A \sin k(x - a), & a/2 < x \leq a, \end{cases} \tag{4.278}$$

The energy eigenvalues can be found from the discontinuity condition of the first derivative of the wave function, which in turn can be obtained by integrating (4.275) from $a/2-\varepsilon$ to $a/2+\varepsilon$ and then letting $\varepsilon \to 0$:

$$\lim_{\varepsilon\to 0}\left(\left.\frac{d\psi_R(x)}{dx}\right|_{x=a/2+\varepsilon} - \left.\frac{d\psi_L(x)}{dx}\right|_{x=a/2-\varepsilon}\right)_{\varepsilon=0} + \frac{2mV_0}{\hbar^2}\psi\left(\frac{a}{2}\right) = 0. \tag{4.279}$$

Substituting (4.278) into (4.279) we obtain

$$-kA\cos\left[k\left(\frac{a}{2}-a\right)\right] - kA\cos\left(k\frac{a}{2}\right) + A\frac{2mV_0}{\hbar^2}\sin\left(k\frac{a}{2}\right) = 0 \tag{4.280}$$

or

$$\tan\left(k\frac{a}{2}\right) = \frac{\hbar^2 k}{mV_0} \quad\Longrightarrow\quad \tan\left(\sqrt{\frac{ma^2|E|}{2\hbar^2}}\right) = \sqrt{\frac{2\hbar^2|E|}{mV_0^2}}. \tag{4.281}$$

This is a transcendental equation for the energy; its solutions, which can be obtained numerically or graphically, yield the values of E.

Problem 4.8
Using the uncertainty principle, show that the lowest energy of an oscillator is $\hbar\omega/2$.

Solution
The motion of the particle is confined to the region $-a/2 \le x \le a/2$; that is, $\Delta x \simeq a$. Then as a result of the uncertainty principle, the lowest value of this particle's momentum is $\hbar/(2\Delta x) \simeq \hbar/(2a)$. The total energy as a function of a is

$$E(a) \simeq \frac{1}{2m}\left(\frac{\hbar}{2a}\right)^2 + \frac{1}{2}m\omega^2 a^2. \tag{4.282}$$

The minimization of E with respect to a,

$$0 = \left.\frac{dE}{da}\right|_{a=a_0} = -\frac{\hbar^2}{4ma_0^3} + m\omega^2 a_0, \tag{4.283}$$

gives $a_0 = \sqrt{\hbar/2m\omega}$ and hence $E(a_0) \simeq \hbar\omega/2$; this is equal to the exact value of the oscillator's zero-point energy.

Problem 4.9
Find the energy levels of a particle of mass m moving in a one-dimensional potential:

$$V(x) = \begin{cases} +\infty, & x \le 0, \\ \frac{1}{2}m\omega^2 x^2, & x > 0. \end{cases}$$

Solution
This is an asymmetric harmonic oscillator potential in which the particle moves only in the region $x > 0$. The only acceptable solutions are those for which the wave function vanishes at $x = 0$. These solutions must be those of an ordinary (symmetric) harmonic oscillator that have odd parity, since the wave functions corresponding to the symmetric harmonic oscillator are

either even (n even) or odd (n odd), and only the odd solutions vanish at the origin, $\psi_{2n+1}(0) = 0$ ($n = 0, 1, 2, 3, \ldots$). Therefore, the energy levels of this asymmetric potential must be given by those corresponding to the odd n energy levels of the symmetric potential, i.e.,

$$E_n = \left[(2n+1) + \frac{1}{2}\right]\hbar\omega = \left(2n + \frac{3}{2}\right)\hbar\omega \qquad (n = 0, 1, 2, 3, \ldots). \tag{4.284}$$

Problem 4.10
Consider the box potential

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & \text{elsewhere.} \end{cases}.$$

(a) Estimate the energies of the ground state as well as those of the first and the second excited states for (i) an electron enclosed in a box of size $a = 10^{-10}$ m (express your answer in electron volts; you may use these values: $\hbar c = 200\,\text{MeV fm}$, $m_e c^2 = 0.5\,\text{MeV}$); (ii) a 1 g metallic sphere which is moving in a box of size $a = 10\,\text{cm}$ (express your answer in joules).

(b) Discuss the importance of the quantum effects for both of these two systems.

(c) Use the uncertainty principle to estimate the velocities of the electron and the metallic sphere.

Solution
The energy of a particle of mass m in a box having perfectly rigid walls is given by

$$E_n = \frac{n^2 h^2}{8ma^2}, \qquad n = 1, 2, 3, \ldots, \tag{4.285}$$

where a is the size of the box.

(a) (i) For the electron in the box of size 10^{-10} m, we have

$$\begin{aligned} E_n &= \frac{\hbar^2 c^2}{m_e c^2 a^2}\frac{4\pi^2 n^2}{8} \equiv \frac{4 \times 10^4\,(\text{MeV fm})^2}{0.5\,\text{MeV} \times 10^{10}\,\text{fm}^2}\frac{\pi^2}{2}n^2 \\ &= 4\pi^2 n^2\,\text{eV} \simeq 39n^2\,\text{eV}. \end{aligned} \tag{4.286}$$

Hence $E_1 = 39\,\text{eV}$, $E_2 = 156\,\text{eV}$, and $E_3 = 351\,\text{eV}$.

(ii) For the sphere in the box of side 10 cm we have

$$E_n = \frac{(6.6 \times 10^{-34}\,\text{J s})^2}{10^{-3}\,\text{kg} \times 10^{-2}\,\text{m}^2}n^2 = 43.6 \times 10^{-63} n^2\,\text{J} \tag{4.287}$$

Hence $E_1 = 43.6 \times 10^{-63}$ J, $E_2 = 174.4 \times 10^{-63}$ J, and $E_3 = 392.4 \times 10^{-63}$ J.

(b) The differences between the energy levels are

$$(E_2 - E_1)_{electron} = 117\,\text{eV}, \qquad (E_3 - E_2)_{electron} = 195\,\text{eV}, \tag{4.288}$$

$$(E_2 - E_1)_{sphere} = 130.8 \times 10^{-63}\,\text{J}, \qquad (E_3 - E_2)_{sphere} = 218 \times 10^{-63}\,\text{J}. \tag{4.289}$$

These results show that:

- The spacings between the energy levels of the electron are quite large; the levels are far apart from each other. Thus, the quantum effects are important.

- The energy levels of the sphere are practically indistinguishable; the spacings between the levels are negligible. The energy spectrum therefore forms a continuum; hence the quantum effects are not noticeable for the sphere.

(c) According to the uncertainty principle, the speed is proportional to $v \sim \hbar/(ma)$. For the electron, the typical distances are atomic, $a \simeq 10^{-10}$ m; hence

$$v \sim \frac{\hbar c}{mc^2 a} c \sim \frac{200\,\text{MeV fm}}{0.5\,\text{MeV} \times 10^5\,\text{fm}} c \simeq 4 \times 10^{-3} c = 1.2 \times 10^6 \text{m s}^{-1}, \tag{4.290}$$

where c is the speed of light. The electron therefore moves quite fast; this is expected since we have confined the electron to move within a small region.

For the sphere, the typical distances are in the range of 1 cm:

$$v \sim \frac{\hbar}{ma} \sim \frac{6.6 \times 10^{-34}\,\text{J s}}{10^{-3}\,\text{kg} \times 10^{-2}\,\text{m}} \simeq 6.6 \times 10^{-29} \text{m s}^{-1} \tag{4.291}$$

At this speed the sphere is practically at rest.

Problem 4.11

(a) Verify that the matrices representing the operators $\hat{X}$ and $\hat{P}$ in the N-space for a harmonic oscillator obey the correct commutation relation $[\hat{X},\ \hat{P}] = i\hbar$.

(b) Show that the energy levels of the harmonic oscillator can be obtained by inserting the matrices of $\hat{X}$ and $\hat{P}$ into the Hamiltonian $\hat{H} = \hat{P}^2/(2m) + \frac{1}{2}m\omega^2\hat{X}^2$.

Solution

(a) Using the matrices of $\hat{X}$ and $\hat{P}$ in (4.181) and (4.182), we obtain

$$\hat{X}\hat{P} = i\frac{\hbar}{2}\begin{pmatrix} 1 & 0 & -\sqrt{2} & \cdots \\ 0 & 1 & 0 & \cdots \\ \sqrt{2} & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad \hat{P}\hat{X} = i\frac{\hbar}{2}\begin{pmatrix} -1 & 0 & -\sqrt{2} & \cdots \\ 0 & -1 & 0 & \cdots \\ \sqrt{2} & 0 & -1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}; \tag{4.292}$$

hence

$$\hat{X}\hat{P} - \hat{P}\hat{X} = i\hbar\begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \tag{4.293}$$

or $[\hat{X},\ \hat{P}] = i\hbar I$, where I is the unit matrix.

(b) Again, using the matrices of $\hat{X}$ and $\hat{P}$ in (4.181) and (4.182), we can verify that

$$\hat{X}^2 = \frac{\hbar}{2m\omega}\begin{pmatrix} 1 & 0 & \sqrt{2} & \cdots \\ 0 & 3 & 0 & \cdots \\ \sqrt{2} & 0 & 5 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad \hat{P}^2 = -\frac{m\hbar\omega}{2}\begin{pmatrix} -1 & 0 & \sqrt{2} & \cdots \\ 0 & -3 & 0 & \cdots \\ \sqrt{2} & 0 & -5 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}; \tag{4.294}$$

hence

$$\frac{\hat{P}^2}{2m} + \frac{1}{2}m\omega^2\hat{X}^2 = \frac{\hbar\omega}{2}\begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 3 & 0 & \cdots \\ 0 & 0 & 5 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{4.295}$$

The form of this matrix is similar to the result we obtain from an analytical treatment, $E_n = \hbar\omega(2n+1)/2$, since

$$H_{n'n} = \langle n'|\hat{H} \mid n\rangle = \frac{\hbar\omega}{2}(2n+1)\delta_{n'n}. \tag{4.296}$$

Problem 4.12
Calculate the probability of finding a particle in the classically forbidden region of a harmonic oscillator for the states $n = 0, 1, 2, 3, 4$. Are these results compatible with their classical counterparts?

Solution
The classical turning points are defined by $E_n = V(x_n)$ or by $\hbar\omega(n+\frac{1}{2}) = \frac{1}{2}m\omega^2 x_n^2$; that is, $x_n = \pm\sqrt{\hbar/(m\omega)}\sqrt{2n+1}$. Thus, the probability of finding a particle in the classically forbidden region for a state $\psi_n(x)$ is

$$P_n = \int_{-\infty}^{-|x_n|} |\psi_n(x)|^2\,dx + \int_{|x_n|}^{+\infty} |\psi_n(x)|^2\,dx = 2\int_{|x_n|}^{+\infty} |\psi_n(x)|^2\,dx, \tag{4.297}$$

where $\psi_n(x)$ is given in (4.172), $\psi_n(x) = 1/\sqrt{\sqrt{\pi}2^n n! x_0}e^{-x^2/2x_0^2}H_n(x/x_0)$, where x_0 is given by $x_0 = \sqrt{\hbar/(m\omega)}$. Using the change of variable $y = x/x_0$, we can rewrite P_n as

$$P_n = \frac{2}{\sqrt{\pi}2^n n!}\int_{\sqrt{2n+1}}^{+\infty} e^{-y^2}H_n^2(y)\,dy, \tag{4.298}$$

where the Hermite polynomials $H_n(y)$ are listed in (4.120). The integral in (4.298) can be evaluated only numerically. Using the numerical values

$$\int_1^\infty e^{-y^2}dy = 0.1394, \qquad \int_{\sqrt{3}}^\infty y^2 e^{-y^2}dy = 0.0495, \tag{4.299}$$

$$\int_{\sqrt{5}}^\infty \left(4y^2-2\right)^2 e^{-y^2}dy = 0.6740, \qquad \int_{\sqrt{7}}^\infty \left(8y^3-12y\right)^2 e^{-y^2}dy = 3.6363, \tag{4.300}$$

$$\int_{\sqrt{9}}^\infty \left(16y^4-48y^2+12\right)^2 e^{-y^2}dx = 26.86, \tag{4.301}$$

we obtain

$$P_0 = 0.1573, \qquad P_1 = 0.1116, \qquad P_2 = 0.095\,069, \tag{4.302}$$

$$P_3 = 0.085\,48, \qquad P_4 = 0.078\,93. \tag{4.303}$$

This shows that the probability decreases as n increases, so it would be very small for very large values of n. It is therefore unlikely to find the particle in the classically forbidden region when the particle is in a very highly excited state. This is what we expect, since the classical approximation is recovered in the limit of high values of n.

Problem 4.13
Consider a particle of mass m moving in the following potential

$$V(x) = \begin{cases} \infty, & x \leq 0, \\ -V_0, & 0 < x < a, \\ 0, & x \geq a, \end{cases}$$

where $V_0 > 0$.

(a) Find the wave function.

(b) Show how to obtain the energy eigenvalues from a graph.

(c) Calculate the minimum value of V_0 (in terms of m, a, and $\hbar$) so that the particle will have one bound state; then calculate it for two bound states. From these two results, try to obtain the lowest value of V_0 so that the system has n bound states.

Solution

(a) As shown in Figure 4.16, the wave function in the region $x < 0$ is zero, $\psi(x) = 0$. In the region $x > 0$ the Schrödinger equation for the bound state solutions, $-V_0 < E < 0$, is given by

$$\left(\frac{d^2}{dx^2} + k_1^2\right)\psi_1(x) = 0 \qquad (0 < x < a), \tag{4.304}$$

$$\left(\frac{d^2}{dx^2} - k_2^2\right)\psi_2(x) = 0 \qquad (x > a), \tag{4.305}$$

where $k_1^2 = 2m(V_0 + E)/\hbar^2$ and $k_2^2 = -2mE/\hbar^2$. On one hand, the solution of (4.304) is oscillatory, $\psi_1(x) = A \sin k_1 x + B \cos k_1 x$, but since $\psi_1(0) = 0$ we must have $B = 0$. On the other hand, eliminating the physically unacceptable solutions which grow exponentially for large values of x, the solution of (4.305) is $\psi_2(x) = Ce^{-k_2 x}$. Thus, the wave function is given by

$$\psi(x) = \begin{cases} 0, & x < 0, \\ \psi_1(x) = A \sin k_1 x, & 0 < x < 0, \\ \psi_2(x) = Ce^{-k_2 x}, & x > a. \end{cases} \tag{4.306}$$

(b) To determine the eigenvalues, we need to use the boundary conditions at $x = a$. The condition $\psi_1(a) = \psi_2(a)$ yields

$$A \sin k_1 a = Ce^{-k_2 a}, \tag{4.307}$$

while the continuity of the first derivative, $\psi_1'(a) = \psi_2\prime(a)$, leads to

$$Ak_1 \cos k_1 a = -Ck_2 e^{-k_2 a}. \tag{4.308}$$

Dividing (4.308) by (4.307) we obtain

$$k_1 a \cot k_1 a = -k_2 a. \tag{4.309}$$

Since $k_1^2 = 2m(V_0 + E)/\hbar^2$ and $k_2^2 = -2mE/\hbar^2$, we have

$$(k_1 a)^2 + (k_2 a)^2 = \gamma^2, \tag{4.310}$$

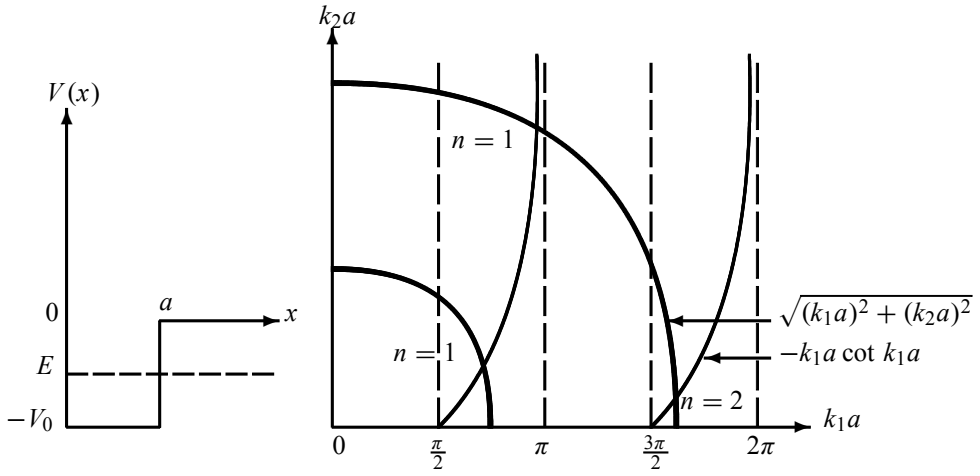


Figure 4.16 Potential $V(x)$ (left curve); the energy levels of $V(x)$ are given graphically by the intersection of the circular curve $\sqrt{(k_1a)^2 + (k_2a)^2}$ with $-k_1a \cot k_1a$ (right curve).

where $\gamma = \sqrt{2mV_0}a/\hbar$.

The transcendental equations (4.309) and (4.310) can be solved graphically. As shown in Figure 4.16, the energy levels are given by the intersection of the circular curve $(k_1a)^2 + (k_2a)^2 = \gamma^2$ with $k_1a \cot k_1a = -k_2a$.

(c) If $\pi/2 < \gamma < 3\pi/2$ there will be only one bound state, the ground state $n = 1$, for there is only one crossing between the curves $(k_1a)^2 + (k_2a)^2 = \gamma^2$ and $k_1a \cot k_1a = -k_2a$. The lowest value of V_0 that yields a single bound state is given by the relation $\gamma = \pi/2$, which leads to $2ma^2V_0/\hbar^2 = \pi^2/4$ or to

$$V_0 = \frac{\pi^2\hbar^2}{8ma^2}. \tag{4.311}$$

Similarly, if $3\pi/2 < \gamma < 5\pi/2$ there will be two crossings between $(k_1a)^2 + (k_2a)^2 = \gamma^2$ and $k_1a \cot k_1a = -k_2a$. Thus, there will be two bound states: the ground state, $n = 1$, and the first excited state, $n = 2$. The lowest value of V_0 that yields two bound states corresponds to $2ma^2V_0/\hbar^2 = 9\pi^2/4$ or to

$$V_0 = \frac{9\pi^2\hbar^2}{8ma^2}. \tag{4.312}$$

We may thus infer the following general result. If $n\pi - \pi/2 < \gamma < n\pi + \pi/2$, there will be n crossings and hence n bound states:

$$n\pi - \frac{\pi}{2} < \frac{\sqrt{2mV_0}}{\hbar}a < n\pi + \frac{\pi}{2} \quad \Longrightarrow \quad \text{there are } n \text{ bound states.} \tag{4.313}$$

The lowest value of V_0 giving n bound states is

$$V_0 = \frac{\pi^2\hbar^2}{8ma^2}(2n-1)^2. \tag{4.314}$$

Problem 4.14

(a) Assuming the potential seen by a neutron in a nucleus to be schematically represented by a one-dimensional, infinite rigid walls potential of length 10 fm, estimate the minimum kinetic energy of the neutron.

(b) Estimate the minimum kinetic energy of an electron bound within the nucleus described in (a). Can an electron be confined in a nucleus? Explain.

Solution

The energy of a particle of mass m in a one-dimensional box potential having perfectly rigid walls is given by

$$E_n = \frac{\pi^2\hbar^2}{2ma^2}n^2, \qquad n = 1, 2, 3, \ldots, \tag{4.315}$$

where a is the size of the box.

(a) Assuming the neutron to be nonrelativistic (i.e., its energy $E \ll m_nc^2$), the lowest energy the neutron can have in a box of size $a = 10$ fm is

$$E_{min} = \frac{\pi^2\hbar^2}{2m_na^2} = \frac{\pi^2(\hbar^2c^2)}{2(m_nc^2)a^2} \simeq 2.04\,\text{MeV}, \tag{4.316}$$

where we have used the fact that the rest mass energy of a neutron is $m_nc^2 \simeq 939.57$ MeV and $\hbar c \simeq 197.3$ MeV fm. Indeed, we see that $E_{min} \ll m_nc^2$.

(b) The minimum energy of a (nonrelativistic) electron moving in a box of size $a = 10$ fm is given by

$$E_{min} = \frac{\pi^2\hbar^2}{2m_ea^2} = \frac{\pi^2(\hbar^2c^2)}{2(m_ec^2)a^2} \simeq 3755.45\,\text{MeV}. \tag{4.317}$$

The rest mass energy of an electron is $m_ec^2 \simeq 0.511$ MeV, so this electron is ultra-relativistic since $E_{min} \gg m_ec^2$. It implies that an electron with this energy cannot be confined within such a nucleus.

Problem 4.15

(a) Calculate the expectation value of the operator $\hat{X}^4$ in the N-representation with respect to the state $|\,n\rangle$ (i.e., $\langle n \mid \hat{X}^4 \mid n\rangle$).

(b) Use the result of (a) to calculate the energy E_n for a particle whose Hamiltonian is $\hat{H} = \hat{P}^2/(2m) + \frac{1}{2}m\omega^2\hat{X}^2 - \lambda\hat{X}^4$.

Solution

(a) Since $\sum_{m=0}^{\infty} |m\rangle\langle m| = 1$ we can write the expectation value of $\hat{X}^4$ as

$$\langle n \mid \hat{X}^4 \mid n\rangle = \sum_{m=0}^{\infty}\langle n \mid \hat{X}^2|m\rangle\langle m|\hat{X}^2 \mid n\rangle = \sum_{m=0}^{\infty}\left|\langle m|\hat{X}^2 \mid n\rangle\right|^2. \tag{4.318}$$

Now since

$$\hat{X}^2 = \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}\right) = \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}^{\dagger 2} + 2\hat{a}^\dagger\hat{a} + 1\right), \tag{4.319}$$

the only terms $\langle m|\hat{X}^2 \mid n\rangle$ that survive are

$$\langle n \mid \hat{X}^2 \mid n\rangle = \frac{\hbar}{2m\omega}\langle n \mid 2\hat{a}^\dagger\hat{a} + 1 \mid n\rangle = \frac{\hbar}{2m\omega}(2n+1), \tag{4.320}$$

$$\langle n-2 \mid \hat{X}^2 \mid n\rangle = \frac{\hbar}{2m\omega}\langle n-2 \mid \hat{a}^2 \mid n\rangle = \frac{\hbar}{2m\omega}\sqrt{n(n-1)}, \tag{4.321}$$

$$\langle n+2 \mid \hat{X}^2 \mid n\rangle = \frac{\hbar}{2m\omega}\langle n+2 \mid \hat{a}^{\dagger 2} \mid n\rangle = \frac{\hbar}{2m\omega}\sqrt{(n+1)(n+2)}. \tag{4.322}$$

Thus

$$\begin{aligned}\langle n \mid \hat{X}^4 \mid n\rangle &= \left|\langle n \mid \hat{X}^2 \mid n\rangle\right|^2 + \left|\langle n-2 \mid \hat{X}^2 \mid n\rangle\right|^2 + \left|\langle n+2 \mid \hat{X}^2 \mid n\rangle\right|^2 \\ &= \frac{\hbar^2}{4m^2\omega^2}\left[(2n+1)^2 + n(n-1) + (n+1)(n+2)\right] \\ &= \frac{\hbar^2}{4m^2\omega^2}\left(6n^2+6n+3\right).\end{aligned} \tag{4.323}$$

(b) Using (4.323), and since the Hamiltonian can be expressed in terms of the harmonic oscillator, $\hat{H} = \hat{H}_{HO} - \lambda\hat{X}^4$, we immediately obtain the particle energy:

$$E_n = \langle n \mid \hat{H}_{HO} \mid n\rangle - \lambda\langle n \mid \hat{X}^4 \mid n\rangle = \hbar\omega\left(n+\frac{1}{2}\right) - \frac{\lambda\hbar^2}{4m^2\omega^2}\left(6n^2+6n+3\right). \tag{4.324}$$

Problem 4.16
Find the energy levels and the wave functions of two harmonic oscillators of masses m_1 and m_2, having identical frequencies ω, and coupled by the interaction $\frac{1}{2}k(\hat{X}_1 - \hat{X}_2)^2$.

Solution
This problem reduces to finding the eigenvalues for the Hamiltonian

$$\begin{aligned}\hat{H} &= \hat{H}_1 + \hat{H}_2 + \frac{1}{2}K(\hat{X}_1 - \hat{X}_2)^2 \\ &= \frac{1}{2m_1}\hat{P}_1^2 + \frac{1}{2}m_1\omega^2\hat{X}_1^2 + \frac{1}{2m_2}\hat{P}_2^2 + \frac{1}{2}m_2\omega^2\hat{X}_2^2 + \frac{1}{2}K(\hat{X}_1 - \hat{X}_2)^2.\end{aligned} \tag{4.325}$$

This is a two-particle problem. As in classical mechanics, it is more convenient to describe the dynamics of a two-particle system in terms of the center of mass (CM) and relative motions. For this, let us introduce the following operators:

$$\hat{P} = \hat{p}_1 + \hat{p}_2, \qquad \hat{X} = \frac{m_1\hat{x}_1 + m_2\hat{x}_2}{M}, \tag{4.326}$$

$$\hat{p} = \frac{m_2\hat{p}_1 - m_1\hat{p}_2}{M}, \qquad \hat{x} = \hat{x}_1 - \hat{x}_2, \tag{4.327}$$

where $M = m_1 + m_2$ and $\mu = m_1m_2/(m_1+m_2)$ is the reduced mass; $\hat{P}$ and $\hat{X}$ pertain to the CM; $\hat{p}$ and $\hat{x}$ pertain to the relative motion. These relations lead to

$$\hat{p}_1 = \frac{m_1}{M}\hat{P} + \hat{p}, \qquad \hat{p}_2 = \frac{m_2}{M}\hat{P} - \hat{p}, \tag{4.328}$$

$$\hat{x}_1 = \frac{m_2}{M}\hat{x} + \hat{X}, \qquad \hat{x}_2 = -\frac{m_1}{M}\hat{x} + \hat{X}. \tag{4.329}$$

Note that the sets (X, P) and (x, p) are conjugate variables separately: $[\hat{X}, \hat{P}] = i\hbar$, $[\hat{x}, \hat{p}] = i\hbar$, $[\hat{X}, \hat{p}] = [\hat{x}, \hat{P}] = 0$. Taking $\hat{p}_1$, $\hat{p}_2$, $\hat{x}_1$, and $\hat{x}_2$ of (4.328) and (4.329) and inserting them into (4.325), we obtain

$$\begin{aligned}\hat{H} &= \frac{1}{2m_1}\left(\frac{m_1}{M}\hat{P} + \hat{p}\right)^2 + \frac{1}{2}m_1\omega^2\left(\frac{m_2}{M}\hat{x} + \hat{X}\right)^2 \\ &\quad + \frac{1}{2m_2}\left(\frac{m_2}{M}\hat{P} - \hat{p}\right)^2 + \frac{1}{2}m_2\omega^2\left(-\frac{m_1}{M}\hat{x} + \hat{X}\right)^2 + \frac{1}{2}K\hat{x}^2 \\ &= \hat{H}_{CM} + \hat{H}_{rel}, \end{aligned} \tag{4.330}$$

where

$$\hat{H}_{CM} = \frac{1}{2M}\hat{P}^2 + \frac{1}{2}M\omega^2\hat{X}^2, \qquad \hat{H}_{rel} = \frac{1}{2\mu}\hat{p}^2 + \frac{1}{2}\mu\Omega^2\hat{x}^2, \tag{4.331}$$

with $\Omega^2 = \omega^2 + k/\mu$. We have thus reduced the Hamiltonian of these two coupled harmonic oscillators to the sum of two independent harmonic oscillators, one with frequency ω and mass M and the other of mass μ and frequency $\Omega = \sqrt{\omega^2 + k/\mu}$. That is, by introducing the CM and relative motion variables, we have managed to eliminate the coupled term from the Hamiltonian.

The energy levels of this two-oscillator system can be inferred at once from the suggestive Hamiltonians of (4.331):

$$E_{n_1 n_2} = \hbar\omega\left(n_1 + \frac{1}{2}\right) + \hbar\Omega\left(n_2 + \frac{1}{2}\right). \tag{4.332}$$

The states of this two-particle system are given by the product of the two states $|N\rangle = |n_1\rangle|n_2\rangle$; hence the total wave function, $\psi_n(X, x)$, is equal to the product of the center of mass wave function, $\psi_{n_1}(X)$, and the wave function of the relative motion, $\psi_{n_2}(x)$: $\psi_n(X, x) = \psi_{n_1}(X)\psi_{n_2}(x)$. Note that both of these wave functions are harmonic oscillator functions whose forms can be found in (4.172):

$$\psi_n(X, x) = \frac{1}{\sqrt{\pi}\sqrt{2^{n_1}2^{n_2}n_1!n_2!x_{0_1}x_{0_2}}} e^{-X^2/2x_{0_1}^2} e^{-x^2/2x_{0_2}^2} H_{n_1}\left(\frac{X}{x_{0_1}}\right) H_{n_2}\left(\frac{x}{x_{0_2}}\right), \tag{4.333}$$

where $n = (n_1,\ n_2)$, $x_{0_1} = \sqrt{\hbar/(M\omega)}$, and $x_{0_2} = \sqrt{\hbar/(\mu\Omega)}$.

Problem 4.17
Consider a particle of mass m and charge q moving under the influence of a one-dimensional harmonic oscillator potential. Assume it is placed in a constant electric field $\mathcal{E}$. The Hamiltonian of this particle is therefore given by $\hat{H} = \hat{P}^2/(2m) + \frac{1}{2}m\omega^2\hat{X}^2 - q\mathcal{E}\hat{X}$. Derive the energy expression and the wave function of the nth excited state.

Solution
To find the eigenenergies of the Hamiltonian

$$\hat{H} = \frac{1}{2m}\hat{P}^2 + \frac{1}{2}m\omega^2\hat{X}^2 - q\mathcal{E}\hat{X}, \tag{4.334}$$

it is convenient to use the change of variable $y = \hat{X} - q\mathcal{E}/(m\omega^2)$. Thus the Hamiltonian becomes

$$\hat{H} = \frac{1}{2m}\hat{P}^2 + \frac{1}{2}m\omega^2\hat{y}^2 - \frac{q^2\mathcal{E}^2}{2m\omega^2}. \tag{4.335}$$

Since the term $q^2\mathcal{E}^2/(2m\omega^2)$ is a mere constant and $\hat{P}^2/(2m) + \frac{1}{2}m\omega^2\hat{y}^2 = \hat{H}_{HO}$ has the structure of a harmonic oscillator Hamiltonian, we can easily infer the energy levels:

$$E_n = \langle n \mid \hat{H} \mid n \rangle = \hbar\omega\left(n + \frac{1}{2}\right) - \frac{q^2\mathcal{E}^2}{2m\omega^2}. \tag{4.336}$$

The wave function is given by $\psi_n(y) = \psi_n(x - q\mathcal{E}/(m\omega^2))$, where $\psi_n(y)$ is given in (4.172):

$$\psi_n(y) = \frac{1}{\sqrt{\sqrt{\pi}2^n n! x_0}} e^{-y^2/2x_0^2} H_n\left(\frac{y}{x_0}\right). \tag{4.337}$$

Problem 4.18
Consider a particle of mass m that is bouncing vertically and elastically on a smooth reflecting floor in the Earth's gravitational field

$$V(z) = \begin{cases} mgz, & z > 0, \\ +\infty, & z \leq 0, \end{cases},$$

where g is a constant (the acceleration due to gravity). Find the energy levels and wave function of this particle.

Solution
We need to solve the Schrödinger equation with the boundary conditions $\psi(0) = 0$ and $\psi(+\infty) = 0$:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(z)}{dz^2} + mgz\psi(z) = E\psi(z) \implies \frac{d^2\psi(z)}{dz^2} - \frac{2m}{\hbar^2}(mgz - E)\psi(z) = 0. \tag{4.338}$$

With the change of variable $x = (\hbar^2/(2m^2g))^{2/3}(2m/\hbar^2)(mgz - E)$, we can reduce this equation to

$$\frac{d^2\phi(x)}{dx^2} - x\phi(x) = 0. \tag{4.339}$$

This is a standard differential equation; its solution (which vanishes at $x \to= +\infty$, i.e., $\phi(+\infty) = 0$) is given by

$$\phi(x) = B\mathrm{Ai}(x) \qquad \text{where} \qquad \mathrm{Ai}(x) = \frac{1}{\pi}\int_0^\infty \cos\left(\frac{1}{3}t^3 + xt\right)dt, \tag{4.340}$$

where $\mathrm{Ai}(x)$ is called the *Airy function*.

When $z = 0$ we have $x = -(2/(mg^2\hbar^2))^{1/3}E$. The boundary condition $\psi(0) = 0$ yields $\phi[-(2/(mg^2\hbar^2))^{1/3}E] = 0$ or $\mathrm{Ai}[-(2/(mg^2\hbar^2))^{1/3}E] = 0$. The Airy function has zeros only at certain values of R_n: $\mathrm{Ai}(R_n) = 0$ with $n = 0, 1, 2, 3, \ldots$. The roots R_n of the Airy function can be found in standard tables. For instance, the first few roots are $R_0 = -2.338$, $R_1 = -4.088$, $R_2 = -5.521$, $R_3 = -6.787$.

The boundary condition $\psi(0) = 0$ therefore gives a *discrete* set of energy levels which can be expressed in terms of the roots of the Airy function:

$$\mathrm{Ai}\left[-\left(\frac{2}{mg^2\hbar^2}\right)^{1/3}E\right] = 0 \quad \implies \quad -\left(\frac{2}{mg^2\hbar^2}\right)^{1/3}E_n = R_n; \tag{4.341}$$

hence

$$E_n = -\left(\frac{1}{2}mg^2\hbar^2\right)^{1/3} R_n, \qquad \psi_n(z) = B_n \mathrm{Ai}\left[-\left(\frac{2m^2g^2}{\hbar^2}\right)^{1/3} z - R_n\right]. \tag{4.342}$$

The first few energy levels are

$$E_0 = 2.338\left(\frac{1}{2}mg^2\hbar^2\right)^{1/3}, \qquad E_1 = 4.088\left(\frac{1}{2}mg^2\hbar^2\right)^{1/3}, \tag{4.343}$$

$$E_2 = 5.521\left(\frac{1}{2}mg^2\hbar^2\right)^{1/3}. \qquad E_3 = 6.787\left(\frac{1}{2}mg^2\hbar^2\right)^{1/3}. \tag{4.344}$$

4.11 Exercises

Exercise 4.1
A particle of mass m is subjected to a potential

$$V(x) = \begin{cases} 0, & |x| < a/2, \\ \infty, & |x| > a/2. \end{cases}$$

(a) Find the ground, first, and second excited state wave functions.
(b) Find expressions for E_1, E_2, and E_3.
(c) Plot the probability densities $P_2(x,t)$ and $P_3(x,t)$.
(d) Find $\langle X\rangle_2$, $\langle X\rangle_3$, $\langle P\rangle_2$, and $\langle P\rangle_3$.
(e) Evaluate $\Delta x\,\Delta p$ for the states $\psi_2(x,t)$ and $\psi_3(x,t)$.

Exercise 4.2
Consider a system whose wave function at $t = 0$ is

$$\psi(x,0) = \frac{3}{\sqrt{30}}\phi_0(x) + \frac{4}{\sqrt{30}}\phi_1(x) + \frac{1}{\sqrt{6}}\phi_4(x),$$

where $\phi_n(x)$ is the wave function of the nth excited state of an infinite square well potential of width a and whose energy is $E_n = \pi^2\hbar^2n^2/(2ma^2)$.

(a) Find the average energy of this system.

(b) Find the state $\psi(x,t)$ at a later time t and the average value of the energy. Compare the result with the value obtained in (a).

Exercise 4.3
An electron with a kinetic energy of 10 eV at large negative values of x is moving from left to right along the x-axis. The potential energy is

$$V(x) = \begin{cases} 0 & (x \le 0), \\ 20\,\mathrm{eV} & (x > 0). \end{cases}$$

(a) Write the time-independent Schrödinger equation in the regions $x \le 0$ and $x > 0$.
(b) Describe the shapes for $\psi(x)$ for $x \le 0$ and $x > 0$.

(c) Calculate the electron wavelength (in meters) in $-20\,\text{m} < x < -10\,\text{m}$ and $x > 10\,\text{m}$.

(d) Write down the boundary conditions at $x = 0$.

(e) Calculate the ratio of the probabilities for finding the electron near $x = 10^{-10}\,\text{m}$ and $x = 0$.

Exercise 4.4

A particle is moving in the potential well

$$V(x) = \begin{cases} 0, & -a \le x \le -b, \\ V_0, & -b \le x \le b, \\ 0, & b \le x \le a, \\ +\infty & \text{elsewhere} \end{cases},$$

where V_0 is positive. In this problem consider $E < V_0$. Let $\psi_1(x)$ and $\psi_2(x)$ represent the two lowest energy solutions of the Schrödinger equation; call their energies E_1 and E_2, respectively.

(a) Calculate E_1 and E_2 in units of eV for the case where $mc^2 = 1\,\text{GeV}$, $a = 10^{-14}\,\text{m}$, and $b = 0.4 \times 10^{-14}\,\text{m}$; take $\hbar c \simeq 200\,\text{MeV fm}$.

(b) A particular solution of the Schrödinger equation can be constructed by superposing $\psi_1(x)e^{iE_1t/\hbar}$ and $\psi_2(x)e^{iE_2t/\hbar}$. Construct a wave packet ψ which at $t = 0$ is (almost) entirely to the left-hand side of the well and describe its motion in time; find the period of oscillations between the two terms of ψ.

Exercise 4.5

A particle moves in the potential

$$V(x) = \frac{\hbar^2}{2m}\left[\frac{4}{225}\sinh^2 x - \frac{2}{5}\cosh x\right].$$

(a) Sketch $V(x)$ and locate the position of the two minima.

(b) Show that $\psi(x) = (1+4\cosh x)\exp\left(-\frac{2}{15}\cosh x\right)$ is a solution of the time-independent Schrödinger equation for the particle. Find the corresponding energy level and indicate its position on the sketch of $V(x)$.

(c) Sketch $\psi(x)$ and show that it has the proper behavior at the classical turning points and in the classically forbidden regions.

Exercise 4.6

Show that for a particle of mass m which moves in a one-dimensional infinite potential well of length a, the uncertainties product $\Delta x_n \Delta p_n$ is given by $\Delta x_n \Delta p_n \simeq n\pi\hbar/\sqrt{12}$.

Exercise 4.7

A particle of mass m is moving in an infinite potential well

$$V(x) = \begin{cases} V_0, & 0 < x < a, \\ \infty, & \text{elsewhere.} \end{cases}$$

(a) Solve the Schrödinger equation and find the energy levels and the corresponding normalized wave functions.

(b) Calculate $\langle\hat{X}\rangle_5$, $\langle\hat{P}\rangle_5$, $\langle\hat{X}^2\rangle_5$, and $\langle\hat{P}^2\rangle_5$ for the fourth excited state and infer the value of $\Delta x\,\Delta p$.

Exercise 4.8
Consider the potential step

$$V(x) = \begin{cases} 6\,\text{eV}, & x < 0, \\ 0, & x > 0. \end{cases}$$

(a) An electron of energy 8 eV is moving from left to right in this potential. Calculate the probability that the electron will (i) continue moving along its initial direction after reaching the step and (ii) get reflected at the potential step.

(b) Now suppose the electron is moving from right to left with an energy 3 eV. (i) Estimate the order of magnitude of the distance the electron can penetrate the barrier. (ii) Repeat part (i) for a 70 kg person initially moving at $4\,\text{m}\,\text{s}^{-1}$ and running into a wall which can be represented by a potential step of height equal to four times this person's energy before reaching the step.

Exercise 4.9
Consider a system whose wave function at time $t = 0$ is given by

$$\psi(x,0) = \frac{5}{\sqrt{50}}\phi_0(x) + \frac{4}{\sqrt{50}}\phi_1(x) + \frac{3}{\sqrt{50}}\phi_2(x),$$

where $\phi_n(x)$ is the wave function of the nth excited state for a harmonic oscillator of energy $E_n = \hbar\omega(n + 1/2)$.

(a) Find the average energy of this system.

(b) Find the state $\psi(x,t)$ at a later time t and the average value of the energy; compare the result with the value obtained in (a).

(c)Find the expectation value of the operator $\hat{X}$ with respect to the state $\psi(x,t)$ (i.e., find $\langle\psi(x,t)|\hat{X}|\psi(x,t)\rangle$).

Exercise 4.10
Calculate $\langle n \mid \hat{X}^2 \mid m\rangle$ and $\langle m \mid \hat{X}^4 \mid n\rangle$ in the N-representation; $\mid n\rangle$ and $\mid m\rangle$ are harmonic oscillator states.

Exercise 4.11
Consider the dimensionless Hamiltonian $\hat{H} = \frac{1}{2}\hat{P}^2 + \frac{1}{2}\hat{X}^2$, with $\hat{P} = -id/dx$.

(a) Show that the wave functions $\psi_0(x) = e^{-x^2/2}/\sqrt{\sqrt{\pi}}$ and $\psi_1(x) = \sqrt{2/\sqrt{\pi}}xe^{-x^2/2}$ are eigenfunctions of $\hat{H}$ with eigenvalues $1/2$ and $3/2$, respectively.

(b) Find the values of the coefficients α and β such that

$$\psi_2(x) = \frac{1}{\sqrt{2\sqrt{\pi}}}\left(\alpha x^2 - 1\right)e^{-x^2/2} \quad \text{and} \quad \psi_3(x) = \frac{1}{\sqrt{6\sqrt{\pi}}}x\left(1 + \beta x^2\right)e^{-x^2/2}$$

are orthogonal to $\psi_0(x)$ and $\psi_1(x)$, respectively. Then show that $\psi_2(x)$ and $\psi_3(x)$ are eigenfunctions of $\hat{H}$ with eigenvalues $5/2$ and $7/2$, respectively.

Exercise 4.12
Consider the dimensionless Hamiltonian $\hat{H} = \frac{1}{2}\hat{P}^2 + \frac{1}{2}\hat{X}^2$ (with $\hat{P} = -id/dx$) whose wave function at time $t = 0$ is given by

$$\Psi(x,0) = \frac{1}{\sqrt{2}}\psi_0(x) + \frac{1}{\sqrt{8}}\psi_1(x) + \frac{1}{\sqrt{10}}\psi_2(x),$$

where $\psi_0(x) = \frac{1}{\sqrt{\sqrt{\pi}}} e^{-x^2/2}$, $\psi_1(x) = \sqrt{\frac{2}{\sqrt{\pi}}} x e^{-x^2/2}$, and $\psi_2(x) = \frac{1}{\sqrt{2\sqrt{\pi}}} \left(2x^2 - 1\right) e^{-x^2/2}$.

(a) Calculate $\Delta x_n \Delta p_n$ for $n = 0, 1$ where $\Delta x_n = \sqrt{\langle \psi_n | \hat{X}^2 | \psi_n \rangle - \langle \psi_n | \hat{X} | \psi_n \rangle^2}$.

(b) Calculate $\hat{a}^\dagger \psi_0(x)$, $\hat{a} \psi_0(x)$, $\hat{a}^\dagger \psi_1(x)$, $\hat{a} \psi_1(x)$, and $\hat{a} \psi_2(x)$, where the operators $\hat{a}^\dagger$ and $\hat{a}$ are defined by $\hat{a} = (\hat{X} + d/dx)/\sqrt{2}$ and $\hat{a}^\dagger = (\hat{X} - d/dx)/\sqrt{2}$.

Exercise 4.13
Consider a particle of mass m that is moving in a one-dimensional infinite potential well with walls at $x = 0$ and $x = a$ which is initially (i.e., at $t = 0$) in the state

$$\psi(x, 0) = \frac{1}{\sqrt{2}} \left[\phi_1(x) + \phi_3(x)\right],$$

where $\phi_1(x)$ and $\phi_3(x)$ are the ground and second excited states, respectively.

(a) What is the state vector $\psi(x, t)$ for $t > 0$ in the Schrödinger picture.

(b) Find the expectation values $\langle \hat{X} \rangle$, $\langle \hat{P} \rangle$, $\langle \hat{X}^2 \rangle$, and $\langle \hat{P}^2 \rangle$ with respect to $| \psi \rangle$.

(c) Evaluate $\Delta x \Delta p$ and verify that it satisfies the uncertainty principle.

Exercise 4.14
If the state of a particle moving in a one-dimensional harmonic oscillator is given by

$$| \psi \rangle = \frac{1}{\sqrt{17}} | 0 \rangle + \frac{3}{\sqrt{17}} | 1 \rangle - \frac{2}{\sqrt{17}} | 2 \rangle - \sqrt{\frac{3}{17}} | 3 \rangle,$$

where $| n \rangle$ represents the normalized nth energy eigenstate, find the expectation values of the number operator, $\hat{N}$, and of the Hamiltonian operator.

Exercise 4.15
Find the number of bound states and the corresponding energies for the finite square well potential when (a) $R = 7$ (i.e., $\sqrt{ma^2 V_0/(2\hbar^2)} = 7$) and (b) $R = 3\pi$.

Exercise 4.16
A ball of mass $m = 0.2\,\text{kg}$ bouncing on a table located at $z = 0$ is subject to the potential

$$V(z) = \begin{cases} V_0 & (z < 0), \\ mgz & (z > 0), \end{cases}$$

where $V_0 = 3\,\text{J}$ and g is the acceleration due to gravity.

(a) Describe the spectrum of possible energies (i.e., continuous, discrete, or nonexistent) as E increases from large negative values to large positive values.

(b) Estimate the order of magnitude for the lowest energy state.

(c) Describe the general shapes of the wave functions $\psi_0(z)$ and $\psi_1(z)$ corresponding to the lowest two energy states and sketch the corresponding probability densities.

Exercise 4.17
Consider a particle of mass m moving in a one-dimensional harmonic oscillator potential, with $\hat{X} = \sqrt{\hbar/(2m\omega)}(\hat{a} + \hat{a}^\dagger)$ and $\hat{P} = i\sqrt{m\hbar\omega/2}(\hat{a}^\dagger - \hat{a})$.

(a) Calculate the product of the uncertainties in position and momentum for the particle in the fifth excited state, i.e., $(\Delta X \Delta P)_5$.

(b) Compare the result of (a) with the uncertainty product when the particle is in its lowest energy state. Explain why the two uncertainty products are different.

Exercise 4.18
A particle of mass m in an infinite potential well of length a has the following initial wave function at $t = 0$:

$$\psi(x, 0) = \frac{2}{\sqrt{7a}} \sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{6}{7a}} \sin\left(\frac{2\pi x}{a}\right) + \frac{2}{\sqrt{7a}} \sin\left(\frac{3\pi x}{a}\right).$$

(a) If we measure energy, what values will we find and with what probabilities? Calculate the average energy.

(b) Find the wave function $\psi(x, t)$ at any later time t. Determine the probability of finding the particle at a time t in the state $\varphi(x, t) = 1/\sqrt{a} \sin(3\pi x/a) \exp(-i E_3 t/\hbar)$.

(c) Calculate the probability density $\rho(x, t)$ and the current density $\vec{J}(x, t)$. Verify that $\partial\rho/\partial t + \vec{\nabla} \cdot \vec{J}(x, t) = 0$.

Exercise 4.19
Consider a particle in an infinite square well whose wave function is given by

$$\psi(x) = \begin{cases} Ax(a^2 - x^2), & 0 < x < a, \\ 0, & \text{elsewhere,} \end{cases}$$

where A is a real constant.

(a) Find A so that $\psi(x)$ is normalized.

(b) Calculate the position and momentum uncertainties, Δx and Δp, and the product $\Delta x \, \Delta p$.

(c) Calculate the probability of finding $5^2\pi^2\hbar^2/(2ma^2)$ for a measurement of the energy.

Exercise 4.20
The relativistic expression for the energy of a free particle is $E^2 = m_0^2 \, c^4 + p^2 \, c^2$.

(a) Write down the corresponding relativistic Schrödinger equation, by quantizing this energy expression (i.e., replacing E and p with their corresponding operators). This equation is called the Klein–Gordon equation.

(b) Find the solutions corresponding to a free particle moving along the x-axis.

Exercise 4.21
(a) Write down the classical (gravitational) energy E_c of a particle of mass m at rest a height h_0 above the ground (take the zero potential energy to be located at the ground level).

(b) Use the uncertainty principle to estimate the ground state energy E_0 of the particle introduced in (a); note that the particle is subject to gravity. Compare E_0 to E_c.

(c) If $h_0 = 3\,\text{m}$ obtain the numerical values of E_c and the quantum mechanical correction $(E_0 - E_c)$ for a neutron and then for a particle of mass $m = 0.01\,\text{kg}$. Comment on the importance of the quantum correction in both cases.

Exercise 4.22
Find the energy levels and the wave functions of two noninteracting particles of masses m_1 and m_2 that are moving in a common infinite square well potential

$$V(x_i) = \begin{cases} 0, & 0 \le x_i \le a, \\ +\infty, & \text{elsewhere,} \end{cases}$$

where x_i is the position of the ith particle (i.e., x_i denotes $x = x_1$ or x_2).

Exercise 4.23
A particle of mass m is subject to a repulsive delta potential $V(x) = V_0\delta(x)$, where $V_0 > 0$ (V_0 has the dimensions of Energy×Distance). Find the reflection and transmission coefficients, R and T.

Exercise 4.24
A particle of mass m is scattered by a double-delta potential $V(x) = V_0\delta(x-a) + V_0\delta(x+a)$, where $V_0 > 0$.

(a) Find the transmission coefficient for the particle at an energy $E > 0$.

(b) When V_0 is very large (i.e., $V_0 \to \infty$), find the energies corresponding to the resonance case (i.e., $T = 1$) and compare them with the energies of an infinite square well potential having a width of $2a$.

Exercise 4.25
A particle of mass m is subject to an antisymmetric delta potential $V(x) = V_0\delta(x+a) - V_0\delta(x-a)$, where $V_0 > 0$.

(a) Show that there is always one and only one bound state, and find the expression that gives its energy.

(b) Find the transmission coefficient T.

Exercise 4.26
A particle of mass m is subject to a delta potential

$$V(x) = \begin{cases} \infty, & x \le 0, \\ V_0\delta(x-a), & x > 0, \end{cases}$$

where $V_0 > 0$.

(a) Find the wave functions corresponding to the cases $0 < x < a$ and $x > a$.

(b) Find the transmission coefficient.

Exercise 4.27
A particle of mass m, besides being confined to move in an infinite square well potential of size $2a$ with walls at $x = -a$ and $x = a$, is subject to an attractive delta potential

$$V(x) = \begin{cases} V_0\delta(x), & -a < x < a, \\ \infty, & \text{elsewhere}, \end{cases}$$

where $V_0 > 0$.

(a) Find the particle's wave function corresponding to *even* solutions when $E > 0$.

(b) Find the energy levels corresponding to *even* solutions.

Exercise 4.28
A particle of mass m, besides being confined to move in an infinite square well potential of size $2a$ with walls at $x = -a$ and $x = a$, is subject to an attractive delta potential

$$V(x) = \begin{cases} V_0\delta(x), & -a < x < a, \\ \infty, & \text{elsewhere}, \end{cases}$$

where $V_0 > 0$.

(a) Find the particle's wave function corresponding to *odd* solutions when $E > 0$.

(b) Find the energy levels corresponding to *odd* solutions.

Exercise 4.29
Consider a particle of mass m that is moving under the influence of an attractive delta potential

$$V(x) = \begin{cases} -V_0\delta(x), & x > -a, \\ \infty, & x < -a, \end{cases}$$

where $V_0 > 0$. Discuss the existence of bound states in terms of V_0 and a.

Exercise 4.30
Consider a system of two identical harmonic oscillators (with an angular frequency ω).

(a) Find the energy levels when the oscillators are independent (non-interacting).

(b) Find the energy levels when the oscillators are coupled by an interaction $-\lambda\hat{X}_1\hat{X}_2$, where λ is a constant.

(c) Assuming that $\lambda \ll m\omega^2$ (weak coupling limit), find an approximate value to first order in $\lambda/m\omega^2$ for the energy expression derived in part (b).

Exercise 4.31
A particle is initially in its ground state in an infinite one-dimensional potential box with sides at $x = 0$ and $x = a$. If the wall of the box at $x = a$ is *suddenly* moved to $x = 3a$, calculate the probability of finding the particle in

(a) the ground state of the new box and

(b) the first excited state of the new box.

(c) Now, calculate the probability of finding the particle in the first excited state of the new box, assuming the particle was initially in the first excited state of the old box.

Exercise 4.32
A particle is initially in its ground state in a one-dimensional harmonic oscillator potential, $\hat{V}(x) = \frac{1}{2}k\hat{x}^2$. If the spring constant is *suddenly* doubled, calculate the probability of finding the particle in the ground state of the new potential.

Exercise 4.33
Consider an electron in an infinite potential well

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ +\infty, & \text{elsewhere}, \end{cases}$$

where $a = 10^{-10}$ m.

(a) Calculate the energy levels of the three lowest states (the results should be expressed in eV) and the corresponding wavelengths of the electron.

(b) Calculate the frequency of the radiation that would cause the electron to jump from the ground to the third excited energy level.

(c) When the electron de-excites, what are the frequencies of the emitted photons?

(d) Specify the probability densities for all these three states and plot them.

Exercise 4.34
Consider an electron which is confined to move in an infinite square well of width $a = 10^{-10}m$.

(a) Find the exact energies of the 11 lowest states (express them in eV).

(b) Solve the Schrödinger equation *numerically* and find the energies of the 11 lowest states and compare them with the exact results obtained in (a). Plot the wave functions of the five lowest states.